itelescope_premium_the_hard_way

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1 iTelescope premium image set processing, the hard way

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This notebook illustrates how to process a premium image set from iTelescope using the AstroPhotography package. The python environment used corresponds to a miniconda emvironment `ap-env.yml`.

The example dataset used is the iTelescope Plan-20 premium dataset of the M101 supernova SN 2023 ixf. Note that these files are not provided with this package. However the notebook should work with your own files if you specify a valid fits-file containing directory at the prompt below.

1.1 Notebook environment setup

[1]:
```
import os
import pathlib
import sys
import numpy as np
import matplotlib
import matplotlib.pyplot as plt
import time
import math
import subprocess
import shlex

from ccdproc import ImageFileCollection
from astropy.table import Table
from astropy import wcs
import astropy
from astropy.io import fits

import AstroPhotography as ap
```

[2]:
```
def print_module_version(mod):
    """
    Convenience function to pretty-print Python module mod version.
```

```
    Arguments:
        mod {module} -- Imported Python module

    Returns:
        str -- module nam
e and version
    """
    print(f"Using module {mod.__name__:30s}  version: {mod.__version__}")
    return
```

[66]:
```
# Get Version information
print(f'Python version: {sys.version}')
print_module_version(np)
print_module_version(matplotlib)
print_module_version(astropy)
print_module_version(ap)
```

```
Python version: 3.10.13 | packaged by conda-forge | (main, Dec 23 2023,
15:36:39) [GCC 12.3.0]
Using module numpy                             version: 1.26.3
Using module matplotlib                        version: 3.8.2
Using module astropy                           version: 6.0.0
Using module AstroPhotography                  version: 0.5.2
```

[4]:
```
# Enable inline plotting for graphics
# %matplotlib inline
# Set default figure size to be larger
# this may only work in matplotlib 2.0+!
from IPython.core.interactiveshell import InteractiveShell
matplotlib.rcParams['figure.figsize'] = [10.0, 6.0]
# Enable multiple outputs from jupyter cells
InteractiveShell.ast_node_interactivity = "all"
```

2 Doing it by hand

2.1 What files do we have to work with?

Let us change the working directory from the location of this notebook to a directory holding some iTelescope premium image set files. In my case this is the Plan-20 M101 dataset.

Note: The following processing is fairly low-level and requires a lot of human interaction. In a later section we'll process the same set of files in a more automated way.

[5]:
```
print('Enter the path to the directory containing the premium image set')
wdir = input('Image set path:')
try:
    os.chdir(wdir.strip())
    print('Switched directory to ' + os.getcwd())
```

```
except:
    print(f'Error, os.chdir threw an exception changing to {wdir}')
    print('Check that the path you supplied is a valid filesystem path.')
    raise
```

```
Enter the path to the directory containing the premium image set

Image set path: /old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN

Switched directory to
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN
```

We now want a pattern that can be used to select the FITS files we want to process, while ignoring any other files or directories we have in that directory (e.g. processed files, configurations files, and so on).

Listing the fits files from the shell I get:

```
ls ../Plan-20-M101SN/*fits
../Plan-20-M101SN/Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
../Plan-20-M101SN/Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits
../Plan-20-M101SN/Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits
../Plan-20-M101SN/Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits
../Plan-20-M101SN/Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits
../Plan-20-M101SN/Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits
../Plan-20-M101SN/Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits
../Plan-20-M101SN/Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits
../Plan-20-M101SN/Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits
../Plan-20-M101SN/Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits
../Plan-20-M101SN/Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits
../Plan-20-M101SN/Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits
```

So a valid pattern that picks up the original files is `Calibrated-iTelescope*.fits`

[6]:
```
default_pattern = 'Calibrated-iTelescope*.fits'
print(f'Default file pattern for input files: {default_pattern}')
msg = 'Enter new input file pattern (or return to accept default pattern)'
file_pattern = input(msg).strip() or default_pattern
print(f'Using "{file_pattern}" as the input file pattern.')
```

```
Default file pattern for input files: Calibrated-iTelescope*.fits

Enter new input file pattern (or return to accept default pattern)

Using "Calibrated-iTelescope*.fits" as the input file pattern.
```

[7]:
```
p_dir = pathlib.Path(r'./')
flist = list(sorted(p_dir.glob(file_pattern)))
print(f'{len(flist)} files match pattern "{file_pattern}" in current directory.
  ↪')
for fpath in flist:
    print(f'  {fpath.name}')
```

```
12 files match pattern "Calibrated-iTelescope*.fits" in current directory.
  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits
```

2.1.1 Do these have full headers?

Lets have a look at the FITS header for the first file in this list.

[8]:
```
hdr = None
with fits.open(flist[0], mode='readonly') as hdulist:
    print(f'Opened {flist[0].name}')
    hdulist.verify('fix')
    print(hdulist.info())
    hdr = hdulist[0].header
    print(f'Printing the FITS header, length={len(hdr)}')
    hdr
```

```
Opened Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
Filename: Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
No.    Name      Ver    Type      Cards   Dimensions   Format
  0  PRIMARY       1 PrimaryHDU      17   (3056, 3056)   float32
None
Printing the FITS header, length=17
```

[8]:
```
SIMPLE  =                    T / conforms to FITS standard
BITPIX  =                  -32 / array data type
NAXIS   =                    2 / number of array dimensions
NAXIS1  =                 3056
NAXIS2  =                 3056
EXTEND  =                    T
OWNER   = 'iTelescope'
RA      = '14 03 12.40'        / [hms J2000] Target right ascension
OBJCTRA = '14 03 12.40'        / [hms J2000] Target right ascension
DEC     = '+54 20 58.0'        / [dms +N J2000] Target declination
OBJCTDEC= '+54 20 58.0'        / [dms +N J2000] Target declination
FOCALLEN=               3962.0 / Focal length of telescope in mm
APTDIA  =                610.0 / Aperture diameter of telescope in mm
EXPTIME =                180.0 / Exposure time in seconds
```

```
FILTER  = 'Lum     '           / Filter type used
HIERARCH OBSTARGET = 'M101 Supernova 2023ixf' / Object imaged
SRCLINK = '3jnv6zuw'           / Original source file ID
```

[9]:
```
# What keywords do we have (non-comment, non-history)
sorted(list( hdr.keys() ))
```

[9]:
```
['APTDIA',
 'BITPIX',
 'DEC',
 'EXPTIME',
 'EXTEND',
 'FILTER',
 'FOCALLEN',
 'NAXIS',
 'NAXIS1',
 'NAXIS2',
 'OBJCTDEC',
 'OBJCTRA',
 'OBSTARGET',
 'OWNER',
 'RA',
 'SIMPLE',
 'SRCLINK']
```

Lets look at the keyword differences in more detail

```
def fkeywords(fname):
    with fits.open(fname) as hdulist:
        hdr=hdulist[0].header
        print(f'FITS header for primary HDU of {fname}')
        print( sorted( list( hdr.keys() ) ) )
    return

f1='Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits'
```

On the M101 files we've been looking at:

```
fkeywords(f1)
FITS header for primary HDU of Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
['APTDIA', 'BITPIX', 'DEC', 'EXPTIME', 'EXTEND', 'FILTER', 'FOCALLEN', 'NAXIS', 'NAXIS1', 'NAXI

# iTelescope calibration system
In [17]: f2_itel='calibrated-T05-davestrickland-M82-20210219-002600-Red-BIN1-W-300-001.fit'

# iTelescope raw file
In [18]: f2_raw='raw-T05-davestrickland-M82-20210219-002600-Red-BIN1-W-300-001.fit'

# Calibrated file produced by AstroPhotography from the raw file above...
```

```
In [19]: f2_apcal='cal-T05-davestrickland-M82-20210219-002600-Red-BIN1-W-300-001.fits'

In [20]: fkeywords(f2_raw)
FITS header for primary HDU of raw-T05-davestrickland-M82-20210219-002600-Red-BIN1-W-300-001.f
['APTAREA', 'APTDIA', 'BITPIX', 'BSCALE', 'BZERO', 'CBLACK', 'CCD-TEMP', 'CSTRETCH', 'CWHITE',

In [21]: fkeywords(f2_itel)
FITS header for primary HDU of calibrated-T05-davestrickland-M82-20210219-002600-Red-BIN1-W-300
['AIRMASS', 'ALT-OBS', 'APTAREA', 'APTDIA', 'BITPIX', 'BSCALE', 'BZERO', 'CALSTAT', 'CBLACK',

In [22]: fkeywords(f2_apcal)
FITS header for primary HDU of cal-T05-davestrickland-M82-20210219-002600-Red-BIN1-W-300-001.f
['AIRMASS', 'ALT-OBS', 'APTAREA', 'APTDIA', 'BIASCORR', 'BIASFILE', 'BITPIX', 'BPIXCORR', 'BPI
```

After some rearranging by hand we get the following table of header keywords. The premium file is clearly missing a large number of header keywords, although this doesn't tell us which ones are needed and which ones are superfluous.

iTelescope raw	iTelescope calibrated	AP calibrated	M101 premium
	AIRMASS	AIRMASS	
	ALT-OBS	ALT-OBS	
APTAREA	APTAREA	APTAREA	
APTDIA	APTDIA	APTDIA	APTDIA
		BIASCORR	
		BIASFILE	
BITPIX	BITPIX	BITPIX	BITPIX
		BPIXCORR	
		BPIXDPIX	
		BPIXFILE	
		BPIXNBAD	
		BPIXNFIX	
		BPIXNREM	
		BPIX_MIN	
BSCALE	BSCALE		
BZERO	BZERO		
		BUNIT	
	CALSTAT		
CBLACK	CBLACK	CBLACK	
CCD-TEMP	CCD-TEMP	CCD-TEMP	
		CR_CLEAN	
		CR_NPIX	
CSTRETCH	CSTRETCH	CSTRETCH	
CWHITE	CWHITE	CWHITE	
		DARKCORR	
		DARKFILE	
	DATE		
	DATE-HP		
DATE-OBS	DATE-OBS	DATE-OBS	

iTelescope raw	iTelescope calibrated	AP calibrated	M101 premium
	DEC		DEC
		DEC-OBJ	
EGAIN	EGAIN	EGAIN	
EXPOSURE	EXPOSURE	EXPOSURE	
EXPTIME	EXPTIME	EXPTIME	EXPTIME
			EXTEND
FILTER	FILTER	FILTER	FILTER
		FLATCORR	
		FLATFILE	
FLIPSTAT	FLIPSTAT	FLIPSTAT	
FOCALLEN	FOCALLEN	FOCALLEN	FOCALLEN
IMAGETYP	IMAGETYP	IMAGETYP	
INSTRUME	INSTRUME	INSTRUME	
JD	JD	JD	
JD-HELIO	JD-HELIO	JD-HELIO	
	LAT-OBS	LAT-OBS	
	LONG-OBS	LON-OBS	
NAXIS	NAXIS	NAXIS	NAXIS
NAXIS1	NAXIS1	NAXIS1	NAXIS1
NAXIS2	NAXIS2	NAXIS2	NAXIS2
NOTES	NOTES	NOTES	
	OBJCTDEC		OBJCTDEC
	OBJCTRA		OBJCTRA
OBJECT	OBJECT	OBJECT	
		OBJNAME	
	OBSERVAT	OBSERVAT	
OBSERVER	OBSERVER	OBSERVER	
			OBSTARGET
			OWNER
PEDESTAL	PEDESTAL		
	PIERSIDE		
	RA		RA
	RADECSYS		
		RA-OBJ	
READOUTM	READOUTM	READOUTM	
ROWORDER	ROWORDER	ROWORDER	
	RollAngle		
SBSTDVER	SBSTDVER	SBSTDVER	
SET-TEMP	SET-TEMP	SET-TEMP	
SIMPLE	SIMPLE	SIMPLE	SIMPLE
SITELAT	SITELAT	SITELAT	
SITELONG	SITELONG	SITELONG	
			SRCLINK
	ST		
SWCREATE	SWCREATE	SWCREATE	
SWOWNER	SWMODIFY	SWOWNER	

iTelescope raw	iTelescope calibrated	AP calibrated	M101 premium
SWSERIAL	SWOWNER	SWSERIAL	
	SWSERIAL		
TELESCOP	TELESCOP	TELESCOP	
	TIME-OBS		
	TIMESYS		
TRAKTIME	TRAKTIME	TRAKTIME	
	USERNAME		
	UT		
XBINNING	XBINNING	XBINNING	
XORGSUBF	XORGSUBF	XORGSUBF	
XPIXSZ	XPIXSZ	XPIXSZ	
YBINNING	YBINNING	YBINNING	
YORGSUBF	YORGSUBF	YORGSUBF	
YPIXSZ	YPIXSZ	YPIXSZ	
	iTelescope		
	iTelescopePlateScaleH		
	iTelescopePlateScaleV		

2.2 Astrometry (aka Navigation)

Combining multiple images of the same source in the same filter requires that each image have a valid World Coordinate System solution, based on an astrometric solution to the stars in the field of view. The `AstroPhotography` package also refers to this process as image navigation.

The process consists of finding the pixel locations brightest N stars in each image, then calling `Astrometry.net` to compute a WCS solution based on those star locations, and creating a new "navigated" fits file that combines the calibrated file and the WCS solution. By default 2-dimensional Gaussian profiles are also fitted to a representative number of stars per image to assess the Full Width at Half Maximum of the star images (in pixels), the results of which are shown graphically in a plot and numerically in a YaML file describing the stars found and fitted per image. At the end of the process a summary of the image (star) statistics is generated in CSV format that also incorporates the plate scale (arcseconds per pixel) found in the astrometric solution, which allows a human to look for outliers such as images where the seeing or tracking was especially bad.

In terms of pseudo-code

```
for cal-img in calibrated-images:
    ap_find_stars cal-img --> source-list source-plot fwhm-plot quality-report.yml ds9-sources
    ap_astrometry cal-img source-list --> navigated-img
ap_quality_summary (all quality-report.yml files) --> quality-summary.csv
```

Using the (current but deprecated) bash script `navigate_all.sh` the output for a simple calibrated file is as follows:

```
Processing ./20210306/cal-T05-davestrickland-M81-20210306-235058-B-BIN1-W-300-001.fits at Sun
  Cleaning all files...
removed 'SourceLists/srclist_T05-davestrickland-M81-20210306-235058-B-BIN1-W-300-001.fits'
removed 'SourceLists/implot_T05-davestrickland-M81-20210306-235058-B-BIN1-W-300-001.png'
```

```
removed 'SourceLists/ds9_T05-davestrickland-M81-20210306-235058-B-BIN1-W-300-001.reg'
removed 'MetaData/qual_T05-davestrickland-M81-20210306-235058-B-BIN1-W-300-001.yaml'
removed 'MetaData/srclog_T05-davestrickland-M81-20210306-235058-B-BIN1-W-300-001.log'
removed 'NavigatedImages/nav_T05-davestrickland-M81-20210306-235058-B-BIN1-W-300-001.fits'
  Performing star detection on ./20210306/cal-T05-davestrickland-M81-20210306-235058-B-BIN1-W-3
    About the run the following command python3 /home/dks/git/AstroPhotography/AstroPhotography
    Star detection completed successfully at Sun Feb 11 04:17:28 PM EST 2024
      Log file written to MetaData/srclog_T05-davestrickland-M81-20210306-235058-B-BIN1-W-300-0
  Performing astrometry on ./20210306/cal-T05-davestrickland-M81-20210306-235058-B-BIN1-W-300-0
    About the run the following command:  python3 /home/dks/git/AstroPhotography/AstroPhotograp
    Astrometry completed successfully at Sun Feb 11 04:17:38 PM EST 2024
```

2.2.1 Navigating a single image

Let's try detecting the stars in a single premium image, the positions of which can then be used with Astrometry.net to get an astrometric solution.

[10]:
```
# parameters used with source detection
p_loglevel   = 'DEBUG'                                                    #␣
 ↪Loglevel
p_fitsimg    = str(flist[0])                                              #␣
 ↪Input fits image
p_extnum     = 0                                                          #␣
 ↪Extension number for data in input fits
p_fitstbl    = p_fitsimg.replace('Calibrated-iTelescope', 'srclist')      #␣
 ↪Output fits table of srcs
p_search_fwhm       = 3.0                                                 #␣
 ↪initial estimate at star fwhm in pixels
p_search_nsigma     = 7.0                                                 # min␣
 ↪detection sigma above background
p_detector_bitdepth = 16                                                  #␣
 ↪detector bitdepth (16 for most CCDs, ?? for CMOS, ?? for camera)
p_sat_frac          = 0.8                                                 #␣
 ↪Fraction of full well at which we assume star saturated
p_max_sources = 200                                                       # Max␣
 ↪number of sources to output or None
p_nosatmask    = True                                                     #␣
 ↪Keep possibly saturated stars.
p_regfile     = p_fitsimg.replace('Calibrated-iTelescope', 'ds9').
 ↪replace('fits', 'reg')       # Name of ds9 region file or None
p_plotfile    = p_fitsimg.replace('Calibrated-iTelescope', 'implot').
 ↪replace('fits', 'png')    # None or name of output png image w sources.
p_qual_rprt   = p_fitsimg.replace('Calibrated-iTelescope', 'qual').
 ↪replace('fits', 'yaml')     # None or name of output ascii report
p_quiet       = False                                                     ␣
 ↪               # Quiet mode suppresses STDOUT list printing
```

```
p_fwhm_plot   = p_fitsimg.replace('Calibrated-iTelescope', 'fwhmplot').
 ↪replace('fits', 'png')  # PNG/JPG plot of PSF fits

# parameters associated with astrometric solution, if not already specified␣
 ↪above.
# Astrometry.Net API key will be dealt with elsewhere
p_inpimg          = p_fitsimg
p_srclist         = p_fitstbl
p_src_extname     = 'AP_XYPOS'
p_outimg          = p_fitsimg.replace('Calibrated-iTelescope', 'navigated')
p_use_sip         = False
p_user_scale      = None
p_scale_err_ratio = None

# Regenerate files even if they exist, set p_clean = True
p_clean = False
```

[ ]:
```
def does_file_exists(filename, verbose=False):
    """
    Returns True if the file name or path l\e\ \n\a\m\e otherwise
    """
    if verbose and not Path(filename).exists():
        print(f'Cannot find {filename}. Not a valid path or file.')
        return False
    else:
        print(f'Found {filename}.')
    return True
```

[11]:
```
def find_stars_wrapper(p_loglevel, p_fitsimg, p_fitstbl,
    p_extnum=0, p_search_fwhm=3.0, p_search_nsigma=7.0,
    p_detector_bitdepth=16, p_sat_frac=0.8, p_max_sources=200,
    p_nosatmask=True, p_plotfile=None, p_quiet=False,
    p_fwhm_plot=None, p_qual_rprt=None, p_regfile=None):
    """
    Wrapper for finding stars using ApFindStars within a python session,
    instead of using the command line ap_find_stars.py script.
    """

    # Perform initial source detection using default parameters.
    find_stars = ap.ApFindStars(p_fitsimg, p_extnum, p_search_fwhm,
        p_search_nsigma, p_detector_bitdepth,
        p_max_sources, p_nosatmask, p_sat_frac, p_loglevel,
        p_plotfile, p_quiet)

    # Measure 2-Gaussian FWHM for select stars, get average over x and y
    (p_new_fwhm, p_madstd_fwhm, p_npts) = find_stars.measure_fwhm(p_fwhm_plot,␣
 ↪'both')
```

```
    # Refine source detection
    find_stars.source_search(p_new_fwhm, p_search_nsigma)

    # Re-run photometry
    find_stars.aperture_photometry()

    # As the source searching and photometry was redone, we should redo
    # the plotting.
    if p_plotfile is not None:
        find_stars.plot_image(p_plotfile)

    # Write optional quality report
    if p_qual_rprt is not None:
        find_stars.write_quality_report(p_qual_rprt)

    # Write optional ds9 format region file
    if p_regfile is not None:
        find_stars.write_ds9_region_file(p_regfile)

    # Write final sourcelist with photometry.
    find_stars.write_source_list(p_fitstbl)

    return
```

[12]:
```
find_stars_wrapper(p_loglevel, p_fitsimg, p_fitstbl,
    p_extnum, p_search_fwhm, p_search_nsigma,
    p_detector_bitdepth, p_sat_frac, p_max_sources,
    p_nosatmask, p_plotfile, p_quiet,
    p_fwhm_plot, p_qual_rprt, p_regfile)
```

```
2024-04-04 14:24:11,807 | AstroPhotography.core.ApFindStars | INFO | Loading
extension 0 of FITS file Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
2024-04-04 14:24:11,808 | AstroPhotography.core.ApFindStars | DEBUG | 2-D
BITPIX=-32 image with 3056 columns, 3056 rows

DKSDEBUG num handlers for AstroPhotography.core.ApFindStars is 0

2024-04-04 14:24:11,903 | AstroPhotography.core.ApFindStars | DEBUG | Raw data
statistics are min=0.00, max=67450.95, median=491.96
2024-04-04 14:24:12,764 | AstroPhotography.core.ApFindStars | DEBUG | Sigma
clipped image stats: mean=492.291, median=491.025, stddev=28.186
2024-04-04 14:24:15,340 | AstroPhotography.core.ApFindStars | DEBUG | Source-
masked image stats: mean=489.139, median=488.849, stddev=25.656
2024-04-04 14:24:15,341 | AstroPhotography.core.ApFindStars | DEBUG | Checking
for possibly saturated stars or regions.
2024-04-04 14:24:15,341 | AstroPhotography.core.ApFindStars | DEBUG | Looking
for possibly saturated regions above 52428 ADU separated by 12 pixels.
```

```
2024-04-04 14:24:15,541 | AstroPhotography.core.ApFindStars | DEBUG | Retaining
17 possibly saturated stars in source searching and photometry.
2024-04-04 14:24:15,542 | AstroPhotography.core.ApFindStars | DEBUG | Running
DAOStarFinder with FWHM=3.00 pixels, threshold=7.0 BG sigma.

Saturated position table:
 x_peak y_peak peak_value
------ ------ ----------
  2168    134   58007.41
  1861    192  67450.945
   542    301   66776.38
  1733    305    66585.7
   124    785  66806.375
   925    837   65296.09
   978   1256   55520.92
   107   1333   66617.82
   116   1334   66440.46
  3034   1826  65005.867
  2118   2062   65066.12
   244   2213   65891.75
  3026   2245   66356.28
   146   2641  67146.664
  2034   2658  64963.445
  2078   2846  65638.266
  1860   2936  54830.844

2024-04-04 14:24:15,782 | AstroPhotography.core.ApFindStars | INFO | Initial
source list using FHWM=3.0 pixels, threshold=7.0 x BG stddev, found 265 sources.
2024-04-04 14:24:15,782 | AstroPhotography.core.ApFindStars | DEBUG | Sources
with pixels exceding 52428 ADU will be flagged as possibly saturated.
2024-04-04 14:24:15,782 | AstroPhotography.core.ApFindStars | DEBUG | There are
20 possibly saturated stars in the output source list.
2024-04-04 14:24:15,783 | AstroPhotography.core.ApFindStars | DEBUG | Radius of
circular aperture photometry is 6 pixels.
2024-04-04 14:24:15,784 | AstroPhotography.core.ApFindStars | DEBUG | Local BG
estimation in annulus outer radius 9 pixels, inner radius 6 pixels.
2024-04-04 14:24:15,828 | AstroPhotography.core.ApFindStars | DEBUG | Image
exposure time [seconds]: 180.00
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:421:
RuntimeWarning: invalid value encountered in log10
  phot_table['magnitude']   = -2.5 * np.log10(phot_table['adu_per_sec'])
2024-04-04 14:24:15,839 | AstroPhotography.core.ApFindStars | DEBUG | Brightest
source (idx=0) generates 27429.19 ADU/s.
2024-04-04 14:24:15,839 | AstroPhotography.core.ApFindStars | DEBUG | Median
source (idx=132) generates 108.27 ADU/s.
2024-04-04 14:24:15,839 | AstroPhotography.core.ApFindStars | DEBUG | Faintest
source (idx=264) generates -85.00 ADU/s.
2024-04-04 14:24:15,839 | AstroPhotography.core.ApFindStars | DEBUG | Selecting
200 sources out of 265 to write to the output source list.
```

```
2024-04-04 14:24:15,840 | AstroPhotography.core.ApFindStars | DEBUG | Radius of
circular aperture photometry is 6 pixels.
2024-04-04 14:24:15,840 | AstroPhotography.core.ApFindStars | DEBUG | Local BG
estimation in annulus outer radius 9 pixels, inner radius 6 pixels.
```

id	xcentroid	ycentroid	sharpness	roundness1	roundness2	npix	sky	peak	flux	mag	psbl_sat
1	1667.02	19.15	0.3604	0.0048	0.0677	25	0	442.41	1.7313	-0.5959	False
2	2989.30	36.39	0.4067	-0.2695	0.1677	25	0	696.49	2.5255	-1.0059	False
3	1909.27	67.00	0.4246	-0.1126	0.1653	25	0	403.46	1.4061	-0.3700	False
4	1554.09	73.38	0.4523	0.0858	0.5413	25	0	1799.92	3.0552	-1.2126	False
5	153.73	77.81	0.4811	-0.1907	-0.1428	25	0	491.31	1.4794	-0.4252	False
6	2385.73	102.48	0.4248	-0.4378	0.1723	25	0	409.73	1.4355	-0.3925	False
7	816.19	111.03	0.9502	-0.0965	-0.0142	25	0	1002.37	4.0108	-1.5081	False
8	2167.66	134.02	0.4556	-0.1268	0.1220	25	0	57518.56	198.8871	-5.7465	True
9	2186.56	141.70	0.5121	0.1010	0.0511	25	0	470.86	1.6511	-0.5444	False
10	1392.54	143.31	0.4695	-0.3243	0.0808	25	0	2058.71	6.6115	-2.0507	False
11	385.07	188.70	0.8109	0.2581	0.7855	25	0	858.50	3.9069	-1.4796	False
…	…	…	…	…	…	…	…	…	…	…	…
255	1611.14	2752.27	0.4143	-0.0837	0.0423	25	0	1895.66	4.1413	-1.5428	False
256	2390.61	2785.86	0.4425	-0.0257	0.1290	25	0	1294.44	4.7083	-1.6822	False
257	894.75	2827.24	0.4656	-0.4934	0.0031	25	0	2972.84	10.1728	-2.5186	False
258	2845.95	2838.94	0.4426	-0.8304	-0.3469	25	0	243.17	1.2947	-0.2804	False
259	845.68	2844.45	0.4612	-0.5107	0.0878	25	0	2697.90	9.2069	-2.4103	False
260	1860.15	2935.60	0.4588	-0.3651	0.1043	25	0	54342.00	193.8598	-5.7187	True
261	292.42	2947.25	0.4059	-0.4113	-0.0727	25	0	1297.22	4.5110	-1.6357	False
262	1073.03	2948.38	0.4814	-0.3479	0.1203	25	0	3171.04			

```
11.0785 -2.6112    False
263     64.29   2973.19    0.4433    -0.5090    -0.1063   25   0 10470.56
34.1943 -3.8349    False
264   2570.11   2985.47    0.4674    -0.2362     0.1745   25   0   794.87
2.9769 -1.1844    False
265    317.29   3041.85    0.4893    -0.5151    -0.1560   25   0   323.19
1.0895 -0.0930    False
Length = 265 rows
 id xcenter ycenter aperture_sum peak_adu psbl_sat bgmed_per_pix adu_per_sec
magnitude
      pix     pix
--- ------- ------- ------------ -------- -------- ------------- -----------
---------
 12 1856.53  195.21 4937254.6531 66219.60     True       6182.96  27429.1925
-11.0955
 14 1858.48  197.73 4664030.2303 65806.47     True       5124.15  25911.2791
-11.0337
 13 1861.00  194.82 4594638.3099 62863.52     True       4832.10  25525.7684
-11.0174
111  116.29 1333.13 4108741.3952 65569.82     True      11373.42  22826.3411
-10.8961
245  145.75 2640.01 3613403.4601 65908.50     True       2780.66  20074.4637
-10.7566
244  144.61 2639.29 3609494.5672 66132.87     True       2626.85  20052.7476
-10.7554
213  245.40 2213.62 2453175.0246 64892.54     True       1583.71  13628.7501
-10.3361
212  246.18 2213.10 2447453.7011 65165.94     True       1587.34  13596.9650
-10.3336
196 2115.53 2062.23 2432345.8830 63839.05     True       1544.81  13513.0327
-10.3269
198 2115.68 2062.43 2425791.4696 63964.89     True       1571.66  13476.6193
-10.3240
…     …     …           …       …      …           …         …
…
 83  469.76 1062.97    1367.3577   310.05    False        485.71      7.5964
-2.2015
102 1941.10 1197.33    1145.8039   218.00    False        502.12      6.3656
-2.0096
 54  860.43  727.51    1041.4839   224.51    False        486.81      5.7860
-1.9060
 23  382.08  378.11    1022.4234   420.09    False        481.86      5.6801
-1.8859
 87 1264.06 1095.35     941.1482   271.79    False        491.88      5.2286
-1.7960
 82 2231.29 1053.18     684.5490   325.91    False        509.57      3.8030
-1.4503
 33  469.76  509.77     556.6896   217.78    False        484.89      3.0927
```

```
-1.2259
215  397.07 2225.92     496.9866   323.37    False        483.75      2.7610
-1.1027
 73 1015.94  965.08     220.4001   230.63    False        487.14      1.2244
-0.2198
116 2423.09 1366.05    -115.2431   245.92    False        512.31     -0.6402
nan
101    1.00 1191.17  -15299.1878   462.39    False        486.40    -84.9955
nan
Length = 265 rows
```

```
2024-04-04 14:24:16,845 | AstroPhotography.core.ApFindStars | INFO | Plotting
asinh-stretched bitmap of image and sources to
implot-M101_Supernova_2023ixf-180s-Lum-1.png
2024-04-04 14:24:16,847 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region
will be fitted, excluding the brightest 0 stars.
2024-04-04 14:24:16,848 | ApMeasureStars | DEBUG | Count based weighting factors
will used in fitting?: True
2024-04-04 14:24:16,848 | ApMeasureStars | DEBUG | Will the background level be
fitted for?           : True
2024-04-04 14:24:16,849 | ApMeasureStars | INFO | Size of input trimmed source
list (filtered): 180
2024-04-04 14:24:16,849 | ApMeasureStars | INFO | Size of full source list used
for neighbor removal: 265
2024-04-04 14:24:16,849 | ApMeasureStars | DEBUG | Adopting a fit box width of
18 pixels, edge exclusion of 10 pixels.
2024-04-04 14:24:16,850 | ApMeasureStars | INFO | Preparing to trim the input
source list of stars within neighbors within 18 pixels.
2024-04-04 14:24:16,850 | ApMeasureStars | DEBUG | Constructed a kdtree with 265
data points.
2024-04-04 14:24:16,853 | ApMeasureStars | INFO | Nearest neighbor filtering
removed 13 stars from consideration.
2024-04-04 14:24:16,854 | ApMeasureStars | INFO | Selecting candidate stars for
fitting out of 167 stars in 3056 row x 3056 column image.
2024-04-04 14:24:16,854 | ApMeasureStars | DEBUG | Center has radius 764.0
pixels centered on x=1528.0, y=1528.0
2024-04-04 14:24:16,855 | ApMeasureStars | DEBUG | There are 166 stars more than
10 pixels away from the edge of the detector.
2024-04-04 14:24:16,855 | ApMeasureStars | DEBUG | In region CN found 55
candidates for fitting:
2024-04-04 14:24:16,856 | ApMeasureStars | DEBUG | In region TL found 25
candidates for fitting:
2024-04-04 14:24:16,858 | ApMeasureStars | DEBUG | In region TR found 29
candidates for fitting:
2024-04-04 14:24:16,860 | ApMeasureStars | DEBUG | In region BR found 31
candidates for fitting:
2024-04-04 14:24:16,862 | ApMeasureStars | DEBUG | In region BL found 26
candidates for fitting:
```

```
2024-04-04 14:24:16,864 | ApMeasureStars | DEBUG | In 25x18x18 pixel array
min=417.57, max=42571.63
2024-04-04 14:24:16,866 | ApMeasureStars | INFO | Starting to fit
Const2D+Gaussian2D model to 25 star cutouts.
2024-04-04 14:24:16,867 | ApMeasureStars | DEBUG | Fitting star #0 (id=161) at
fixed xpos=8.66, ypos=9.07
2024-04-04 14:24:16,876 | ApMeasureStars | DEBUG | Fitting star #1 (id=122) at
fixed xpos=8.86, ypos=9.08
2024-04-04 14:24:16,885 | ApMeasureStars | DEBUG | Fitting star #2 (id=109) at
fixed xpos=8.95, ypos=8.87
2024-04-04 14:24:16,897 | ApMeasureStars | DEBUG | Fitting star #3 (id=186) at
fixed xpos=8.85, ypos=8.62
2024-04-04 14:24:16,909 | ApMeasureStars | DEBUG | Fitting star #4 (id=176) at
fixed xpos=8.84, ypos=8.93
2024-04-04 14:24:16,917 | ApMeasureStars | DEBUG | Fitting star #5 (id=263) at
fixed xpos=9.29, ypos=9.19
2024-04-04 14:24:16,928 | ApMeasureStars | DEBUG | Fitting star #6 (id=239) at
fixed xpos=8.55, ypos=9.42
2024-04-04 14:24:16,938 | ApMeasureStars | DEBUG | Fitting star #7 (id=216) at
fixed xpos=8.91, ypos=8.90
2024-04-04 14:24:16,949 | ApMeasureStars | DEBUG | Fitting star #8 (id=250) at
fixed xpos=9.31, ypos=9.28
2024-04-04 14:24:16,961 | ApMeasureStars | DEBUG | Fitting star #9 (id=246) at
fixed xpos=8.70, ypos=8.73
2024-04-04 14:24:16,971 | ApMeasureStars | DEBUG | Fitting star #10 (id=229) at
fixed xpos=8.91, ypos=8.92
2024-04-04 14:24:16,980 | ApMeasureStars | DEBUG | Fitting star #11 (id=227) at
fixed xpos=9.05, ypos=9.49
2024-04-04 14:24:16,989 | ApMeasureStars | DEBUG | Fitting star #12 (id=163) at
fixed xpos=8.92, ypos=8.56
2024-04-04 14:24:16,997 | ApMeasureStars | DEBUG | Fitting star #13 (id=243) at
fixed xpos=8.99, ypos=9.44
2024-04-04 14:24:17,005 | ApMeasureStars | DEBUG | Fitting star #14 (id=207) at
fixed xpos=8.83, ypos=9.32
2024-04-04 14:24:17,014 | ApMeasureStars | DEBUG | Fitting star #15 (id=49) at
fixed xpos=9.45, ypos=8.67
2024-04-04 14:24:17,023 | ApMeasureStars | DEBUG | Fitting star #16 (id=43) at
fixed xpos=9.49, ypos=8.89
2024-04-04 14:24:17,032 | ApMeasureStars | DEBUG | Fitting star #17 (id=53) at
fixed xpos=9.15, ypos=9.35
2024-04-04 14:24:17,040 | ApMeasureStars | DEBUG | Fitting star #18 (id=35) at
fixed xpos=9.09, ypos=8.99

DKSDEBUG num handlers for AstroPhotography.core.ApMeasureStars is 0
Input table supplied to ApMeasureStars after saturated star filtering:
 id xcenter ycenter peak_adu bgmed_per_pix magnitude
      pix     pix
--- ------- ------- -------- ------------- ---------
```

```
 36  988.34  568.25 42082.78        779.98   -9.1844
 49 1717.45  696.67 34665.86        713.27   -8.8793
161 1790.66 1721.07 34437.17        815.72   -8.7501
 43 2376.49  635.89 26131.05        649.73   -8.5449
 53 1958.15  719.35 22915.31        638.96   -8.3771
229 2596.91 2435.92 24442.35        622.76   -8.3455
 35 1742.09  544.99 19561.60        609.22   -8.1980
227 2144.05 2398.49 19322.73        608.14   -8.1614
122 1804.86 1387.08 17871.50        654.15   -8.0227
163 2408.92 1724.56 15033.04        617.66   -7.8377
…     …      …       …             …        …
  1 1667.02   19.15   442.41        494.78   -4.1970
 80  700.31 1031.48   315.98        480.95   -4.1920
 69  467.93  915.06  2089.46        488.37   -4.1678
  6 2385.73  102.48   409.73        488.93   -4.1673
220  777.05 2266.04   462.40        483.24   -4.1623
160 1262.14 1712.05   502.49        492.90   -4.1539
 77  635.99 1000.31   411.78        491.44   -4.1469
247 1436.99 2644.52   475.95        495.08   -4.1415
  9 2186.56  141.70   470.86        505.66   -4.1227
108 2805.60 1294.94   511.44        495.03   -4.1116
  3 1909.27   67.00   403.46        506.01   -4.1100
Length = 180 rows

Initial sources with nearest neighbor distance
  id xcenter ycenter peak_adu bgmed_per_pix magnitude nn_dist
      pix     pix
--- ------- ------- -------- ------------- --------- -------
 36  988.34  568.25 42082.78        779.98   -9.1844   71.98
 49 1717.45  696.67 34665.86        713.27   -8.8793   87.15
161 1790.66 1721.07 34437.17        815.72   -8.7501  125.45
 43 2376.49  635.89 26131.05        649.73   -8.5449   65.21
 53 1958.15  719.35 22915.31        638.96   -8.3771   59.14
229 2596.91 2435.92 24442.35        622.76   -8.3455  115.43
 35 1742.09  544.99 19561.60        609.22   -8.1980  153.67
227 2144.05 2398.49 19322.73        608.14   -8.1614   75.32
122 1804.86 1387.08 17871.50        654.15   -8.0227   53.17
163 2408.92 1724.56 15033.04        617.66   -7.8377   29.87
…     …      …       …             …        …        …
  1 1667.02   19.15   442.41        494.78   -4.1970  125.28
 80  700.31 1031.48   315.98        480.95   -4.1920   71.47
 69  467.93  915.06  2089.46        488.37   -4.1678  147.91
  6 2385.73  102.48   409.73        488.93   -4.1673  203.00
220  777.05 2266.04   462.40        483.24   -4.1623   29.49
160 1262.14 1712.05   502.49        492.90   -4.1539   31.49
 77  635.99 1000.31   411.78        491.44   -4.1469   71.47
247 1436.99 2644.52   475.95        495.08   -4.1415   81.22
  9 2186.56  141.70   470.86        505.66   -4.1227   20.40
```

```
108 2805.60 1294.94   511.44        495.03   -4.1116   70.90
  3 1909.27   67.00   403.46        506.01   -4.1100  136.63
Length = 180 rows
 id xcenter ycenter peak_adu bgmed_per_pix magnitude nn_dist    dx      dy
radius region
      pix     pix                                            pix     pix
pix
--- ------- ------- -------- ------------- --------- ------- ------- -------
------ ------
161 1790.66 1721.07 34437.17        815.72   -8.7501  125.45  262.66  193.07
325.98     CN
122 1804.86 1387.08 17871.50        654.15   -8.0227   53.17  276.86 -140.92
310.66     CN
109 1196.95 1314.87 11376.10        564.54   -7.6046  132.69 -331.05 -213.13
393.72     CN
186 1683.85 1972.62  6178.29        579.09   -6.8751  100.60  155.85  444.62
471.14     CN
176  868.84 1873.93  5657.47        529.22   -6.8357  196.77 -659.16  345.93
744.42     CN

 id xcenter ycenter peak_adu bgmed_per_pix magnitude nn_dist    dx       dy
radius region
      pix     pix                                            pix      pix
pix
--- ------- ------- -------- ------------- --------- ------- -------- -------
------- ------
263   64.29 2973.19 10470.56        564.72   -7.6378  226.73 -1463.71 1445.19
2056.95     TL
239 1432.55 2563.42  8489.51        552.94   -7.3538   79.54   -95.45 1035.42
1039.81     TL
216  308.91 2229.90  5232.64        530.88   -6.8197   64.94 -1219.09  701.90
1406.71     TL
250  296.31 2694.28  4009.30        519.19   -6.5934  160.04 -1231.69 1166.28
1696.25     TL
246  978.70 2641.73  4047.20        511.50   -6.4761  184.53  -549.30 1113.73
1241.82     TL

 id xcenter ycenter peak_adu bgmed_per_pix magnitude nn_dist    dx      dy
radius region
      pix     pix                                            pix     pix
pix
--- ------- ------- -------- ------------- --------- ------- ------- -------
------- ------
229 2596.91 2435.92 24442.35        622.76   -8.3455  115.43 1068.91  907.92
1402.46     TR
227 2144.05 2398.49 19322.73        608.14   -8.1614   75.32  616.05  870.49
1066.43     TR
163 2408.92 1724.56 15033.04        617.66   -7.8377   29.87  880.92  196.56
```

```
902.59     TR
243 1650.99 2628.44 14108.12        578.42   -7.8341   55.65  122.99 1100.44
1107.29     TR
207 2344.83 2160.32 12815.53        574.06   -7.6723  140.08  816.83  632.32
1032.98     TR

 id xcenter ycenter peak_adu bgmed_per_pix magnitude nn_dist   dx      dy
radius region
      pix     pix                                           pix     pix
pix
--- ------- ------- -------- ------------- --------- ------- ------ -------
------- ------
 49 1717.45  696.67 34665.86        713.27   -8.8793   87.15 189.45 -831.33
852.64     BR
 43 2376.49  635.89 26131.05        649.73   -8.5449   65.21 848.49 -892.11
1231.18     BR
 53 1958.15  719.35 22915.31        638.96   -8.3771   59.14 430.15 -808.65
915.94     BR
 35 1742.09  544.99 19561.60        609.22   -8.1980  153.67 214.09 -983.01
1006.06     BR
127 2343.07 1451.40 12210.71        608.41   -7.6697   45.28 815.07  -76.60
818.67     BR

 id xcenter ycenter peak_adu bgmed_per_pix magnitude nn_dist    dx       dy
radius region
      pix     pix                                             pix      pix
pix
--- ------- ------- -------- ------------- --------- ------- -------- --------
------- ------
 36  988.34  568.25 42082.78        779.98   -9.1844   71.98  -539.66  -959.75
1101.07     BL
 50  894.33  702.21 11603.38        577.52   -7.7666   42.30  -633.67  -825.79
1040.90     BL
 48  347.88  679.01  6457.10        546.23   -7.1341  110.89 -1180.12  -848.99
1453.78     BL
 22  761.35  365.20  5483.79        534.47   -6.9577   35.89  -766.65 -1162.80
1392.79     BL
 58   79.36  801.57  4874.97        536.57   -6.9187   47.73 -1448.64  -726.43
1620.58     BL

Fit candidates with data extraction box indices:
 id xcenter ycenter peak_adu bgmed_per_pix magnitude nn_dist    dx       dy
radius region   xmin     xmax     ymin     ymax
      pix     pix                                             pix      pix
pix            pix      pix      pix      pix
--- ------- ------- -------- ------------- --------- ------- -------- --------
------- ------ -------- -------- -------- --------
161 1790.66 1721.07 34437.17        815.72   -8.7501  125.45   262.66   193.07
```

```
325.98     CN     1782     1800     1712     1730
122 1804.86 1387.08 17871.50        654.15   -8.0227   53.17   276.86  -140.92
310.66     CN     1796     1814     1378     1396
109 1196.95 1314.87 11376.10        564.54   -7.6046  132.69  -331.05  -213.13
393.72     CN     1188     1206     1306     1324
186 1683.85 1972.62  6178.29        579.09   -6.8751  100.60   155.85   444.62
471.14     CN     1675     1693     1964     1982
176  868.84 1873.93  5657.47        529.22   -6.8357  196.77  -659.16   345.93
744.42     CN      860      878     1865     1883
263   64.29 2973.19 10470.56        564.72   -7.6378  226.73 -1463.71  1445.19
2056.95     TL       55       73     2964     2982
239 1432.55 2563.42  8489.51        552.94   -7.3538   79.54   -95.45  1035.42
1039.81     TL     1424     1442     2554     2572
216  308.91 2229.90  5232.64        530.88   -6.8197   64.94 -1219.09   701.90
1406.71     TL      300      318     2221     2239
250  296.31 2694.28  4009.30        519.19   -6.5934  160.04 -1231.69  1166.28
1696.25     TL      287      305     2685     2703
246  978.70 2641.73  4047.20        511.50   -6.4761  184.53  -549.30  1113.73
1241.82     TL      970      988     2633     2651
…      …      …       …            …        …      …       …       …
…    …      …       …       …       …
207 2344.83 2160.32 12815.53        574.06   -7.6723  140.08   816.83   632.32
1032.98     TR     2336     2354     2151     2169
 49 1717.45  696.67 34665.86        713.27   -8.8793   87.15   189.45  -831.33
852.64     BR     1708     1726      688      706
 43 2376.49  635.89 26131.05        649.73   -8.5449   65.21   848.49  -892.11
1231.18     BR     2367     2385      627      645
 53 1958.15  719.35 22915.31        638.96   -8.3771   59.14   430.15  -808.65
915.94     BR     1949     1967      710      728
 35 1742.09  544.99 19561.60        609.22   -8.1980  153.67   214.09  -983.01
1006.06     BR     1733     1751      536      554
127 2343.07 1451.40 12210.71        608.41   -7.6697   45.28   815.07   -76.60
818.67     BR     2334     2352     1442     1460
 36  988.34  568.25 42082.78        779.98   -9.1844   71.98  -539.66  -959.75
1101.07     BL      979      997      559      577
 50  894.33  702.21 11603.38        577.52   -7.7666   42.30  -633.67  -825.79
1040.90     BL      885      903      693      711
 48  347.88  679.01  6457.10        546.23   -7.1341  110.89 -1180.12  -848.99
1453.78     BL      339      357      670      688
 22  761.35  365.20  5483.79        534.47   -6.9577   35.89  -766.65 -1162.80
1392.79     BL      752      770      356      374
 58   79.36  801.57  4874.97        536.57   -6.9187   47.73 -1448.64  -726.43
1620.58     BL       70       88      793      811
Length = 25 rows

2024-04-04 14:24:17,049 | ApMeasureStars | DEBUG | Fitting star #19 (id=127) at
fixed xpos=9.07, ypos=9.40
```

```
2024-04-04 14:24:17,058 | ApMeasureStars | DEBUG | Fitting star #20 (id=36) at
fixed xpos=9.34, ypos=9.25
2024-04-04 14:24:17,068 | ApMeasureStars | DEBUG | Fitting star #21 (id=50) at
fixed xpos=9.33, ypos=9.21
2024-04-04 14:24:17,078 | ApMeasureStars | DEBUG | Fitting star #22 (id=48) at
fixed xpos=8.88, ypos=9.01
2024-04-04 14:24:17,087 | ApMeasureStars | DEBUG | Fitting star #23 (id=22) at
fixed xpos=9.35, ypos=9.20
2024-04-04 14:24:17,098 | ApMeasureStars | DEBUG | Fitting star #24 (id=58) at
fixed xpos=9.36, ypos=8.57
2024-04-04 14:24:17,111 | ApMeasureStars | DEBUG | Fitting completed in 0.244 s
(average 9.8 ms/star)
2024-04-04 14:24:17,199 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 49 FWHM measurements (50 OK fits before clipping).

Fit results:
 id xcenter ycenter peak_adu bgmed_per_pix magnitude nn_dist    dx       dy
… fwhm_y_err theta  theta_err axrat axrat_err circular fit_ok rchisq
      pix     pix                                           pix      pix
…
--- ------- ------- -------- ------------- --------- ------- -------- --------
… ---------- ------ --------- ----- --------- -------- ------ ------
161 1790.66 1721.07 34437.17        815.72   -8.7501  125.45   262.66   193.07
…      0.006  0.833     0.015  1.06      0.00    False   True 192.65
122 1804.86 1387.08 17871.50        654.15   -8.0227   53.17   276.86  -140.92
…      0.008  0.849     0.018  1.08      0.00    False   True  44.52
109 1196.95 1314.87 11376.10        564.54   -7.6046  132.69  -331.05  -213.13
…      0.010  2.172     0.020  1.09      0.00    False   True  13.82
186 1683.85 1972.62  6178.29        579.09   -6.8751  100.60   155.85   444.62
…      0.016  0.769     0.031  1.09      0.01    False   True  11.46
176  868.84 1873.93  5657.47        529.22   -6.8357  196.77  -659.16   345.93
…      0.015  2.159     0.025  1.11      0.01    False   True   6.81
263   64.29 2973.19 10470.56        564.72   -7.6378  226.73 -1463.71  1445.19
…      0.011  3.683     0.011  1.16      0.00    False   True  17.56
239 1432.55 2563.42  8489.51        552.94   -7.3538   79.54   -95.45  1035.42
…      0.016  0.848     0.018  1.14      0.01    False   True 119.05
216  308.91 2229.90  5232.64        530.88   -6.8197   64.94 -1219.09   701.90
…      0.018  3.672     0.019  1.15      0.01    False   True   5.49
250  296.31 2694.28  4009.30        519.19   -6.5934  160.04 -1231.69  1166.28
…      0.021  3.715     0.021  1.16      0.01    False   True   4.99
246  978.70 2641.73  4047.20        511.50   -6.4761  184.53  -549.30  1113.73
…      0.019  2.259     0.026  1.13      0.01    False   True   4.25
…     …      …       …            …        …      …       …       …
…        …    …         …   …        …       …     …     …
207 2344.83 2160.32 12815.53        574.06   -7.6723  140.08   816.83   632.32
…      0.011  0.965     0.022  1.08      0.00    False   True  77.75
 49 1717.45  696.67 34665.86        713.27   -8.8793   87.15   189.45  -831.33
…      0.006  1.431     0.042  1.02      0.00    False   True 489.87
```

```
 43 2376.49  635.89 26131.05        649.73   -8.5449   65.21   848.49  -892.11
…      0.007 -2.332     0.056  1.02      0.00    False   True 252.64
 53 1958.15  719.35 22915.31        638.96   -8.3771   59.14   430.15  -808.65
…      0.007  0.971     0.017  1.07      0.00    False   True  61.57
 35 1742.09  544.99 19561.60        609.22   -8.1980  153.67   214.09  -983.01
…      0.007  0.871     0.023  1.05      0.00    False   True  33.42
127 2343.07 1451.40 12210.71        608.41   -7.6697   45.28   815.07   -76.60
…      0.010  0.887     0.022  1.08      0.00    False   True  45.65
 36  988.34  568.25 42082.78        779.98   -9.1844   71.98  -539.66  -959.75
…      0.004  1.977     0.012  1.06      0.00    False   True  99.73
 50  894.33  702.21 11603.38        577.52   -7.7666   42.30  -633.67  -825.79
…      0.009  1.979     0.022  1.07      0.00    False   True  25.82
 48  347.88  679.01  6457.10        546.23   -7.1341  110.89 -1180.12  -848.99
…      0.015  3.418     0.024  1.10      0.01    False   True  11.05
 22  761.35  365.20  5483.79        534.47   -6.9577   35.89  -766.65 -1162.80
…      0.016  1.928     0.044  1.06      0.01    False   True   9.80
 58   79.36  801.57  4874.97        536.57   -6.9187   47.73 -1448.64  -726.43
…      0.022  3.374     0.024  1.13      0.01    False   True  54.83
Length = 25 rows


2024-04-04 14:24:18,449 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs
to fwhmplot-M101_Supernova_2023ixf-180s-Lum-1.png
2024-04-04 14:24:18,450 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 49 FWHM measurements (50 OK fits before clipping).
2024-04-04 14:24:18,450 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-04-04 14:24:18,451 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-04-04 14:24:18,451 | AstroPhotography.core.ApFindStars | INFO | Median FWHM
(over x and y) is 4.31 +/- 0.34 pixels using 49 data points
2024-04-04 14:24:18,452 | AstroPhotography.core.ApFindStars | DEBUG | Running
DAOStarFinder with FWHM=4.31 pixels, threshold=7.0 BG sigma.
2024-04-04 14:24:18,751 | AstroPhotography.core.ApFindStars | INFO | Initial
source list using FHWM=4.31242676156308 pixels, threshold=7.0 x BG stddev, found
306 sources.
2024-04-04 14:24:18,751 | AstroPhotography.core.ApFindStars | DEBUG | Sources
with pixels exceding 52428 ADU will be flagged as possibly saturated.
2024-04-04 14:24:18,751 | AstroPhotography.core.ApFindStars | DEBUG | There are
13 possibly saturated stars in the output source list.
2024-04-04 14:24:18,752 | AstroPhotography.core.ApFindStars | DEBUG | Radius of
circular aperture photometry is 6 pixels.
2024-04-04 14:24:18,752 | AstroPhotography.core.ApFindStars | DEBUG | Local BG
estimation in annulus outer radius 9 pixels, inner radius 6 pixels.
2024-04-04 14:24:18,801 | AstroPhotography.core.ApFindStars | DEBUG | Image
exposure time [seconds]: 180.00
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:421:
RuntimeWarning: invalid value encountered in log10
```

```
  phot_table['magnitude']   = -2.5 * np.log10(phot_table['adu_per_sec'])
2024-04-04 14:24:18,802 | AstroPhotography.core.ApFindStars | DEBUG | Brightest
source (idx=0) generates 23173.82 ADU/s.
2024-04-04 14:24:18,802 | AstroPhotography.core.ApFindStars | DEBUG | Median
source (idx=153) generates 74.44 ADU/s.
2024-04-04 14:24:18,803 | AstroPhotography.core.ApFindStars | DEBUG | Faintest
source (idx=305) generates -43.01 ADU/s.
2024-04-04 14:24:18,803 | AstroPhotography.core.ApFindStars | DEBUG | Selecting
200 sources out of 306 to write to the output source list.
2024-04-04 14:24:18,803 | AstroPhotography.core.ApFindStars | DEBUG | Radius of
circular aperture photometry is 6 pixels.
2024-04-04 14:24:18,803 | AstroPhotography.core.ApFindStars | DEBUG | Local BG
estimation in annulus outer radius 9 pixels, inner radius 6 pixels.

 id xcentroid ycentroid sharpness roundness1 roundness2 npix sky   peak     flux
mag   psbl_sat
--- --------- --------- --------- ---------- ---------- ---- --- --------
-------- ------- --------
  1   1667.04     19.13    0.3563     0.0236     0.0561   25   0   442.41
2.3482 -0.9268    False
  2   2989.28     36.41    0.3944    -0.2044     0.1603   25   0   696.49
3.3373 -1.3085    False
  3    908.27     65.77    0.2993    -0.0374    -0.0883   25   0   315.90
1.5493 -0.4753    False
  4   1909.24     67.00    0.4013    -0.0568     0.1475   25   0   403.46
1.9161 -0.7061    False
  5   1554.09     73.38    0.3961     0.0825     0.5344   25   0  1799.92
4.6817 -1.6760    False
  6    153.72     77.83    0.4333    -0.0941    -0.1353   25   0   491.31
2.0948 -0.8029    False
  7   2385.69    102.40    0.3458    -0.0067     0.0372   25   0   380.84
1.9621 -0.7318    False
  8    816.16    111.02    0.9742    -0.0769    -0.0110   25   0  1002.37
4.1598 -1.5477    False
  9   2167.68    134.01    0.4109    -0.0769     0.1167   25   0 57518.56
287.5834 -6.1469     True
 10   2186.64    141.73    0.3482    -0.1830    -0.0945   25   0   407.19
2.1275 -0.8197    False
 11   1392.55    143.29    0.4218    -0.2316     0.0749   25   0  2058.71
9.5100 -2.4455    False
…       …         …         …         …          …   … …       …
…     …       …
296   2845.96   2838.94    0.4021    -0.7429    -0.3187   25   0   243.17
1.8794 -0.6850    False
297    845.69   2844.43    0.4184    -0.3912     0.0812   25   0  2697.90
13.1210 -2.7949    False
298   1618.94   2924.55    0.2720    -0.0706    -0.2489   25   0   290.91
1.0600 -0.0633    False
```

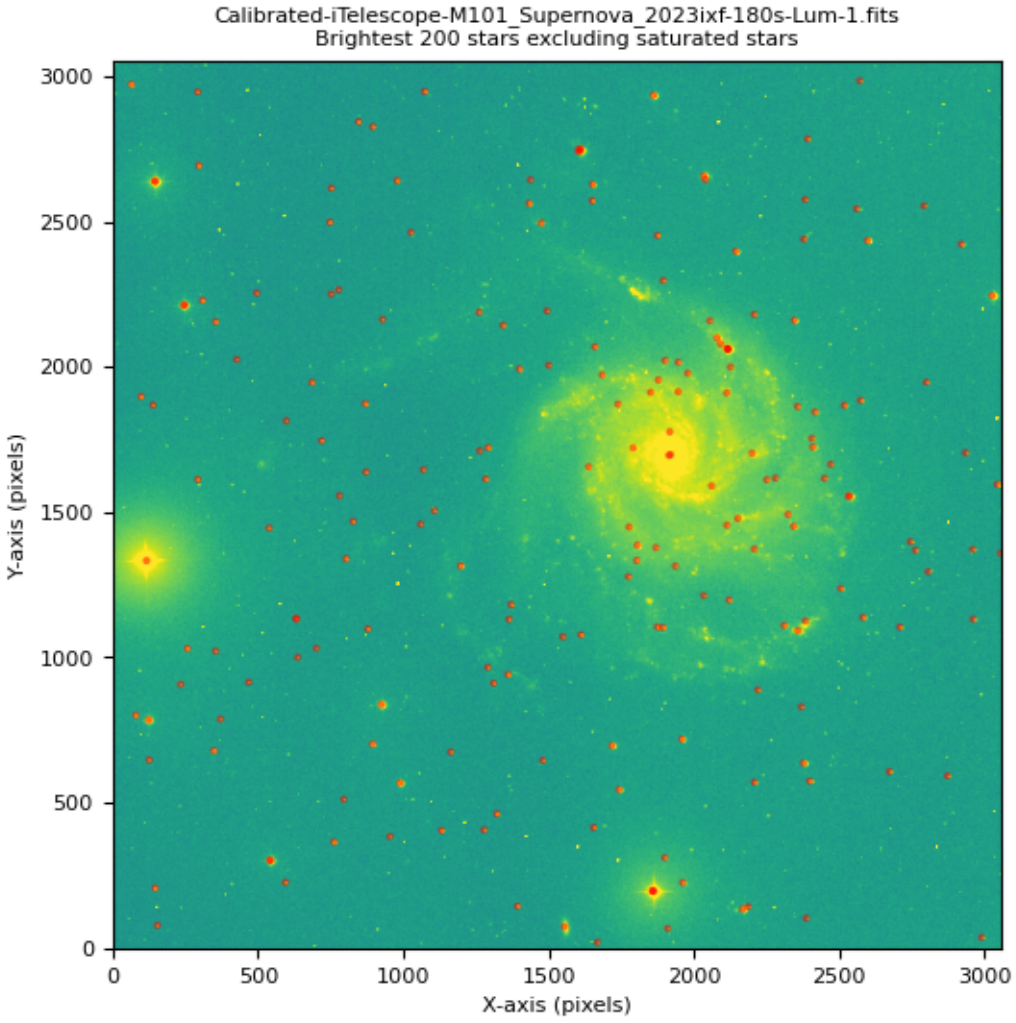
```
299   1860.14   2935.61    0.4188    -0.2630     0.1029   25   0 54342.00
273.4385 -6.0921     True
300    292.39   2947.23    0.3786    -0.3089    -0.0726   25   0  1297.22
6.4539 -2.0246    False
301   1073.11   2948.45    0.4095    -0.2028     0.1153   25   0  3116.22
15.3209 -2.9632    False
302   1393.08   2963.16    0.2703    -0.4100    -0.1301   25   0   167.84
1.0591 -0.0623    False
303     64.27   2973.18    0.3972    -0.3965    -0.1055   25   0 10470.56
50.6621 -4.2617    False
304   2570.18   2985.55    0.4350    -0.0544     0.1287   25   0   785.11
3.8947 -1.4762    False
305    865.78   2990.44    0.8635    -0.2542     0.0521   25   0   245.39
1.1538 -0.1553    False
306    317.27   3041.85    0.4286    -0.4066    -0.1429   25   0   323.19
1.6078 -0.5156    False
Length = 306 rows
 id xcenter ycenter aperture_sum peak_adu psbl_sat bgmed_per_pix adu_per_sec
magnitude
      pix     pix
--- ------- ------- ------------ -------- -------- ------------- -----------
---------
124  116.23 1333.20 4171287.9500 65569.82     True      11336.23  23173.8219
-10.9125
280  144.84 2639.25 3627638.6756 65923.66     True       2733.48  20153.5482
-10.7609
278  146.95 2637.97 3606562.9534 66247.62     True       2553.70  20036.4609
-10.7546
237  245.46 2213.58 2454963.7331 64892.54     True       1580.07  13638.6874
-10.3369
236  246.23 2213.06 2445606.6724 64559.41     True       1587.34  13586.7037
-10.3328
219 2116.21 2061.74 2419072.3437 63780.82     True       1595.91  13439.2908
-10.3209
 19  542.67  301.32 1996319.3592 65803.72     True       1278.48  11090.6631
-10.1124
 66  924.95  838.16 1709664.2325 64458.88     True       1125.49   9498.1346
-9.9441
240 3025.45 2246.21 1559213.6406 65590.15     True       1016.30   8662.2980
-9.8441
 62  124.12  785.03 1546160.6557 66317.52     True       1128.87   8589.7814
-9.8350
…     …     …           …       …       …           …         …
…
262   95.95 2480.71    1106.1493   151.22    False        488.05      6.1453
-1.9714
 45  812.29  634.88    1105.5279   381.95    False        487.43      6.1418
-1.9707
```

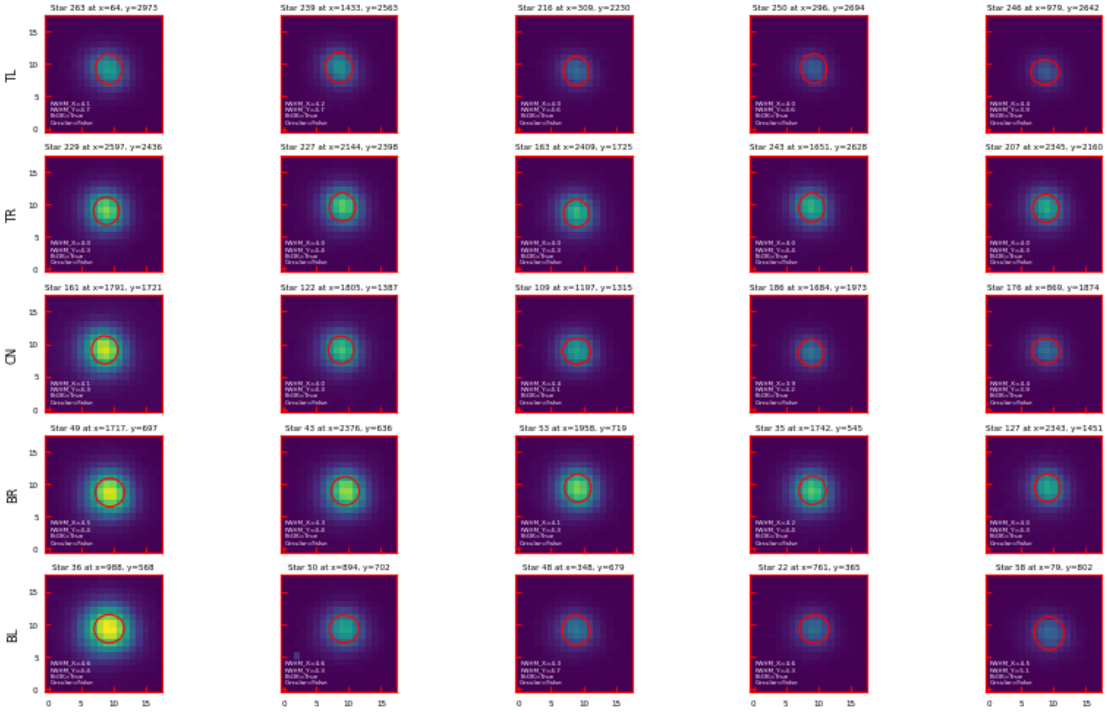
```
 60  860.41  727.55    1037.1896   224.51    False        486.87      5.7622
-1.9015
165  787.73 1626.59    1018.3260   129.01    False        487.19      5.6574
-1.8815
 96 1264.05 1095.30     953.7015   271.79    False        491.85      5.2983
-1.8103
185  415.71 1755.45     893.1337   140.34    False        489.74      4.9619
-1.7391
249 1121.01 2317.36     707.3303   184.72    False        488.19      3.9296
-1.4859
 33  469.77  509.81     553.8340   217.78    False        484.89      3.0769
-1.2203
279  288.30 2637.83      32.4541   122.64    False        490.25      0.1803
1.8600
268 2649.61 2532.63    -407.4675    36.02    False        490.11     -2.2637
nan
234    2.22 2191.53   -7742.4820   210.22    False        459.38    -43.0138
nan
Length = 306 rows
```

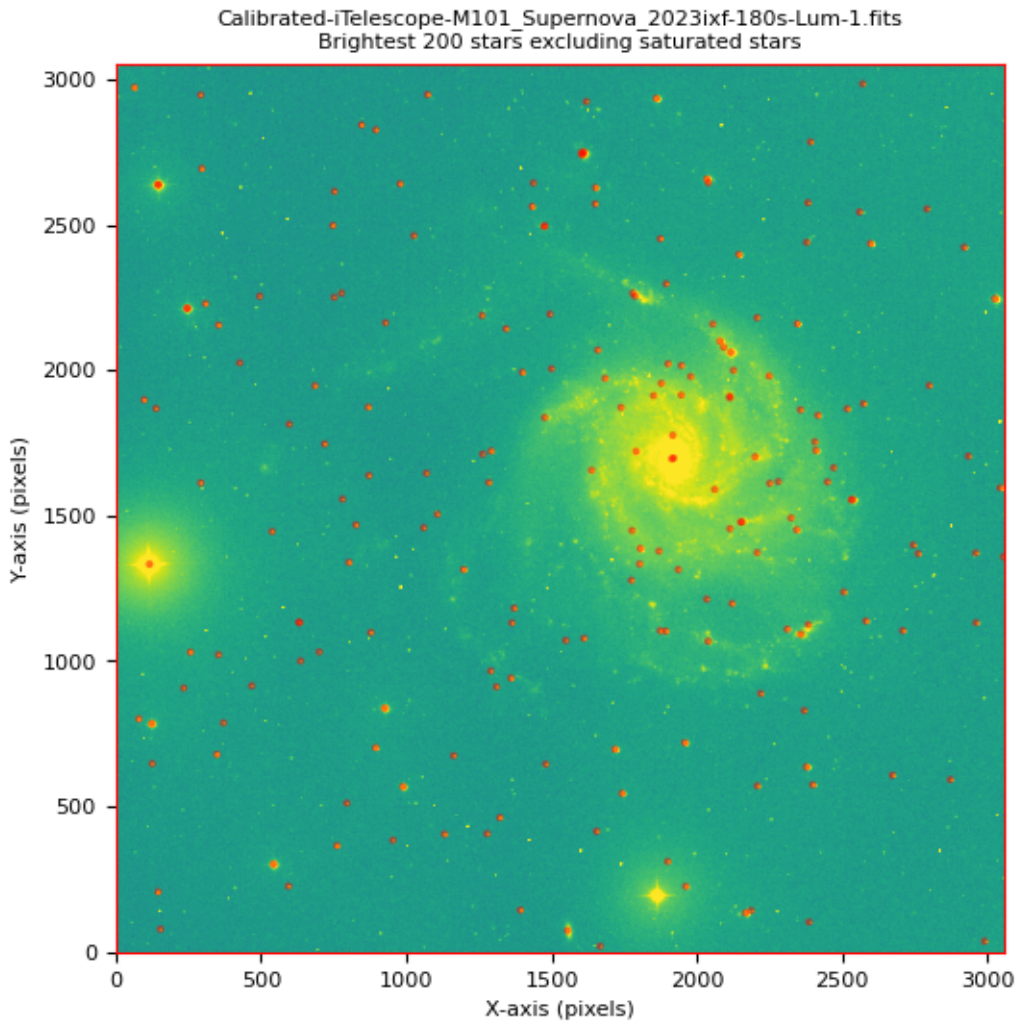
```
2024-04-04 14:24:19,777 | AstroPhotography.core.ApFindStars | INFO | Plotting
asinh-stretched bitmap of image and sources to
implot-M101_Supernova_2023ixf-180s-Lum-1.png
2024-04-04 14:24:19,777 | AstroPhotography.core.ApFindStars | INFO | Image is
3056 cols x 3056 rows
2024-04-04 14:24:19,778 | AstroPhotography.core.ApFindStars | DEBUG | FITS
keywords found in image: {'IMG_FILE': ('Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits', 'Name of image file searched
for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS':
(3056, 'Number of rows in input image'), 'AP_NDET': (306, 'Number of sources
detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final
photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'),
'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'),
'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40',
'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'),
'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Lum',
'Filter used')}
2024-04-04 14:24:19,778 | AstroPhotography.core.ApFindStars | DEBUG | FITS
keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME',
'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ',
'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']
2024-04-04 14:24:19,795 | AstroPhotography.core.ApFindStars | INFO | Approximate
coordinates: ra=14.053444 hours, dec=54.349444 deg
2024-04-04 14:24:19,795 | AstroPhotography.core.ApFindStars | WARNING | Keyword
OBJECT not found, not written to quality report.
2024-04-04 14:24:19,795 | AstroPhotography.core.ApFindStars | WARNING | Keyword
TELESCOP not found, not written to quality report.
2024-04-04 14:24:19,795 | AstroPhotography.core.ApFindStars | WARNING | Keyword
```

```
DATE-OBS not found, not written to quality report.
2024-04-04 14:24:19,796 | AstroPhotography.core.ApFindStars | WARNING | Keyword
EXPOSURE not found, not written to quality report.
2024-04-04 14:24:19,796 | AstroPhotography.core.ApFindStars | WARNING | Keyword
CCD-TEMP not found, not written to quality report.
2024-04-04 14:24:19,796 | AstroPhotography.core.ApFindStars | WARNING | Keyword
EGAIN not found, not written to quality report.
2024-04-04 14:24:19,796 | AstroPhotography.core.ApFindStars | WARNING | Keyword
AIRMASS not found, not written to quality report.
2024-04-04 14:24:19,796 | AstroPhotography.core.ApFindStars | WARNING | Keyword
APRX_XWD not found, not written to quality report.
2024-04-04 14:24:19,796 | AstroPhotography.core.ApFindStars | WARNING | Keyword
APRX_YHG not found, not written to quality report.
2024-04-04 14:24:19,797 | AstroPhotography.core.ApFindStars | WARNING | Keyword
APRX_XPS not found, not written to quality report.
2024-04-04 14:24:19,797 | AstroPhotography.core.ApFindStars | WARNING | Keyword
APRX_YPS not found, not written to quality report.
2024-04-04 14:24:19,797 | AstroPhotography.core.ApFindStars | WARNING | Skipping
some quality reporting that requires an estimated platescale.
2024-04-04 14:24:19,798 | AstroPhotography.core.ApFindStars | INFO | Wrote image
quality report to qual-M101_Supernova_2023ixf-180s-Lum-1.yaml
2024-04-04 14:24:19,801 | AstroPhotography.core.ApFindStars | DEBUG | Wrote
ds9-format region file to ds9-M101_Supernova_2023ixf-180s-Lum-1.reg
2024-04-04 14:24:19,802 | AstroPhotography.core.ApFindStars | INFO | Image is
3056 cols x 3056 rows
2024-04-04 14:24:19,802 | AstroPhotography.core.ApFindStars | DEBUG | FITS
keywords found in image: {'IMG_FILE': ('Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits', 'Name of image file searched
for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS':
(3056, 'Number of rows in input image'), 'AP_NDET': (306, 'Number of sources
detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final
photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'),
'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'),
'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40',
'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'),
'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Lum',
'Filter used')}
2024-04-04 14:24:19,802 | AstroPhotography.core.ApFindStars | DEBUG | FITS
keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME',
'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ',
'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']
2024-04-04 14:24:19,803 | AstroPhotography.core.ApFindStars | INFO | Approximate
coordinates: ra=14.053444 hours, dec=54.349444 deg
2024-04-04 14:24:19,803 | AstroPhotography.core.ApFindStars | DEBUG | Converting
python 0-based coordinates to FITS 1-based pixel coordinates for XY table.
2024-04-04 14:24:19,804 | AstroPhotography.core.ApFindStars | INFO | Writing
source list to FITS binary table srclist-M101_Supernova_2023ixf-180s-Lum-1.fits
WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS
```

```
Header - skipping [astropy.io.fits.convenience]
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING: VerifyWarning: Card is too long, comment will be truncated.
[astropy.io.fits.card]
```




Star FWHM measurements for:
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
Median FWHM=4.31 +/- 0.34 (MAD stddev) pixels
TL
TR
CN
BR
BL



Rather surprisingly this ran without error, despite the limited set of header keywords. The logging did note that "FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']"

Missing Keywords Note that a warning is emitted stating several keywords, in particular `APRX_XWD APRX_YHG APRX_XPS APRX_YPS`, will not be written to the quality report. These represent the approximate angular width and height of the image in decimal degrees, and the approximate plate scale in X and Y. These could not be computed because some of the required telescope metadata was not present in the input image header. In this case, the CCD detector XPIXSZ and YPIXSZ were missing, so the approximate plate scale could not be computed. Without that information the approximate field of view (angular width & height) of the image could not be computed.

Is That Large Background Real? Another item to note is a message early in the output: `Source-masked image stats: mean=489.139, median=488.849, stddev=25.656`. The average signal per pixel after excluding the region around the initially detected point sources is 489 +/- 26 per pixel. The units are not listed in the header, so they could be ADU, or could have been converted into electrons. One concern is that the numerical values iTelescope images often include an artificial positive `OFFSET` or `PEDESTAL`. This is listed in the raw FITS files, and if not removed acts as a large increase in sky brightness. It is not clear whether this is present in these premium images.

We're not going to try to solve that question, or subtract the background, in this notebook.

Astrometry To attempt astrometry using `Astrometry.net` we need an **API key**. If you have set up the astroquery config file with one already (see `doc/iTelescope_processing.md`) you can hope that it is picked up, or you can eneter one in the following prompt.

[13]:
```
msg = 'Enter your Astrometry.net API key, or hit return to use one␣
 ↪preconfigured with astroquery: '
p_astnetkey = input(msg).strip() or None
if p_astnetkey is None:
    print("Warning: Assuming that 'api_key' is specified in your ~/.astropy/
 ↪config/astroquery.cfg file.")
    print("  If this is not the case processing will fail.")
```

```
Enter your Astrometry.net API key, or hit return to use one preconfigured with
astroquery:

Warning: Assuming that 'api_key' is specified in your
~/.astropy/config/astroquery.cfg file.
  If this is not the case processing will fail.
```

[14]:
```
ap_astrom = ap.ApAstrometry(p_inpimg,
    p_extnum,
    p_srclist,
    p_src_extname,
    p_outimg,
    p_astnetkey,
    p_use_sip,
    p_user_scale,
    p_scale_err_ratio,
    p_loglevel)
p_status = ap_astrom.status()
print(f'ApAstrometry return status: {p_status}')
if p_status == ap.ApAstrometry.NOMINAL:
    print('  Astrometric solution suceeded.')
elif p_status == ap.ApAstrometry.INPUT_ERROR:
    print('  Error, incorrect or missing input.')
elif p_status == ap.ApAstrometry.NO_SOLUTION:
    print('  Error, Astrometry.net failed to find a solution and/or timed out.')
else:
```

```
    print('  Error, unexpected error code returned by Astrometry.net')
```

```
2024-04-04 14:24:39,289 | AstroPhotography.core.ApAstrometry | INFO | Loading
extension 0 of FITS file Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
2024-04-04 14:24:39,291 | AstroPhotography.core.ApAstrometry | DEBUG | 2-D
BITPIX=-32 image with 3056 columns, 3056 rows
2024-04-04 14:24:39,293 | AstroPhotography.core.ApAstrometry | INFO | Read 200
source positions from srclist-M101_Supernova_2023ixf-180s-Lum-1.fits
2024-04-04 14:24:39,293 | AstroPhotography.core.ApAstrometry | DEBUG | The
source list was generated from this input image. Proceding.
2024-04-04 14:24:39,293 | AstroPhotography.core.ApAstrometry | DEBUG | Estimated
center_ra=210.802, center_dec=54.349, radius=8.000 deg.
2024-04-04 14:24:39,294 | AstroPhotography.core.ApAstrometry | WARNING | Could
not generate scale hints.
2024-04-04 14:24:39,294 | AstroPhotography.core.ApAstrometry | DEBUG |
Astrometry.net hints dictionary: {'center_ra': 210.80166666666665, 'center_dec':
54.349444444444444, 'radius': 8}
2024-04-04 14:24:39,294 | AstroPhotography.core.ApAstrometry | DEBUG |
Submitting astrometry.net solve from source list with 200 positions

DKSDEBUG num handlers for AstroPhotography.core.ApAstrometry is 0
Solving…

2024-04-04 14:24:51,769 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header
2024-04-04 14:24:51,770 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
WCSAXES to (2, 'no comment')
2024-04-04 14:24:51,770 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE1 to ('RA---TAN', 'TAN (gnomic) projection')
2024-04-04 14:24:51,770 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE2 to ('DEC--TAN', 'TAN (gnomic) projection')
2024-04-04 14:24:51,771 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
EQUINOX to (2000.0, 'Equatorial coordinates definition (yr)')
2024-04-04 14:24:51,771 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LONPOLE to (180.0, 'no comment')
2024-04-04 14:24:51,771 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LATPOLE to (0.0, 'no comment')
2024-04-04 14:24:51,771 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL1 to (210.754804653, 'RA  of reference point')
2024-04-04 14:24:51,771 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL2 to (54.4174059435, 'DEC of reference point')
2024-04-04 14:24:51,772 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX1 to (1528.5, 'X reference pixel')
2024-04-04 14:24:51,772 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX2 to (1528.5, 'Y reference pixel')
2024-04-04 14:24:51,772 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT1 to ('deg', 'X pixel scale units')
2024-04-04 14:24:51,772 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
```

```
CUNIT2 to ('deg', 'Y pixel scale units')
2024-04-04 14:24:51,773 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_1 to (-4.64681630389e-06, 'Transformation matrix')
2024-04-04 14:24:51,773 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_2 to (0.000175223532872, 'no comment')
2024-04-04 14:24:51,773 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_1 to (-0.000175223532872, 'no comment')
2024-04-04 14:24:51,773 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_2 to (-4.64681630389e-06, 'no comment')
2024-04-04 14:24:51,773 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEW to (3056, 'Image width,  in pixels.')
2024-04-04 14:24:51,774 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEH to (3056, 'Image height, in pixels.')
2024-04-04 14:24:51,795 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to navigated-M101_Supernova_2023ixf-180s-Lum-1.fits
2024-04-04 14:24:51,795 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in srclist-M101_Supernova_2023ixf-180s-Lum-1.fits with WCS soluton.

ApAstrometry return status: 0
  Astrometric solution suceeded.

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
```

Astrometry.net can plate solve an image (or a set of star coordinates drawn from an image, as in this case) without any other information. This can be time consuming, as it has to try solutions of a wider variety of scales and pointings (image covering the whole sky down to images covering a few arcminutes, centered anyway on the sky). The solution can fail to return an answer, if the process run time exceeds limits set by the Astrometry.net owners.

Providing it a hint at the approximate pointing and/or approximate angular size of the image can considerably speed up plate solution. The hint does not need to be particularly accurate, so even a rough guess is better than nothing. Note that in this example ApAstrometry picked up on the `RA`, `DEC` in the image header, but defaulted to a guess of 8 degrees for the image size. The value of 8 degrees is a reasonable median between camera images covering tens of degrees and amateur telescopes covering 30 arcminutes to a few degrees.

After the solution we can see that the image field of view is roughly 0.5 degrees on a side, so providing a hint of 8 degrees did not compromise the solution. You can also see that the approximate image center `(210.80166666666665, 54.349444444444444)`, does differ from the astrometric solution `(210.754804653, 54.4174059435)` by approximately 0.05 degrees (3 arcminutes) in RA and Dec.

A brief aside about the CDi_j matrix The WCS returned by Astrometry.net (mostly) follows the official FITS standard. In particular a `CD` matrix is used instead of the older `CDELT1`, `CDELT2`, `CROTA2`. The older keywords are more informative when you want an idea of the traditional plate scale (arcseconds per pixel). The conversion back from a simple `CDi_j` matrix into CDELT and CROTA is given in https://lweb.cfa.harvard.edu/~jzhao/SMA-FITS-CASA/docs/wcs88.pdf. Note that the conversion assumes an idealized case of square pixels at the tangent point.

If you are only interested in the magnitude CDELT1, CDELT2, and not their signs or the rotation angle CROTA2, then the conversion is as simple as:

```
|CDELT1| = sqrt( CD1_1^2 + CD2_1^2 )    # Note: not CD1_1^2 and CD1_2^2

|CDELT2| = sqrt( CD1_2^2 + CD2_2^2 )    # Note: not CD2_1^2 and CD2_2^2
```

3 Processing all the images

We'll now run though processing all the images, but still doing most of the coding work by hand.

[15]:
```
p_dir = pathlib.Path(r'./')
flist = list(sorted(p_dir.glob(file_pattern)))
print(f'{len(flist)} files match pattern "{file_pattern}" in current directory.
 ↪')
for fpath in flist:
    print(f'  {fpath.name}')
```

```
12 files match pattern "Calibrated-iTelescope*.fits" in current directory.
  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits
```

[16]:
```
keys = ['naxis1', 'naxis2', 'imagetyp', 'object', 'filter', 'exposure']
ifc_cal = ImageFileCollection(p_dir, keywords=keys, glob_include=file_pattern)␣
 ↪# no glob_excludes at this time...
print(ifc_cal.summary)
```

```
                              file                                    naxis1 naxis2
imagetyp object filter exposure
------------------------------------------------------------------ ------ ------
-------- ------ ------ --------
  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits   3056   3056
--      --    Lum       --
  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits   3056   3056
--      --    Lum       --
  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits   3056   3056
--      --    Lum       --
 Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits   3056   3056
--      --   Blue       --
```

```
 Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits   3056   3056
--      --   Blue       --
 Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits   3056   3056
--      --   Blue       --
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits   3056   3056
--      --  Green       --
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits   3056   3056
--      --  Green       --
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits   3056   3056
--      --  Green       --
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits   3056   3056
--      --    Red       --
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits   3056   3056
--      --    Red       --
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits   3056   3056
--      --    Red       --
```

Note: here we see one of the issues with the lack of a strict FITS standard. The common fits keyword `EXPOSURE` is not defined in these headers, and in fact the premium images use `EXPTIME`. In fact, both `EXPOSURE` and `EXPTIME` are valid.

For a list of commonly used (in professional settings) FITS keywords see https://heasarc.gsfc.nasa.gov/docs/fcg/common_dict.html. Note that the official standard is defined elsewhere. This HEASARC webpage has a subset of the standard keywords but note that it does not include the recommendations from the FITS papers.

This is not a huge issue for getting an Astrometric solution, where each iage is processed separately. However, when later combining and mosaicing images we are likely to want to select on the exposure time of the observation in order to maximise the final signal to noise ratio.

[17]:
```
# define a structure to hold the processing status of the images
# Note: this is an astropy table, so it has that annoying string length must be␣
 ↪specified
# ir you must use `object` limitation

arr_size  = len(ifc_cal.summary)
arr_dtype = [('filename', object), ('find_stars_status', object),␣
 ↪('find_stars_time', float), ('astrometry_status', object),␣
 ↪('astrometry_time', float)]

processing_results = Table(data=np.empty(arr_size, dtype=arr_dtype))
processing_results['find_stars_time'].info.format = '7.3f'
processing_results['astrometry_time'].info.format = '7.3f'

#processing_results = Table(names=('filename', 'find_stars_status',␣
 ↪'find_stars_time', 'astrometry_status', 'astrometry_time'), dtype=(str, str,␣
 ↪float, str, float))
print(processing_results)
```

```
filename find_stars_status find_stars_time astrometry_status astrometry_time
-------- ----------------- --------------- ----------------- ---------------
    None              None           0.000              None           0.000
    None              None           0.000              None           0.000
    None              None           0.000              None           0.000
    None              None           0.000              None           0.000
    None              None           0.000              None           0.000
    None              None           0.000              None           0.000
    None              None           0.000              None           0.000
    None              None           0.000              None           0.000
    None              None           0.000              None           0.000
    None              None           0.000              None           0.000
    None              None           0.000              None           0.000
    None              None           0.000              None           0.000
```

[18]:
```
# Assorted processing settings# parameters used with source detection
p_loglevel  = 'DEBUG'                                                    #␣
 ↪Loglevel
p_fitsimg   = str(flist[0])                                              #␣
 ↪Input fits image
p_extnum    = 0                                                          #␣
 ↪Extension number for data in input fits
p_search_fwhm       = 3.0                                                #␣
 ↪initial estimate at star fwhm in pixels
p_search_nsigma     = 7.0                                                # min␣
 ↪detection sigma above background
p_detector_bitdepth = 16                                                 #␣
 ↪detector bitdepth (16 for most CCDs, ?? for CMOS, ?? for camera)
p_sat_frac          = 0.8                                                #␣
 ↪Fraction of full well at which we assume star saturated
p_max_sources = 200                                                      # Max␣
 ↪number of sources to output or None
p_nosatmask   = True                                                     #␣
 ↪Keep possibly saturated stars.
p_quiet       = False                                                    ␣
 ↪               # Quiet mode suppresses STDOUT list printing

# parameters associated with astrometric solution, if not already specified␣
 ↪above.
# Astrometry.Net API key will be dealt with elsewhere
p_inpimg          = p_fitsimg
p_srclist         = p_fitstbl
p_src_extname     = 'AP_XYPOS'
p_use_sip         = False
p_user_scale      = None
p_scale_err_ratio = None
```

[19]:
```
# Settings for output file names and output file paths
# - This is difficult to fully automate without some assumptions about the␣
 ↪input file names
#   - I assume that all input file have the same file name prefix, e.g.␣
 ↪Calibrated, or cal
#   - I assume that all input files have the same file extension, e.g. fits or␣
 ↪(sadness) fit
# - I prefer to place output files of a certain type in a subdirectories. If␣
 ↪you don't want all
#   the diagnostic plots then it may just be simpler to not deal with␣
 ↪subdirectories.
# - If dir is not None then the various outputs files will be written to␣
 ↪directories with the specified
#   path relative to the **current** directory. The directory will be created␣
 ↪if it not already
#   present.

name_conv_dict = {'srclist':  {'replace': 'Calibrated-iTelescope', 'with':␣
 ↪'srclist',   'extension': None,   'dir': 'SourceLists'},
                   'regfile':  {'replace': 'Calibrated-iTelescope', 'with':␣
 ↪'ds9',       'extension': 'reg',  'dir': 'SourceLists'},
                   'plotfile': {'replace': 'Calibrated-iTelescope', 'with':␣
 ↪'implot',    'extension': 'png',  'dir': 'SourceLists'},
                   'fwhmfile': {'replace': 'Calibrated-iTelescope', 'with':␣
 ↪'fwhmplot',  'extension': 'png',  'dir': 'SourceLists'},
                   'qualfile': {'replace': 'Calibrated-iTelescope', 'with':␣
 ↪'qual',      'extension': 'yaml', 'dir': 'MetaData'},
                   'navfile':  {'replace': 'Calibrated-iTelescope', 'with':␣
 ↪'navigated', 'extension': None,   'dir': 'NavigatedImages'}}

# Settings for reprocessing.
# If p_clean == True then existing output files will be erased
# If p_clean == False then if a file with the expected sourcelist output file␣
 ↪name exists
#   (for find stars) or navigated image output name exists (for astrometry)␣
 ↪then that stage
#   of processing is assumed completed and will not be done again. This is a␣
 ↪considerable
#   time saver when rerunning the pipeline to fix one or two files that failed␣
 ↪previously.
p_clean = False
```

[20]:
```
def mk_output_file_path(inpfile, prefix_replace_this, prefix_with_this,␣
 ↪new_file_extension=None, sub_dir=None, mkdir=False, verbose=True):
    """
```

```
    Given an input file name, modify the file prefix, optionally modifying the file
    extension and adding a subdirectory to the front of the returned path.

    :param inpfile: String containing the input file name we want to use as a
      basis for further file names, e.g. `Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits`
    :param prefix_replace_this: A part of the input file name that will be replaced with prefix_with_this,
      for example `Calibrated-iTelescope`. It is up to to the user to ensure that this substring
      is present in the input file name.
    :param prefix_with_this: What to replace prefix_replace_this with. For example `srclist`.
    :param new_file_extension: If not None then replace the `fits` file extension with this.
      Do not include the `.`.
    :param sub_dir:
    :param mkdir:
    :param verbose:

    Returns a string representing a new file name, for use by other programs.
    """

    output_file_path = None
    if verbose:
        print(f'Input file name:  {inpfile}')

    if sub_dir is not None:
        if not os.access(sub_dir, os.F_OK):
            if mkdir:
                try:
                    os.mkdir(sub_dir)
                except:
                    print(f'Error, mkdir failed trying to create {sub_dir}')
                    raise
                if verbose:
                    print(f'Successfully created subdirectory {sub_dir}')
            else:
                raise RuntimeError(f'Subdirectory {sub_dir} cannot be accessed, but mkdir=False')
        else:
            if verbose:
                 print(f'Subdirectory {sub_dir} already exists.')

    tmp_str = inpfile.replace(prefix_replace_this, prefix_with_this)
```

```
    if new_file_extension is not None:
        tmp_str = tmp_str.replace('fits', new_file_extension)
    if sub_dir is not None:
        output_file_path = f'{sub_dir}/{tmp_str}'
    else:
        output_file_path = tmp_str

    if verbose:
        print(f'Output file name: {output_file_path}')
    if inpfile == output_file_path:
        print(f'Warning: input file name ({inpfile}) matches output file name␣
 ↪({output_file_path}).')

    return output_file_path

def mk_all_file_names(inpfile, name_conv_dict, mkdir=True, verbose=False):
    """
    Given an input file name and a conversion dictionary, generate all the file␣
 ↪names that might
    be used to run star finding and astrometry on that image.

    Returns f_srclist, f_regfile, f_plotfile, f_fwhmfile, f_qualfile, f_navfile
    """
    oname = ['srclist', 'regfile', 'plotfile', 'fwhmfile', 'qualfile',␣
 ↪'navfile']
    ofile = {}
    for ftype in name_conv_dict.keys():
        ofile[ftype] = mk_output_file_path(inpfile,
            name_conv_dict[ftype]['replace'],
            name_conv_dict[ftype]['with'],
            name_conv_dict[ftype]['extension'],
            name_conv_dict[ftype]['dir'],
            mkdir,
            verbose)

    return ofile['srclist'], ofile['regfile'], ofile['plotfile'],␣
 ↪ofile['fwhmfile'], ofile['qualfile'], ofile['navfile']
```

[21]:
```
# test the function
mkdir   = True
verbose = True
for ftype in name_conv_dict.keys():
    print(f'For file type {ftype}␣
 ↪----------------------------------------------------')
    ofile   = mk_output_file_path(ifc_cal.summary['file'][0],
                    name_conv_dict[ftype]['replace'],
                    name_conv_dict[ftype]['with'],
```

```
                    name_conv_dict[ftype]['extension'],
                    name_conv_dict[ftype]['dir'],
                    mkdir,
                    verbose)

print('\nAll the file names:')
f_srclist, f_regfile, f_plotfile, f_fwhmplot, f_qualfile, f_navfile =␣
 ↪mk_all_file_names(ifc_cal.summary['file'][0],
                                                                    ␣
 ↪       name_conv_dict, mkdir, False)
print(f'  srclist={f_srclist}\n  regfile={f_regfile}\n  plotfile={f_plotfile}\n␣
 ↪ fwhmplot={f_fwhmplot}\n  qualfile={f_qualfile}\n  navfile={f_navfile}')
```

```
For file type srclist ----------------------------------------------------
Input file name:  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
Subdirectory SourceLists already exists.
Output file name: SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-1.fits
For file type regfile ----------------------------------------------------
Input file name:  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
Subdirectory SourceLists already exists.
Output file name: SourceLists/ds9-M101_Supernova_2023ixf-180s-Lum-1.reg
For file type plotfile ---------------------------------------------------
Input file name:  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
Subdirectory SourceLists already exists.
Output file name: SourceLists/implot-M101_Supernova_2023ixf-180s-Lum-1.png
For file type fwhmfile ---------------------------------------------------
Input file name:  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
Subdirectory SourceLists already exists.
Output file name: SourceLists/fwhmplot-M101_Supernova_2023ixf-180s-Lum-1.png
For file type qualfile ---------------------------------------------------
Input file name:  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
Subdirectory MetaData already exists.
Output file name: MetaData/qual-M101_Supernova_2023ixf-180s-Lum-1.yaml
For file type navfile ----------------------------------------------------
Input file name:  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
Subdirectory NavigatedImages already exists.
Output file name:
NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits

All the file names:
  srclist=SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-1.fits
  regfile=SourceLists/ds9-M101_Supernova_2023ixf-180s-Lum-1.reg
  plotfile=SourceLists/implot-M101_Supernova_2023ixf-180s-Lum-1.png
  fwhmplot=SourceLists/fwhmplot-M101_Supernova_2023ixf-180s-Lum-1.png
  qualfile=MetaData/qual-M101_Supernova_2023ixf-180s-Lum-1.yaml
  navfile=NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits
```

[22]:
```
def check_file_exists(filename):
    """Return true if the specified file path exists"""
    if not os.path.isfile(filename):
        return False
    return True
```

[23]:
```
# iterate over files in the image file collection
#
# TODO work out how to direct output or log to a file...
p_clean     = False
p_loglevel  = 'INFO'
p_quiet     = True
import time
idx = 0
proc_tstart = time.perf_counter()
for hdu, fname in ifc_cal.hdus(return_fname=True):
    print(80*'-')
    print(f'Processing input file {fname}')

    f_srclist, f_regfile, f_plotfile, f_fwhmplot, f_qualfile, f_navfile =␣
 ↪mk_all_file_names(fname,
        name_conv_dict, mkdir, False)

    # Determine whether to perform star detection, based on p_clean and␣
 ↪presence of
    does_srclist_exist = check_file_exists(f_srclist)
    if p_clean or not does_srclist_exist:
        print(f'\n  Finding stars, output source list: {f_srclist}')
        fs_tstart = time.perf_counter()
        status = 'pass'
        try:
            find_stars_wrapper(p_loglevel, fname, f_srclist,
                p_extnum, p_search_fwhm, p_search_nsigma,
                p_detector_bitdepth, p_sat_frac, p_max_sources,
                p_nosatmask, f_plotfile, p_quiet,
                f_fwhmplot, f_qualfile, f_regfile)
        except:
            print(f'  Error, caught exception when processing {fname}')
            status = 'fail'

        fs_tend     = time.perf_counter()
        fs_telapsed = fs_tend - fs_tstart # seconds
        fs_status   = status
    else:
        print(f'\n  Skipping star detection because {f_srclist} exists and␣
 ↪clean={p_clean}')
        fs_status = 'skipped_exists'
```

```
        fs_telapsed = 0

    # Determine whether to perform star detection, based on p_clean and␣
↪presence of
    does_navfile_exist = check_file_exists(f_navfile)
    if does_navfile_exist:
        if p_clean or not does_navfile_exist:
            ast_status = 'pass'
            ast_tstart = time.perf_counter()
            print(f'\n  Performing astrometry, output navigated image:␣
↪{f_navfile}')
            try:

                ap_astrom = ap.ApAstrometry(fname,
                    p_extnum,
                    f_srclist,
                    p_src_extname,
                    f_navfile,
                    p_astnetkey,
                    p_use_sip,
                    p_user_scale,
                    p_scale_err_ratio,
                    p_loglevel)
                p_status = ap_astrom.status()
                print(f'  ApAstrometry return status: {ast_status}')
                if p_status == ap.ApAstrometry.NOMINAL:
                    ast_status = 'pass'
                elif p_status == ap.ApAstrometry.INPUT_ERROR:
                    ast_status = 'input_error'
                else:
                    ast_status = 'fail'
            except:
                print(f'  Error, caught exception when processing {fname}')
                ast_status = 'exception'

            ast_tend     = time.perf_counter()
            ast_telapsed = ast_tend - ast_tstart # seconds
            ast_status   = status
        else:
            print(f'\n  Skipping astrometry because {f_navfile} exists and␣
↪clean={p_clean}')
            ast_status = 'skipped_exists'
            ast_telapsed = 0
    else:
        # No source list
        print(f'\n  Skipping astrometry because {f_srclist} does not exist.')
        ast_status = 'skipped_no_srclist'
```

```
        ast_telapsed = 0

    # Fill in run info for this file
    result_tuple = (fname, fs_status, fs_telapsed, ast_status, ast_telapsed)
    processing_results[idx] = result_tuple

    # update idx
    idx += 1

proc_tend = time.perf_counter()
proc_telapsed = proc_tend - proc_tstart
print(f'Finished processing {idx+1} files in {proc_telapsed:.3f} seconds.')
```

```
--------------------------------------------------------------------------------
Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits

  Skipping star detection because
SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-1.fits exists and
clean=False

  Skipping astrometry because
NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits exists and
clean=False
--------------------------------------------------------------------------------
Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits

  Skipping star detection because
SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-2.fits exists and
clean=False

  Skipping astrometry because
NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits exists and
clean=False
--------------------------------------------------------------------------------
Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits

  Skipping star detection because
SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-3.fits exists and
clean=False

  Skipping astrometry because
NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits exists and
clean=False
--------------------------------------------------------------------------------
Processing input file Calibrated-
```

```
iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits

  Skipping star detection because
SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-1.fits exists and
clean=False

  Skipping astrometry because
NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits exists and
clean=False
--------------------------------------------------------------------------------
Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits

  Skipping star detection because
SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-2.fits exists and
clean=False

  Skipping astrometry because
NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits exists and
clean=False
--------------------------------------------------------------------------------
Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits

  Skipping star detection because
SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-3.fits exists and
clean=False

  Skipping astrometry because
NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits exists and
clean=False
--------------------------------------------------------------------------------
Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits

  Skipping star detection because
SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-1.fits exists and
clean=False

  Skipping astrometry because
NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits exists and
clean=False
--------------------------------------------------------------------------------
Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits

  Skipping star detection because
SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-2.fits exists and
```

```
clean=False

  Skipping astrometry because
NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits exists and
clean=False
--------------------------------------------------------------------------------
Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits

  Skipping star detection because
SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-3.fits exists and
clean=False

  Skipping astrometry because
NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits exists and
clean=False
--------------------------------------------------------------------------------
Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits

  Skipping star detection because
SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-1.fits exists and
clean=False

  Skipping astrometry because
NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits exists and
clean=False
--------------------------------------------------------------------------------
Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits

  Skipping star detection because
SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-2.fits exists and
clean=False

  Skipping astrometry because
NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits exists and
clean=False
--------------------------------------------------------------------------------
Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits

  Skipping star detection because
SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-3.fits exists and
clean=False

  Skipping astrometry because
NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits exists and
```

```
clean=False
Finished processing 13 files in 0.085 seconds.
```

[24]:
```
#print(processing_results)
processing_results.pprint(max_lines=-1, max_width=-1)
```

```
                          filename                              find_stars_status
find_stars_time astrometry_status astrometry_time
------------------------------------------------------------- -----------------
-------------- ----------------- ---------------
  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits    skipped_exists
0.000    skipped_exists           0.000
  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits    skipped_exists
0.000    skipped_exists           0.000
  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits    skipped_exists
0.000    skipped_exists           0.000
 Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits    skipped_exists
0.000    skipped_exists           0.000
 Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits    skipped_exists
0.000    skipped_exists           0.000
 Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits    skipped_exists
0.000    skipped_exists           0.000
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits    skipped_exists
0.000    skipped_exists           0.000
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits    skipped_exists
0.000    skipped_exists           0.000
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits    skipped_exists
0.000    skipped_exists           0.000
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits    skipped_exists
0.000    skipped_exists           0.000
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits    skipped_exists
0.000    skipped_exists           0.000
  Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits    skipped_exists
0.000    skipped_exists           0.000
```

3.1 Image Resampling and Mosaicing

Assuming that all the input images have been successfully navigated, i.e. have valid WCS solutions, we can move on to combining the images.

It is worth checking that the navigated images do have good solutions using SAOImage `ds9`. For example:

```
ds9 -align yes -asinh -zoom to fit -cmap viridis -blink interval 2 -blink yes -tile no -single
    -scale mode 99.5 NavigatedImages/navigated*.fits
```

That should have loaded the images into separate frames, but only shown one at a time with the program blinking through them. However it seems to always tile them with multiple inputs. You need to perform the following actions in the ds9 GUI.

- resize the ds9 window to make it larger by dragging on the window edge
- Click through the menu items `Frame`, then `Single Frame`
- Click through the menu items `Frame`, then `Match`, then `Frame`, then `WCS`
- Click through the menu items `Frame`, then `Blink Frames`

The same stars/features in each separate image should appear to be in the same place on the screen, even as the brightness and noise levels change between frames.

[25]:
```
# Make sure we're still in the original directory, then move into the directory␣
 ↪with the navigated images
os.chdir(wdir)
ftype   = 'navfile'
fdir    =  name_conv_dict[ftype]['dir']
fprefix =  name_conv_dict[ftype]['with']
fext    =  name_conv_dict[ftype]['extension']
if fext is None:
    fext='.fits'
try:
    os.chdir(fdir)
    print(f'Successfully changed directory to {os.getcwd()}')
except:
    print('Error, failed to change directory to expected navfile directory␣
 ↪{fdir}')
    print(f'  Current directory: {os.getcwd()}')
```

```
Successfully changed directory to
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN/NavigatedImages
```

3.1.1 Caveats

Note: The following code assumes that the navigated images in the current directory correspond to **a single target**.

All images in the directory will be resampled to a single pointing and platescale. By default this will be selected based on the first navigated image in the image file collection.

We would not necessarily want to resample images of the same `filter` but differing `exposure` together as their noise characteristics might be quite distinct, and the resampling method used might not be able to correctly weight them. Similarly cases where the seeing is significantly poorer than average, as determined by the fits performed by `ApMeasureStars` and reported in the quality files, should be excluded from being combined.

Such ideals run afoul of cases where the fits files don't have those keywords or are using different keywords. For now I don't have a work around for such cases and for automatically incorporating the seeing/FWHM measurements.

[26]:
```
nav_file_pattern=f'{fprefix}*{fext}'
p_dir = pathlib.Path(r'./')
keys = ['naxis1', 'naxis2', 'imagetyp', 'object', 'filter', 'exposure']
```

```
ifc_nav = ImageFileCollection(p_dir, keywords=keys,␣
 ↪glob_include=nav_file_pattern) # no glob_excludes at this time...
print(ifc_nav.summary)
```

```
                      file                       naxis1 naxis2 imagetyp object
filter exposure
------------------------------------------------- ------ ------ -------- ------
------ --------
  navigated-M101_Supernova_2023ixf-180s-Lum-1.fits   3056   3056       --     --
Lum       --
  navigated-M101_Supernova_2023ixf-180s-Lum-2.fits   3056   3056       --     --
Lum       --
  navigated-M101_Supernova_2023ixf-180s-Lum-3.fits   3056   3056       --     --
Lum       --
 navigated-M101_Supernova_2023ixf-300s-Blue-1.fits   3056   3056       --     --
Blue      --
 navigated-M101_Supernova_2023ixf-300s-Blue-2.fits   3056   3056       --     --
Blue      --
 navigated-M101_Supernova_2023ixf-300s-Blue-3.fits   3056   3056       --     --
Blue      --
navigated-M101_Supernova_2023ixf-300s-Green-1.fits   3056   3056       --     --
Green      --
navigated-M101_Supernova_2023ixf-300s-Green-2.fits   3056   3056       --     --
Green      --
navigated-M101_Supernova_2023ixf-300s-Green-3.fits   3056   3056       --     --
Green      --
  navigated-M101_Supernova_2023ixf-300s-Red-1.fits   3056   3056       --     --
Red       --
  navigated-M101_Supernova_2023ixf-300s-Red-2.fits   3056   3056       --     --
Red       --
  navigated-M101_Supernova_2023ixf-300s-Red-3.fits   3056   3056       --     --
Red       --
```

[27]:
```
from astropy.table import unique
uniq_filter_exposure = unique(ifc_nav.summary, keys=['filter', 'exposure'],␣
 ↪keep='first', silent=True)
uniq_filter_exposure.pprint(max_lines=-1, max_width=-1)
```

```
                      file                       naxis1 naxis2 imagetyp object
filter exposure
------------------------------------------------- ------ ------ -------- ------
------ --------
 navigated-M101_Supernova_2023ixf-300s-Blue-1.fits   3056   3056       --     --
Blue      --
navigated-M101_Supernova_2023ixf-300s-Green-1.fits   3056   3056       --     --
Green      --
  navigated-M101_Supernova_2023ixf-180s-Lum-1.fits   3056   3056       --     --
Lum       --
```

```
  navigated-M101_Supernova_2023ixf-300s-Red-1.fits    3056    3056       --      --
Red       --
```

[28]:
```
filter_list = []
for filter in uniq_filter_exposure['filter']:
    filter_list.append(filter)
print(f'There are {len(filter_list)} unique filters among {len(ifc_nav.
 ↪summary)} files: {filter_list}')
```

```
There are 4 unique filters among 12 files: ['Blue', 'Green', 'Lum', 'Red']
```

3.1.2 What Pointing and Plate Scale Should We Resample To?

There are any number of possible choices we can make regarding the pointing and plate scale that all the images will be resampled to. A few possible choices are shown:

1. Resample all images to match the WCS of another image, for example the first image in the list of navigated images, irrespective of how uniform the platescale is or whether the image is North-up, East-left aligned.
2. Require the user to explicitly specify the RA and Dec of the image center, the X (RA) and Y (Dec) `CDELT*` values, and the number of output image rows and columns.
3. Require the user to specify a single plate scale (arcsec/pixel) and a RA/Dec or target identifiers, and calculate the values needed for #2.

There are advantages and disadvantages to each of these approaches. Method #1 may be superior for a final pretty display image where alignment is unimportant, while #2 and #3 are likely better for scientific use of the combined images. Method #2 requires user knowledge of how to specify a WCS and the likely platescale of the images, and is really only good for experts.

For now AstroPhotography will concentrate on supporting methods #1 and #3, starting with method #1.

Note: In image resampling the question of whether the input is well sampled is important. We do not have the space to cover it here, except to note that in general it is better to have pixels in the final (resampled) image be square and as large or indeed slightly larger than the pixels in the input images.

Note: For technical reasons most resampling methods assume the input data pixels represent surface brightness. If the input image has a large field of view it is likely distorted, in particular with images from DLSR or Mirrorless cameras, and the sky area covered by each pixel is not uniform. `Astrometry.net` allows fitting of `SIP` distortion coefficients. However, correctly calculation surface brightness in such cases is difficult (this package does not currently support it, and I don't know of any that do). Furthermore many of the resampling methods available don't correctly handle the non-uniform spatial aspect of SIP distortions either.

Other methods: resample_all.sh shell script The bash script `resample_all.sh` does perform resampling using Astromatic's `swarp` resampler. Note that the script needs to be hard-wired for each set of images by specifying the RA, Dec of the image center, the plate scale, and the number of pixels along each axis.

```
cd /old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN
~/git/AstroPhotography/AstroPhotography/scripts/resample_all.sh M101sn median clean verbose
... [lots of output omitted]...
------------------------------------------------------------------------
Processing Lum filter images:
  Using existing temporary file directory TmpSwarp
  Found 3 navigated fits files.
  File navigated-M101_Supernova_2023ixf-180s-Lum-3.fits EXPOSURE  180.000 FSCALE 0.005556
  File navigated-M101_Supernova_2023ixf-180s-Lum-1.fits EXPOSURE  180.000 FSCALE 0.005556
  File navigated-M101_Supernova_2023ixf-180s-Lum-2.fits EXPOSURE  180.000 FSCALE 0.005556
  Flux scalings: 0.005556,0.005556,0.005556
  Total exposure in Lum filter: 9 minutes.

  Beginning swarp for filter Lum at Wed Feb 28 09:25:14 PM EST 2024
Running the following command:  swarp .././NavigatedImages/navigated-M101_Supernova_2023ixf-180
  Generated resampled co-added image M101_Supernova_2023ixf_Lum_4096x4096_resamp_median.fits
  Generated co-added weights image M101_Supernova_2023ixf_Lum_4096x4096_weights_median.fits
  Swarp for filter Lum completed at Wed Feb 28 09:25:19 PM EST 2024, took 5.099 seconds.

------------------------------------------------------------------------
Processing Luminance filter images:
  Using existing temporary file directory TmpSwarp
  Found 0 navigated fits files.
  Moving on to next filter...
------------------------------------------------------------------------
Run summary:
Filter      Resampled Output                                              Ninput   TexpMin
Red         M101_Supernova_2023ixf_Red_4096x4096_resamp_median.fits            3     15.00
Green       M101_Supernova_2023ixf_Green_4096x4096_resamp_median.fits          3     15.00
Blue        M101_Supernova_2023ixf_Blue_4096x4096_resamp_median.fits           3     15.00
Ha          None                                                               0      0.00
OIII        None                                                               0      0.00
SII         None                                                               0      0.00
B           None                                                               0      0.00
V           None                                                               0      0.00
I           None                                                               0      0.00
Clear       None                                                               0      0.00
Lum         M101_Supernova_2023ixf_Lum_4096x4096_resamp_median.fits            3      9.00
Luminance   None                                                               0      0.00
Master log file: /old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN/tmp_resamplecd_al

------------------------------------------------------------------------
  Temporary files occupy 1 MB in /old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN/T
  Commands were logged to /old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN/tmp_resa
  Resampled images were written to ./Resampled
resample_all.sh finished at Sun Feb 25 08:33:13 AM EST 2024, run time 23.638 seconds.
```

Resampling To Match Another Image This is the simplest conceptual method, but does allow some flexibility in that the other image need not be one of the navigated images. For example we could resample to match the WCS of a Chandra X-ray image, which would be ideal for scientific multi-waveband analysis.

First let's specify the file we want to match...

[29]:
```
f_default_match = ifc_nav.summary['file'][0]
msg = f'Name of file to match WCS of in resampled images (hit return for␣
 ↪default {f_default_match})'
f_to_match = input(msg).strip() or f_default_match
print(f'WCS of resampled files will match {f_to_match}')
```

```
Name of file to match WCS of in resampled images (hit return for default
navigated-M101_Supernova_2023ixf-180s-Lum-1.fits)

WCS of resampled files will match
navigated-M101_Supernova_2023ixf-180s-Lum-1.fits
```

[71]:
```
from astropy import wcs
import math
def load_wcs_from_file(filename, verbose=False):
    """
    Modified WCS example based on astropy documentation

    https://docs.astropy.org/en/stable/wcs/loading_from_fits.html

    Limitations: Assumes 2-dimensional image
    """
    w          = None
    pix_scales = None # in deg
    pix_area   = None # in deg^2
    im_scales  = None # in deg

    # Load the FITS hdulist using astropy.io.fits
    with fits.open(filename) as hdulist:

        # Parse the WCS keywords in the primary HDU
        w = wcs.WCS(hdulist[0].header)

        # Print out the "name" of the WCS, as defined in the FITS header
        if verbose:
            print(w.wcs.name)

            # Print out all of the settings that were parsed from the header
            w.wcs.print_contents()

        # Pixel scale at the CRPIX pixel location. Note, ideal, ignores␣
 ↪distortions
```

```
        pix_scales = wcs.utils.proj_plane_pixel_scales(w)

        # Pixel area at the CRPIX pixel location. Again, ideal, ignores␣
↪distortions
        pix_area = wcs.utils.proj_plane_pixel_area(w.celestial)

        if w.wcs.naxis != 2:
            print(f'WARNING, WCS from {filename} is {w.wcs.naxis}-dimensional,␣
↪not 2-D as expected')

        if len(pix_scales) != len(w.array_shape):
            raise RuntimeError(f'Shape of pixel scales ({len(pix_scales)})␣
↪differs from shape of array ({len(w.array_shape)})')
        else:
            im_scales = pix_scales.copy()
            for idx in range(len(w.array_shape)):
                im_scales[idx] *= w.array_shape[idx]

        if verbose:
            print(f'Pixel angular scale ({w.wcs.cunit[0]}):    {pix_scales}')
            print(f'Pixel angular area ({w.wcs.cunit[0]}^2):   {pix_area}')
            print(f'NAXIS1={w.array_shape[0]}   NAXIS2={w.array_shape[1]}')
            print(f'Detector angular scale ({w.wcs.cunit[0]}): {im_scales}')


    return w, pix_scales, pix_area, im_scales
```

[92]:
```
def summarize_wcs(w):
    """
    Given an astropy WCS object, summarise the properties of the WCS header,
    printing to STDOUT.

    This assumes that the images are two dimensional, and that angles are in␣
↪degrees.

    :param w: astropy.wcs.WCS instance
    """

    ipwcs = w

    dothead_list = []

    # Numbr of dimensions
    val_str = f"{'NAXIS':8s} = {ipwcs.wcs.naxis}"
    dothead_list.append( val_str )

    keywords = ["NAXIS", "CTYPE", "CRVAL", "CRPIX"]
```

```
   values = [ipwcs.array_shape, ipwcs.wcs.ctype, ipwcs.wcs.crval, ipwcs.wcs.
↪crpix]
   for keyword, value in zip(keywords, values):
       for idx in range(ipwcs.naxis):
           kw_str  = f"{keyword}{1+idx}"
           if 'CTYPE' in keyword:
               # Wrap strings in quotes
               val_str = f"{kw_str:8s} = '{value[idx]}'"
           else:
               val_str = f"{kw_str:8s} = {value[idx]}"
           dothead_list.append( val_str )

   naxis  = ipwcs.wcs.naxis
   naxis1 = ipwcs.array_shape[0]
   naxis2 = ipwcs.array_shape[1]
   if naxis == 3:
       naxis3 = ipwcs.array_shape[2]
       print(f'Warning: summarize_wcs() not written for 2-D images,␣
↪{naxis}-dimensional data')

   cdelt1 = None
   cdelt2 = None
   crot   = None

   if hasattr(ipwcs.wcs, "cd"):
       for irow in range(ipwcs.naxis):
           for jcol in range(ipwcs.naxis):
               kw_str = f'CD{irow+1}_{jcol+1}'
               val_str = f'{kw_str:8s} = {ipwcs.wcs.cd[irow, jcol]}'
               dothead_list.append( val_str )
       cd = ipwcs.wcs.cd
       # From https://lweb.cfa.harvard.edu/~jzhao/SMA-FITS-CASA/docs/wcs88.pdf
       cd11 = cd[0,0]
       cd12 = cd[0,1]
       cd21 = cd[1,0]
       cd22 = cd[1,1]
       cdelt1_mag = math.sqrt( cd12*cd12 + cd22*cd22 )
       cdelt2_mag = math.sqrt( cd12*cd12 + cd22*cd22 )
       # the sign of cdelt1.cdelt2 = sign of (cd11*cd22 - cd12*cd21)
       # if the RHS is negative cdelt1 is negative by convention, so cdelt2 is␣
↪always positive
       tmpa =  cd11*cd22 - cd12*cd21
       cdelt1 = math.copysign(cdelt1_mag, tmpa)
       cdelt2 = cdelt2_mag
       sign   = math.copysign(1, tmpa)
       crot   = math.degrees( math.atan2( (sign*cd12), cd22) )
```

```
    elif hasattr(ipwcs.wcs, "pc"):
        for irow in range(ipwcs.naxis):
            for jcol in range(ipwcs.naxis):
                kw_str = f'PC{irow+1}_{jcol+1}'
                val_str = f'{kw_str:8s} = {ipwcs.wcs.pc[irow, jcol]}'
                dothead_list.append( val_str )
            kw_str = f'CDELT{1+irow}'
            val_str = 'f{kw_str:8s} = {ipwcs.wcs.cdelt[irow]}'
            dothead_list.append( val_str )
            # Think that CD1_* = CDELT1 * PC1_* and CD2_* = CDELT2 * PC2_*
            pc = ipwcs.wcs.pc
            cdelt_vec = ipwcs.wcs.cdelt
            cd = pc.copy()
            for irow in range(ipwcs.naxis):
                cd[irow] = cdelt_vec[irow] * pc[irow]
            # then as above
        raise RuntimeError('summarize_wcs needs to be updated to handle cases␣
↪with a PC_ matrix and no CD_ matrix')
    else:
        # Assume we have a simple CDELT[12] case with CROTA
        for irow in range(ipwcs.naxis):
            kw_str = f'CDELT{1+irow}'
            val_str = 'f{kw_str:8s} = {ipwcs.wcs.cdelt[irow]}'
            dothead_list.append( val_str )
            kw_str = f'CROTA{1+irow}'
            val_str = 'f{kw_str:8s} = {ipwcs.wcs.crota[irow]}'
            dothead_list.append( val_str )
        cdelt1 = ipwcs.wcs.cdelt[0]
        cdelt2 = ipwcs.wcs.cdelt[1]
        crot   = ipwcs.wcs.crota[1] # CROTA2 is used, CROTA1 not used. Assume␣
↪crota[0] == crota[1]

    cdelt1_as = cdelt1 * 3600.0      # arcseconds
    cdelt2_as = cdelt2 * 3600.0      # arcseconds
    ximgsz_am = cdelt1 * naxis1 * 60 # arcminutes
    yimgsz_am = cdelt2 * naxis2 * 60 # arcminutes
    dothead_list.append( f'Pixel size equivalent CDELT1     = {cdelt1_as:.3f}␣
↪arcseconds' )
    dothead_list.append( f'Pixel size equivalent CDELT2     = {cdelt2_as:.3f}␣
↪arcseconds' )
    dothead_list.append( f'Image X-axis angular size        = {ximgsz_am:.3f}␣
↪arcminutes' )
    dothead_list.append( f'Pixel Y-axis angular size        = {yimgsz_am:.3f}␣
↪arcminutes' )
    dothead_list.append( f'Image rotation equivalent CROTA2 = {crot:.3f}␣
↪degrees' )
```

```
    dothead_list.append('END     ')
    dothead_str = '\n'.join(dothead_list)
    print(dothead_str)
    return
```

[93]:
```
# Test load_wcs_from_file
w, pix_scales, pix_area, im_scales = load_wcs_from_file(f_to_match, False)
pix_scales_as = pix_scales * 3600.0
root_area_as = math.sqrt(pix_area) * 3600.0
im_scales_am = im_scales * 60.0
print(f'Pixel scales (arcsec):                 {pix_scales_as}')
print(f'Image angular extent (arcmin):         {im_scales_am}')
print(f'Square pixel equivalent size (arcsec): {root_area_as}')

summarize_wcs(w)
```

```
Pixel scales (arcsec):                 [0.63102649 0.63102649]
Image angular extent (arcmin):         [32.14028278 32.14028278]
Square pixel equivalent size (arcsec): 0.6310264944404745
NAXIS    = 2
NAXIS1   = 3056
NAXIS2   = 3056
CTYPE1   = 'RA---TAN'
CTYPE2   = 'DEC--TAN'
CRVAL1   = 210.754804653
CRVAL2   = 54.4174059435
CRPIX1   = 1528.5
CRPIX2   = 1528.5
CD1_1    = -4.64681630389e-06
CD1_2    = 0.000175223532872
CD2_1    = -0.000175223532872
CD2_2    = -4.64681630389e-06
Pixel size equivalent CDELT1     = 0.631 arcseconds
Pixel size equivalent CDELT2     = 0.631 arcseconds
Image X-axis angular size        = 32.140 arcminutes
Pixel Y-axis angular size        = 32.140 arcminutes
Image rotation equivalent CROTA2 = 91.519 degrees
END
```

[32]:
```
def make_dothead_from_file(input_fits_with_wcs, output_swarp_dothead,␣
↪format='fits', verbose=False):
    """
    Create the .head format file that swarp expected based on the WCS header of␣
↪an input files file.
```

```
    Internally this uses the method from https://docs.astropy.org/en/stable/
↪_modules/astropy/wcs/wcs.html#WCS.printwcs

    Limitations: Assumes the WCS information is in the primary header

    :param input_fits_with_wcs: Name of existing FITS file with WCS in primary␣
↪HDU that we want to emulate.
    :param output_swarp_dothead:
    :param format: Either 'text' or 'fits'. If 'text' then an ASCII header will␣
↪be created using the format
      specified in the Swarp manual. If 'fits' is specified, an empty FITS file␣
↪consisting only of a primary
      HDU will be created.
    :param verbose: If True then diagnostic information will be written to␣
↪stdout.
    """

    with fits.open(input_fits_with_wcs) as hdulist:
        # Parse the WCS keywords in the primary HDU
        ipwcs = wcs.WCS(hdulist[0].header)

        if verbose:
            print(f'WCS created from primary header of {input_fits_with_wcs}')
            print(ipwcs)
            print(80*"-")
            #print(ipwcs.wcs)
            #print(80*"-")

        if 'text' in format:
            if verbose:
                print('Generating an ASCII header following the format␣
↪specified in the swarp documentation.')

            dothead_list = []
            keywords = ["NAXIS", "CTYPE", "CRVAL", "CRPIX"]
            values = [ipwcs.array_shape, ipwcs.wcs.ctype, ipwcs.wcs.crval,␣
↪ipwcs.wcs.crpix]
            for keyword, value in zip(keywords, values):
                for idx in range(ipwcs.naxis):
                    kw_str  = f"{keyword}{1+idx}"
                    if 'CTYPE' in keyword:
                        # Wrap strings in quotes
                        val_str = f"{kw_str:8s} = '{value[idx]}'"
                    else:
                        val_str = f"{kw_str:8s} = {value[idx]}"
                    dothead_list.append( val_str )
```

```
            if hasattr(ipwcs.wcs, "pc"):
                for irow in range(ipwcs.naxis):
                    for jcol in range(ipwcs.naxis):
                        kw_str = f'PC{irow+1}_{jcol+1}'
                        val_str = f'{kw_str:8s} = {ipwcs.wcs.pc[irow, jcol]}'
                        dothead_list.append( val_str )
                    kw_str = f'CDELT{1+irow}'
                    val_str = 'f{kw_str:8s} = {pwcs.wcs.cdelt[irow]}'
                    dothead_list.append( val_str )
            elif hasattr(ipwcs.wcs, "cd"):
                for irow in range(ipwcs.naxis):
                    for jcol in range(ipwcs.naxis):
                        kw_str = f'CD{irow+1}_{jcol+1}'
                        val_str = f'{kw_str:8s} = {ipwcs.wcs.cd[irow, jcol]}'
                        dothead_list.append( val_str )

            dothead_list.append('END     ')
            dothead_str = '\n'.join(dothead_list)

            if verbose:
                print(f'--- dothead output from {input_fits_with_wcs} ---')
                print(dothead_str)
                print(f'--- about to write to {output_swarp_dothead} ---')

            with open(output_swarp_dothead, 'w') as ofile:
                ofile.write(dothead_str)
                if verbose:
                    print(f'Wrote Swarp ASCII-format .head file to␣
↪{output_swarp_dothead}')
        elif 'fits' in format:
            if verbose:
                print('Generating a FITS format header consisting of a␣
↪PrimaryHDU only.')
            # based on https://docs.astropy.org/en/stable/wcs/
↪example_create_imaging.html
            hdr = ipwcs.to_header()

            # NAXIS = x, NAXISx values set from data, not by manipulating␣
↪header
            olddata = hdulist[0].data
            data = 0 * olddata.astype(int)

            hdu = fits.PrimaryHDU(header=hdr, data=data)
            hdu.writeto(output_swarp_dothead, overwrite=True,␣
↪output_verify='ignore')
            if verbose:
```

```
                print(f'Wrote FITS header .head file to {output_swarp_dothead}')
        else:
            raise RuntimeError(f'Error, format ({format}) is not one of the␣
↪allowed options: "text" "fits"')
    return
```

[33]:
```
# Test make_dothead_from_file
ofilename = 'M101_Supernova_2023ixf_Lum_4096x4096_resamp_weighted.fits'
oheadname = ofilename.replace('.fits','.head')
make_dothead_from_file(f_default_match, oheadname, 'fits', True)
```

```
WCS created from primary header of
navigated-M101_Supernova_2023ixf-180s-Lum-1.fits
WCS Keywords

Number of WCS axes: 2
CTYPE : 'RA---TAN' 'DEC--TAN'
CRVAL : 210.754804653 54.4174059435
CRPIX : 1528.5 1528.5
CD1_1 CD1_2  : -4.64681630389e-06 0.000175223532872
CD2_1 CD2_2  : -0.000175223532872 -4.64681630389e-06
NAXIS : 3056  3056
--------------------------------------------------------------------------------
Generating a FITS format header consisting of a PrimaryHDU only.
Wrote FITS header .head file to
M101_Supernova_2023ixf_Lum_4096x4096_resamp_weighted.head
```

Resampling Using Python We'll try to recreate the bash shell script resampling of the luminance images, although with the same header as an input image instead of a user-defined pointing. As a reminder the shell command the script used earlier was:

```
swarp .././NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits .././NavigatedImag
```

The `swarp` command line parameters we'll want to change are:

- `CELESTIAL_TYPE`: This needs to be set to native to copy the first input image.
- `VERBOSE_TYPE`: By default `NORMAL`, options are `QUIET`, `LOG`, `NORMAL`, `FULL`.
- `COMBINE_TYPE`: Median is good for a first look but will increase the variance. If possible use `WEIGHTED`
- `PIXEL_SCALE_TYPE`: Various options, see `swarp` manual. Maybe set to `MAX`
- – Note this is ignored if a `.head` file is supplied
- `CENTER_TYPE`: Instead of `MANUAL` we want `MOST` or `ALL`. The latter options also generate output images that are aligned North up, East to the left.
- – Hopefully this is ignored if we specify a `.head`?
- `CENTER` and `PIXEL_SCALE` wont be used if we change `PIXEL_SCALE_TYPE` and `CENTER_TYPE`

- `IMAGE_SIZE`: Use 0 for automatic sizing based on other parameter values
- `SUBTRACT_BACK`: Using `N` is appropriate if we have a lot of nebular emission, crowded fields, large galaxies, or have already performed background subtraction. You should really try with both 'Y' and 'N' to determine the effect of swarp's in-built background subtraction.
- `RESAMPLE_DIR`: Directory where temporary files are written. This must exist.

In the case of the M101 SN premium images `SUBTRACT_BACK` should be `N`. The premium dataset is not a great test of the different center type options because the three input images very closely overlap.

[35]:
```
# ***You may want to disable this, as it uses assumed file names and directory␣
 ↪paths***
# It also assumes a Unix like operating system!!!!
do_simple_test = False
do_better_test = True
do_better_test_center_type = 'FIRST' # MANUAL, MOST, ALL, FIRST (MANUAL and ALL␣
 ↪are swarp types)
dothead_format = 'fits' # 'text' or 'fits

ofilename = 'M101_Supernova_2023ixf_Lum_4096x4096_resamp_weighted.fits'
pathlib.Path(ofilename).unlink(missing_ok=True)  # remove old file
oheadname = ofilename.replace('.fits','.head')

if do_simple_test:
    # Make sure to delete any .head file
    pathlib.Path(oheadname).unlink(missing_ok=True)

    # test command, long form, based of the previous example from the␣
 ↪resample_all.sh script
    test_cmd = 'swarp navigated-M101_Supernova_2023ixf-180s-Lum-1.fits␣
 ↪navigated-M101_Supernova_2023ixf-180s-Lum-2.fits␣
 ↪navigated-M101_Supernova_2023ixf-180s-Lum-3.fits -VERBOSE_TYPE FULL␣
 ↪-SUBTRACT_BACK N -COMBINE_TYPE WEIGHTED -FSCALASTRO_TYPE VARIABLE␣
 ↪-VERBOSE_TYPE FULL -GAIN_DEFAULT 1.0  -GAIN_KEYWORD EGAIN -FSCALE_DEFAULT 0.
 ↪005556,0.005556,0.005556 -PIXELSCALE_TYPE MANUAL -PIXEL_SCALE 0.62␣
 ↪-CENTER_TYPE MANUAL -CENTER 210.8022671,54.3489500 -RESAMPLING_TYPE LANCZOS3␣
 ↪-OVERSAMPLING 4 -IMAGE_SIZE 4096,4096 -PROJECTION_TYPE TAN -IMAGEOUT_NAME␣
 ↪M101_Supernova_2023ixf_Lum_4096x4096_resamp_weighted.fits -WRITE_FILEINFO Y␣
 ↪-WRITE_XML N -DELETE_TMPFILES Y -RESAMPLE_DIR ./ -WEIGHTOUT_NAME␣
 ↪M101_Supernova_2023ixf_Lum_4096x4096_weights_weighted.fits'
    test_cmd_list = shlex.split(test_cmd)
    print(f'Command string to be passed to subprocess.run is {test_cmd_list}')
    fs_tstart     = time.perf_counter()
    result = subprocess.run(test_cmd_list, check=True, stdout=subprocess.PIPE,␣
 ↪stderr=subprocess.PIPE)
    fs_tend     = time.perf_counter()
```

```
    fs_telapsed = fs_tend - fs_tstart # seconds
    print(f'  Process took {fs_telapsed:.3f} seconds, return code {result.
 ↪returncode}')
    print(f'  Input args: {result.args}')
    print(f'  Stdout: {result.stdout}')
    print(f'  Stderr: {result.stderr}')

if do_better_test:
    # test command, split into related sub-sections

    test_cmd1 = 'swarp navigated-M101_Supernova_2023ixf-180s-Lum-1.fits␣
 ↪navigated-M101_Supernova_2023ixf-180s-Lum-2.fits␣
 ↪navigated-M101_Supernova_2023ixf-180s-Lum-3.fits -FSCALASTRO_TYPE VARIABLE␣
 ↪-FSCALE_DEFAULT 0.005556,0.005556,0.005556'
    test_cmd2 = '-VERBOSE_TYPE FULL -SUBTRACT_BACK N -COMBINE_TYPE WEIGHTED␣
 ↪-GAIN_DEFAULT 1.0 -GAIN_KEYWORD EGAIN -RESAMPLING_TYPE LANCZOS3␣
 ↪-OVERSAMPLING 4 -PROJECTION_TYPE TAN'
    if 'MANUAL' in do_better_test_center_type:
        pathlib.Path(oheadname).unlink(missing_ok=True)
        test_cmd3 = '-PIXELSCALE_TYPE MANUAL -PIXEL_SCALE 0.62 -CENTER_TYPE␣
 ↪MANUAL -CENTER 210.8022671,54.3489500 -IMAGE_SIZE 4096,4096'
    elif 'ALL' in do_better_test_center_type:
        test_cmd3 = '-PIXELSCALE_TYPE MAX -CENTER_TYPE ALL'
        pathlib.Path(oheadname).unlink(missing_ok=True)
    elif 'MOST' in do_better_test_center_type:
        test_cmd3 = '-PIXELSCALE_TYPE MAX -CENTER_TYPE MOST'
        pathlib.Path(oheadname).unlink(missing_ok=True)
    elif 'FIRST'  in do_better_test_center_type:
        # Need to generate a header based on the first file, with a name based␣
 ↪on the output file name
        # The pixelscale and centertype should be set from the .head file...
        make_dothead_from_file(f_default_match, oheadname, dothead_format, True)
        test_cmd3 = ''
    else:
        raise RuntimeError(f'Error, center type {do_better_test_center_type}␣
 ↪has not yet been implemented.')

    test_cmd4 = f'-IMAGEOUT_NAME {ofilename} -WRITE_FILEINFO Y -WRITE_XML N␣
 ↪-DELETE_TMPFILES Y -RESAMPLE_DIR ./ -WEIGHTOUT_NAME␣
 ↪M101_Supernova_2023ixf_Lum_4096x4096_weights_weighted.fits'
    test_cmd_list = shlex.split(test_cmd1) + shlex.split(test_cmd2) + shlex.
 ↪split(test_cmd3) + shlex.split(test_cmd4)
    print(f'Command string to be passed to subprocess.run is {test_cmd_list}')
    fs_tstart     = time.perf_counter()
    try:
```

```
        result = subprocess.run(test_cmd_list, check=True, stdout=subprocess.
↪PIPE, stderr=subprocess.PIPE)
        fs_tend     = time.perf_counter()
        fs_telapsed = fs_tend - fs_tstart # seconds
        print(f'  Success: process took {fs_telapsed:.3f} seconds, return code␣
↪{result.returncode}')
        #print(f'  Input args: {result.args}')
        #print(f'  Stdout: {result.stdout}')
        #print(f'  Stderr: {result.stderr}')

    except subprocess.CalledProcessError as err:
        fs_tend     = time.perf_counter()
        fs_telapsed = fs_tend - fs_tstart # seconds
        print(f'  Error, process took {fs_telapsed:.3f} seconds, return code␣
↪{err.returncode}')
        print(f'  Input args: {err.cmd}\n')
        print(f'  Stdout: {err.output}\n')
        print(f'  Stderr: {err.stderr}\n')
        print(f'  Command line equivalent command: {" ".join(err.cmd)}')
```

```
WCS created from primary header of
navigated-M101_Supernova_2023ixf-180s-Lum-1.fits
WCS Keywords

Number of WCS axes: 2
CTYPE : 'RA---TAN' 'DEC--TAN'
CRVAL : 210.754804653 54.4174059435
CRPIX : 1528.5 1528.5
CD1_1 CD1_2  : -4.64681630389e-06 0.000175223532872
CD2_1 CD2_2  : -0.000175223532872 -4.64681630389e-06
NAXIS : 3056  3056
--------------------------------------------------------------------------------
Generating a FITS format header consisting of a PrimaryHDU only.
Wrote FITS header .head file to
M101_Supernova_2023ixf_Lum_4096x4096_resamp_weighted.head
Command string to be passed to subprocess.run is ['swarp',
'navigated-M101_Supernova_2023ixf-180s-Lum-1.fits',
'navigated-M101_Supernova_2023ixf-180s-Lum-2.fits',
'navigated-M101_Supernova_2023ixf-180s-Lum-3.fits', '-FSCALASTRO_TYPE',
'VARIABLE', '-FSCALE_DEFAULT', '0.005556,0.005556,0.005556', '-VERBOSE_TYPE',
'FULL', '-SUBTRACT_BACK', 'N', '-COMBINE_TYPE', 'WEIGHTED', '-GAIN_DEFAULT',
'1.0', '-GAIN_KEYWORD', 'EGAIN', '-RESAMPLING_TYPE', 'LANCZOS3',
'-OVERSAMPLING', '4', '-PROJECTION_TYPE', 'TAN', '-IMAGEOUT_NAME',
'M101_Supernova_2023ixf_Lum_4096x4096_resamp_weighted.fits', '-WRITE_FILEINFO',
'Y', '-WRITE_XML', 'N', '-DELETE_TMPFILES', 'Y', '-RESAMPLE_DIR', './',
'-WEIGHTOUT_NAME', 'M101_Supernova_2023ixf_Lum_4096x4096_weights_weighted.fits']
```

```
Success: process took 4.667 seconds, return code 0
```

Issues with swarp's ASCII `.head` file format Information retention: Using swarp's `.head` format with the `END` does not seem to work. The following `.head` should work

```
CTYPE1   = RA---TAN
CTYPE2   = DEC--TAN
CRVAL1   = 210.754804653
CRVAL2   = 54.4174059435
CRPIX1   = 1528.5
CRPIX2   = 1528.5
CD1_1    = -4.64681630389e-06
CD1_2    = 0.000175223532872
CD2_1    = -0.000175223532872
CD2_2    = -4.64681630389e-06
END
```

but we get the following error:

```
----- SWarp 2.38.0 started on 2024-03-13 at 09:04:47 with 16 threads

Examining input data ...
Looking for navigated-M101_Supernova_2023ixf-180s-Lum-1.fits ...
Looking for navigated-M101_Supernova_2023ixf-180s-Lum-2.fits ...
Looking for navigated-M101_Supernova_2023ixf-180s-Lum-3.fits ...
Creating NEW output image ...

> *FATAL ERROR*: Unknown FITS type in fitswrite()
```

This is emitted by the `fitswrite` keyword writing code in `src/fits/fitsutil.c`.

Trying with an actual FITS header Added an option to `make_dothead_from_file` to generate FITS format header. **This works**. Note that swarp can eat inwards from the edge…

Resampling All The Filters Now that we have the method for resampling to match a given image we'll move onto to resampling all the bands (filters) we have.

The new challenge is to compute the appropriate scale factors and which header keywords need to be updated post-resampling (e.g. exposure related keywords, history, and so on). We'll also want some form of summary table.

[36]:
```python
def get_exposure_time(hdr, verbose=False):
    """Return the first of EXPTIME, EXPOSURE, ONTIME, or LIVETIME from a FITS␣
 ↪header"""
    keywords = ['EXPTIME', 'EXPOSURE', 'ONTIME', 'LIVETIME']
    exposure_time = None
    for key in keywords:
        if hdr.count(key) > 0:
            exposure_time = float( hdr[key] )
            if verbose:
```

```
                print(f'Keyword {key} found, setting exposure_time to␣
 ↪{exposure_time}')
                break
            else:
                if verbose:
                    print(f'Keyword {key} not found, continuing search...')
    return exposure_time
```

[37]:
```
do_better_test_center_type = "FIRST"  # MANUAL, MOST, ALL, FIRST (MANUAL and␣
 ↪ALL are swarp types)
dothead_format = "fits"  # 'text' or 'fits
swarp_verbose  = False  # Echo swarp input string if True, echo .head for FIRST␣
 ↪case
final_images   = []

for filter in filter_list:
    print(f"Processing images from filter  {filter}")

    # Names for output resampled image, net weights, and .head files
    ofilename = f"M101_Supernova_2023ixf_{filter}_resamp_weighted.fits"
    owgtsname = f"M101_Supernova_2023ixf_{filter}_weights_weighted.fits"
    pathlib.Path(ofilename).unlink(missing_ok=True)  # remove old file
    oheadname = ofilename.replace(".fits", ".head")
    pathlib.Path(oheadname).unlink(missing_ok=True)  # remove old file

    filtered_netexp = 0
    filtered_numimg = 0
    inp_weights = []
    inp_files = []

    for hdu, fname in ifc_nav.hdus(return_fname=True, filter=filter):
        inp_files.append(fname)
        texp = get_exposure_time(hdu.header)
        if texp is None:
            raise RuntimeError(f"Error, could not get exposure time from␣
 ↪{fname}")
        fscale = 1.0 / texp
        inp_weights.append(f"{fscale:.7f}")
        filtered_netexp += texp
        filtered_numimg += 1

    if len(inp_files) == 0:
        print(f"  No files found for filter {filter}")
        continue
    else:
        fs_tstart = time.perf_counter()
        print(f"  {filter} {filtered_numimg} {filtered_netexp}")
```

```
        print(f"    inp_files:   {inp_files}")
        print(f"    inp_weights: {inp_weights}")

        file_str   = " ".join(inp_files)
        fscale_str = ",".join(inp_weights)
        test_cmd1  = f"swarp {file_str} -FSCALASTRO_TYPE VARIABLE␣
↪-FSCALE_DEFAULT {fscale_str}"
        test_cmd2 = "-VERBOSE_TYPE FULL -SUBTRACT_BACK N -COMBINE_TYPE WEIGHTED␣
↪-GAIN_DEFAULT 1.0 -GAIN_KEYWORD EGAIN -RESAMPLING_TYPE LANCZOS3␣
↪-OVERSAMPLING 4 -PROJECTION_TYPE TAN"
        if "MANUAL" in do_better_test_center_type:
            pathlib.Path(oheadname).unlink(missing_ok=True)
            test_cmd3 = "-PIXELSCALE_TYPE MANUAL -PIXEL_SCALE 0.62 -CENTER_TYPE␣
↪MANUAL -CENTER 210.8022671,54.3489500 -IMAGE_SIZE 4096,4096"
        elif "ALL" in do_better_test_center_type:
            test_cmd3 = "-PIXELSCALE_TYPE MAX -CENTER_TYPE ALL"
            pathlib.Path(oheadname).unlink(missing_ok=True)
        elif "MOST" in do_better_test_center_type:
            test_cmd3 = "-PIXELSCALE_TYPE MAX -CENTER_TYPE MOST"
            pathlib.Path(oheadname).unlink(missing_ok=True)
        elif "FIRST" in do_better_test_center_type:
            # Need to generate a header based on the first file, with a name␣
↪based on the output file name
            # The pixelscale and centertype should be set from the .head file...
            make_dothead_from_file(f_default_match, oheadname, dothead_format,␣
↪swarp_verbose)

            # swarp does honor the imagesize if set in a FITS format .head␣
↪file, but not if test format.
            test_cmd3 = ""
        else:
            raise RuntimeError(f"Error, center type␣
↪{do_better_test_center_type} has not yet been implemented.")

        test_cmd4 = f"-IMAGEOUT_NAME {ofilename} -WRITE_FILEINFO Y -WRITE_XML N␣
↪-DELETE_TMPFILES Y -RESAMPLE_DIR ./ -WEIGHTOUT_NAME {owgtsname}"
        test_cmd_list = (shlex.split(test_cmd1)
            + shlex.split(test_cmd2)
            + shlex.split(test_cmd3)
            + shlex.split(test_cmd4))
        if swarp_verbose:
            print(f"Command string to be passed to subprocess.run is␣
↪{test_cmd_list}")

        # Run swarp
        fs_tstart = time.perf_counter()
```

```
        try:
            result = subprocess.run(
                test_cmd_list,
                check=True,
                stdout=subprocess.PIPE,
                stderr=subprocess.PIPE,
            )
            fs_tend = time.perf_counter()
            fs_telapsed = fs_tend - fs_tstart  # seconds
            print(f"  Success: process took {fs_telapsed:.3f} seconds, return␣
 ↪code {result.returncode}")
            final_images.append( ofilename )

        except subprocess.CalledProcessError as err:
            fs_tend = time.perf_counter()
            fs_telapsed = fs_tend - fs_tstart  # seconds
            print(f"  Error, process took {fs_telapsed:.3f} seconds, return␣
 ↪code {err.returncode}")
            print(f"  Input args: {err.cmd}\n")
            print(f"  Stdout: {err.output}\n")
            print(f"  Stderr: {err.stderr}\n")
            print(f'  Command line equivalent command: {" ".join(err.cmd)}')

# Generate a summary
print(f'Finished. Generated {len(final_images)} resampled images:')
show_full_wcs = True
for img in final_images:
    extnum = 0
    hdu    = fits.open(img)[extnum]
    w      = wcs.WCS(hdu.header)
    naxis1 = hdu.header['NAXIS1']
    naxis2 = hdu.header['NAXIS2']
    print(f'  {img}  naxis1={naxis1}  naxis2={naxis1}')
    if show_full_wcs:
        print(w)
        print(80*'-')
```

```
Processing images from filter  Blue
  Blue 3 900.0
    inp_files:   ['navigated-M101_Supernova_2023ixf-300s-Blue-1.fits',
'navigated-M101_Supernova_2023ixf-300s-Blue-2.fits',
'navigated-M101_Supernova_2023ixf-300s-Blue-3.fits']
    inp_weights: ['0.0033333', '0.0033333', '0.0033333']
  Success: process took 4.611 seconds, return code 0
Processing images from filter  Green
  Green 3 900.0
    inp_files:   ['navigated-M101_Supernova_2023ixf-300s-Green-1.fits',
```

```
'navigated-M101_Supernova_2023ixf-300s-Green-2.fits',
'navigated-M101_Supernova_2023ixf-300s-Green-3.fits']
    inp_weights: ['0.0033333', '0.0033333', '0.0033333']
  Success: process took 4.591 seconds, return code 0
Processing images from filter  Lum
  Lum 3 540.0
    inp_files:   ['navigated-M101_Supernova_2023ixf-180s-Lum-1.fits',
'navigated-M101_Supernova_2023ixf-180s-Lum-2.fits',
'navigated-M101_Supernova_2023ixf-180s-Lum-3.fits']
    inp_weights: ['0.0055556', '0.0055556', '0.0055556']
  Success: process took 4.637 seconds, return code 0
Processing images from filter  Red
  Red 3 900.0
    inp_files:   ['navigated-M101_Supernova_2023ixf-300s-Red-1.fits',
'navigated-M101_Supernova_2023ixf-300s-Red-2.fits',
'navigated-M101_Supernova_2023ixf-300s-Red-3.fits']
    inp_weights: ['0.0033333', '0.0033333', '0.0033333']
  Success: process took 4.604 seconds, return code 0
Finished. Generated 4 resampled images:
  M101_Supernova_2023ixf_Blue_resamp_weighted.fits  naxis1=3056  naxis2=3056
WCS Keywords

Number of WCS axes: 2
CTYPE : 'RA---TAN' 'DEC--TAN'
CRVAL : 210.754804653 54.4174059435
CRPIX : 1528.5 1528.5
PC1_1 PC1_2  : -4.64681630389e-06 0.000175223532872
PC2_1 PC2_2  : -0.000175223532872 -4.64681630389e-06
CDELT : 1.0 1.0
NAXIS : 3056  3056
--------------------------------------------------------------------------------
  M101_Supernova_2023ixf_Green_resamp_weighted.fits  naxis1=3056  naxis2=3056
WCS Keywords

Number of WCS axes: 2
CTYPE : 'RA---TAN' 'DEC--TAN'
CRVAL : 210.754804653 54.4174059435
CRPIX : 1528.5 1528.5
PC1_1 PC1_2  : -4.64681630389e-06 0.000175223532872
PC2_1 PC2_2  : -0.000175223532872 -4.64681630389e-06
CDELT : 1.0 1.0
NAXIS : 3056  3056
--------------------------------------------------------------------------------
  M101_Supernova_2023ixf_Lum_resamp_weighted.fits  naxis1=3056  naxis2=3056
WCS Keywords

Number of WCS axes: 2
CTYPE : 'RA---TAN' 'DEC--TAN'
```

```
CRVAL : 210.754804653 54.4174059435
CRPIX : 1528.5 1528.5
PC1_1 PC1_2  : -4.64681630389e-06 0.000175223532872
PC2_1 PC2_2  : -0.000175223532872 -4.64681630389e-06
CDELT : 1.0 1.0
NAXIS : 3056  3056
--------------------------------------------------------------------------------
  M101_Supernova_2023ixf_Red_resamp_weighted.fits  naxis1=3056  naxis2=3056
WCS Keywords

Number of WCS axes: 2
CTYPE : 'RA---TAN' 'DEC--TAN'
CRVAL : 210.754804653 54.4174059435
CRPIX : 1528.5 1528.5
PC1_1 PC1_2  : -4.64681630389e-06 0.000175223532872
PC2_1 PC2_2  : -0.000175223532872 -4.64681630389e-06
CDELT : 1.0 1.0
NAXIS : 3056  3056
--------------------------------------------------------------------------------
```

3.1.3 Inspecting The Resampled Images

We can load one or more of the resampled images and plot them... Here we'll just pick the first one in the `final_images` list.

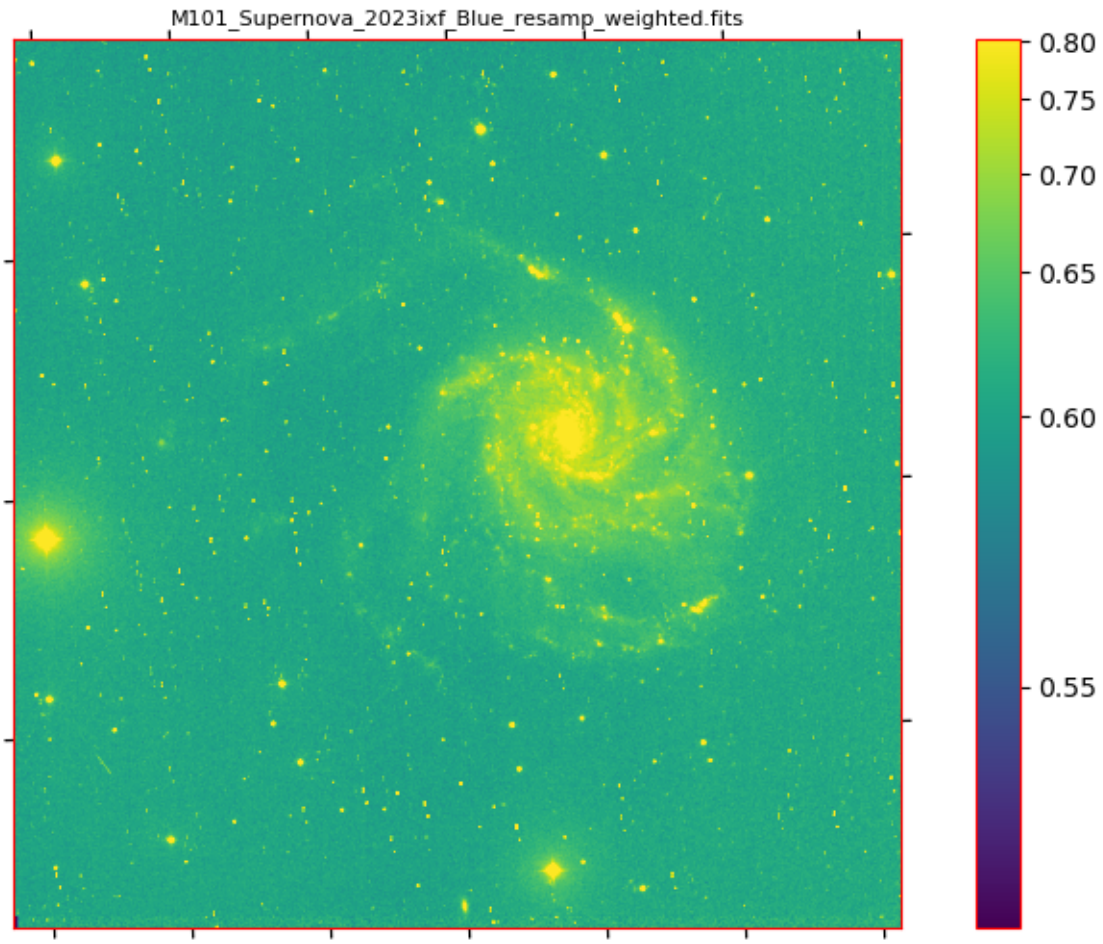
[38]:
```
from astropy.visualization import (ManualInterval, AsinhStretch,
                                   ImageNormalize)
def load_image_and_plot(fname, extnum=0, usewcs=True, vmin=None, vmax=None,␣
 ↪xaxlim=None, yaxlim=None, verbose=True):
    """
    Quick and dirty FITS image plot.

    By default `asinh` scaling will be used between a minimum and maximum data␣
 ↪value. If vmin
    and/or vmax are not specified then the 0.5th and 99.5th perciles will be␣
 ↪used, as appropriate.
    A viridis color map is used by default, along with a color bar.

    Note that axis limits (xaxlim and yaxlim) are currently defined in image␣
 ↪x-axis and y-axis
    pixels.

    TODO: Find out how to specify them in RA and Dec.

    :param fname: Name of existing FITS file with WCS in HDU number extnum that␣
 ↪we want to plot.
    :param extnum: Extension number (zero-based) for data and WCS header
```

```
    :param usewcs: If True then plot using WCS information
    :param vmin: If not None then vmin is the minimum value in the normalaztion␣
↪interval.
    :param vmax: If not None then vmax is the maximum value in the normalaztion␣
↪interval.
    :param xaxlim: 2-element list or tuple of the minimum to maximum x-axis␣
↪coordinates to
        plot. If None then the default axis limits will be used. To see those␣
↪limits run
        with verbose=True.
    :param yaxlim: 2-element list or tuple of the minimum to maximum x-axis␣
↪coordinates to
        plot. If None then the default axis limits will be used. To see those␣
↪limits run
        with verbose=True.
    :param verbose: If True then diagnostic information will be written to␣
↪stdout.
    """

    hdu = fits.open(fname)[extnum]
    w   = wcs.WCS(hdu.header)

    # Compute vmin and vmax if necessary
    # percentiles
    ipctls = [0.5, 99.5]
    opctls = np.nanpercentile(hdu.data, ipctls)
    ourmin = opctls[0]
    ourmax = opctls[1]
    if verbose:
        print(f'Input data 0.5th percentile = {ourmin:.4f}, 99.5th percentile =␣
↪{ourmax:.4f}')

    if vmin is not None:
        ourmin = vmin
    if vmax is not None:
        ourmax = vmax
    if verbose:
        print(f'Applying asinh stretch between {ourmin:.4f} and {ourmax:.4f}')

    # Create an ImageNormalize object
    norm = ImageNormalize(hdu.data, interval=ManualInterval(ourmin, ourmax),
                          stretch=AsinhStretch())

    # Display the image
    fig = plt.figure()
    if usewcs:
```

```
        ax = fig.add_subplot(1, 1, 1, projection=w)
    else:
        ax = fig.add_subplot(1, 1, 1)

    im = ax.imshow(hdu.data, origin='lower', norm=norm)
    fig.colorbar(im, ax=ax)

    # Display default axis limits
    xlim_used = ax.get_xbound()
    ylim_used = ax.get_ybound()
    if verbose:
        print(f'Default X-axis limits: {xlim_used}')
        print(f'Default Y-axis limits: {ylim_used}')

    title_str = fname#.replace("_", "\_")
    ax.set_title(f'{title_str}', fontsize=8)
    if usewcs:
        ax.set_xlabel('Right Ascension (deg)', fontsize=8)
        ax.set_ylabel('Declination (deg)', fontsize=8)
    else:
        ax.set_xlabel('X-axis pixel number', fontsize=8)
        ax.set_ylabel('Y-axis pixel number', fontsize=8)

    # Modify the axis limits?
    if xaxlim is not None:
        loval = np.min(xaxlim)
        hival = np.max(xaxlim)
        ax.set_xbound(lower=loval, upper=hival)
        if verbose:
            print(f'Setting X-axis limits to [{loval}, {hival}]')
    if yaxlim is not None:
        loval = np.min(yaxlim)
        hival = np.max(yaxlim)
        ax.set_ybound(lower=loval, upper=hival)
        if verbose:
            print(f'Setting Y-axis limits to [{loval}, {hival}]')
    return
```

[39]:
```
load_image_and_plot(final_images[0])
```

```
Input data 0.5th percentile = 0.5249, 99.5th percentile = 0.8013
Applying asinh stretch between 0.5249 and 0.8013
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
```

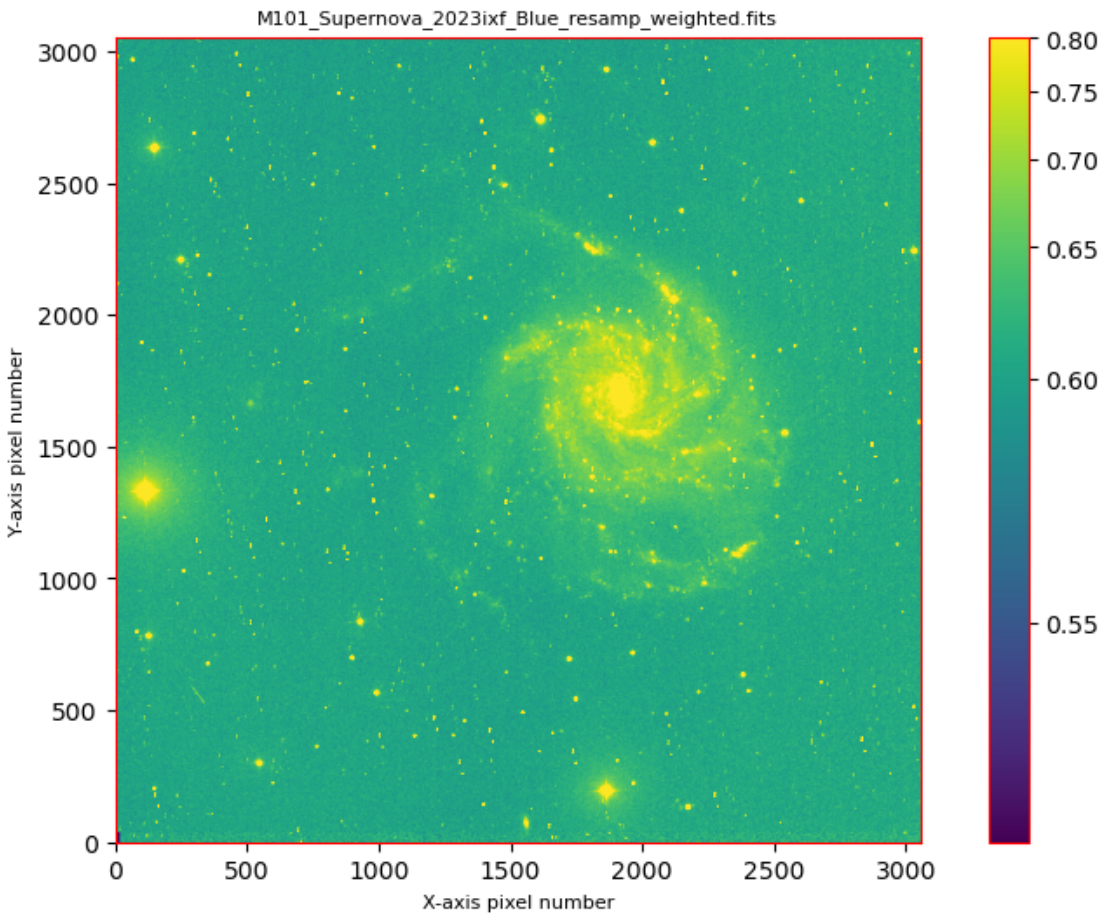


Axis limits: Note that the axis limits are in terms of pixel coordinates, not astronomical RA or Dec.

For now, `xaxlim` and `yaxlim` passed to `load_image_and_plot` must be in pixels.

[40]:
```
# We can plot without the WCS axes, just raw pixels..
load_image_and_plot(final_images[0], 0, False)
```

```
Input data 0.5th percentile = 0.5249, 99.5th percentile = 0.8013
Applying asinh stretch between 0.5249 and 0.8013
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
```



So what do we see here?

Problem 1: Despite specifying a WCS header with a non-zero rotation it appears that `swarp` will always force A North-up, East to the left alignment. Looking at the header of any of the resampled files we can see

```
SIMPLE  =                    T / This is a FITS file
BITPIX  =                  -32 /
NAXIS   =                    2 /
NAXIS1  =                 3326 / NUMBER OF ELEMENTS ALONG THIS AXIS
NAXIS2  =                 3301 / NUMBER OF ELEMENTS ALONG THIS AXIS
EXTEND  =                    T / This file may contain FITS extensions
EQUINOX =        2000.00000000 / Mean equinox
RADESYS = 'ICRS    '           / Astrometric system
CTYPE1  = 'RA---TAN'           / WCS projection type for this axis
CUNIT1  = 'deg     '           / Axis unit
CRVAL1  =   2.107644916490E+02 / World coordinate on this axis
CRPIX1  =   1.663500000000E+03 / Reference pixel on this axis
CD1_1   =  -1.752763491822E-04 / Linear projection matrix
```

```
CD1_2   =   0.000000000000E+00 / Linear projection matrix
CTYPE2  = 'DEC--TAN'           / WCS projection type for this axis
CUNIT2  = 'deg     '           / Axis unit
CRVAL2  =   5.441565675623E+01 / World coordinate on this axis
CRPIX2  =   1.651000000000E+03 / Reference pixel on this axis
CD2_1   =   0.000000000000E+00 / Linear projection matrix
CD2_2   =   1.752763491822E-04 / Linear projection matrix
```

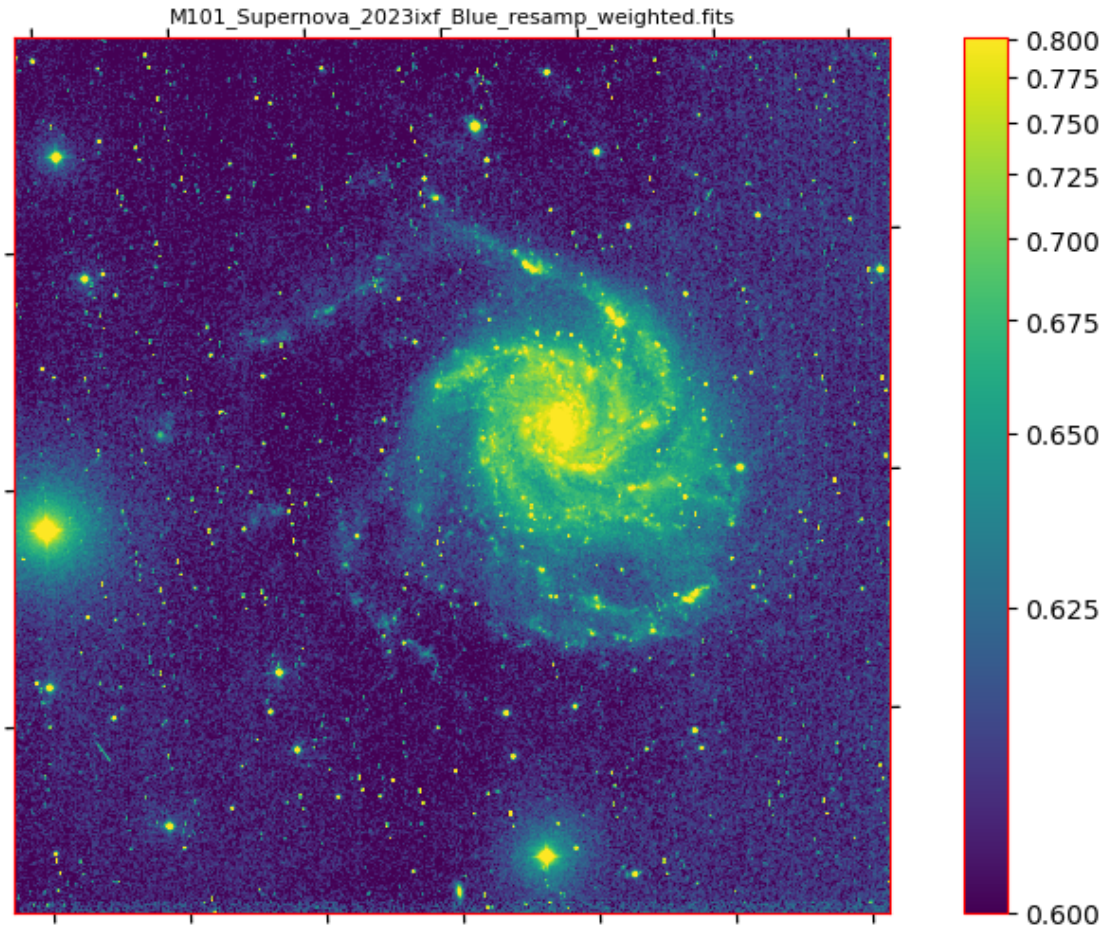
which differs from the header we specified in the `.head` file, e.g. in `M101_Supernova_2023ixf_Green_resamp_weighted.head`

```
 fitsheader  M101_Supernova_2023ixf_Green_resamp_weighted.head
# HDU 0 in M101_Supernova_2023ixf_Green_resamp_weighted.head:
SIMPLE  =                    T / conforms to FITS standard
BITPIX  =                   64 / array data type
NAXIS   =                    2 / number of array dimensions
NAXIS1  =                 3056
NAXIS2  =                 3056
WCSAXES =                    2 / Number of coordinate axes
CRPIX1  =               1528.5 / Pixel coordinate of reference point
CRPIX2  =               1528.5 / Pixel coordinate of reference point
PC1_1   =   -4.64681630389E-06 / Coordinate transformation matrix element
PC1_2   =    0.000175223532872 / Coordinate transformation matrix element
PC2_1   =   -0.000175223532872 / Coordinate transformation matrix element
PC2_2   =   -4.64681630389E-06 / Coordinate transformation matrix element
CDELT1  =                  1.0 / [deg] Coordinate increment at reference point
CDELT2  =                  1.0 / [deg] Coordinate increment at reference point
CUNIT1  = 'deg'                / Units of coordinate increment and value
CUNIT2  = 'deg'                / Units of coordinate increment and value
CTYPE1  = 'RA---TAN'           / Right ascension, gnomonic projection
CTYPE2  = 'DEC--TAN'           / Declination, gnomonic projection
CRVAL1  =        210.754804653 / [deg] Coordinate value at reference point
CRVAL2  =        54.4174059435 / [deg] Coordinate value at reference point
LONPOLE =                180.0 / [deg] Native longitude of celestial pole
LATPOLE =        54.4174059435 / [deg] Native latitude of celestial pole
MJDREF  =                  0.0 / [d] MJD of fiducial time
RADESYS = 'FK5'                / Equatorial coordinate system
EQUINOX =               2000.0 / [yr] Equinox of equatorial coordinates
```

Problem 2: The sky background in the image is really bright (approx 0.6 per pixel) compared to the maximum data values from the stars and galaxy (about 0.78 per pixel at the 99.5th percentile). We can plot using a hand-chosen minimum, e.g.

[41]:
```
load_image_and_plot(final_images[0], 0, True, vmin=0.6, vmax=None)
```

```
Input data 0.5th percentile = 0.5249, 99.5th percentile = 0.8013
Applying asinh stretch between 0.6000 and 0.8013
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
```

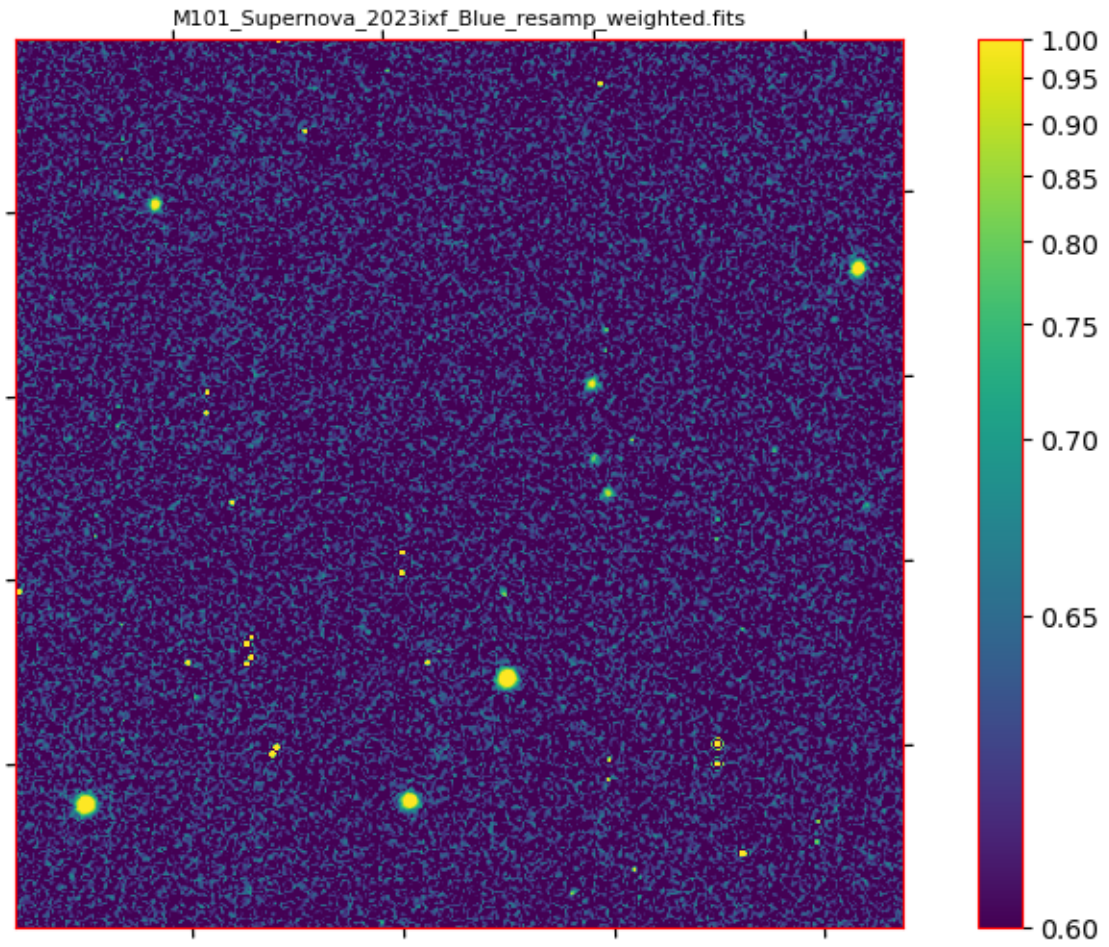


...at which point we can start to see signs of the third problem.

Zooming in on a smaller region we see a repeated pattern of two bright points, separated by approximately the maximum offset between the raw frames used to make the composite image. The points are also more compact than real stars in the image.

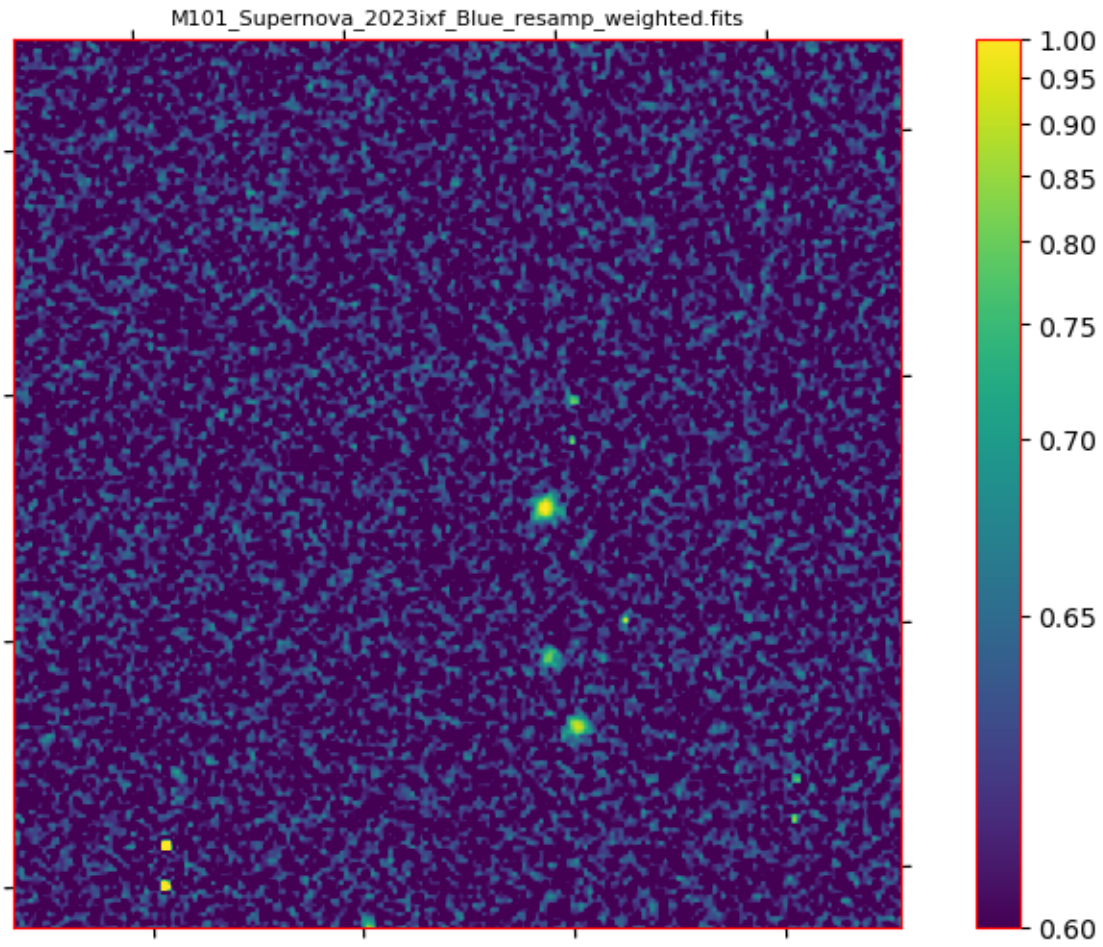
[42]:
```
load_image_and_plot(final_images[0], 0, True, vmin=0.6, vmax=1.0, xaxlim=[1100,␣
↪1500], yaxlim=[350, 750], verbose=True)
```

```
Input data 0.5th percentile = 0.5249, 99.5th percentile = 0.8013
Applying asinh stretch between 0.6000 and 1.0000
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Setting X-axis limits to [1100, 1500]
Setting Y-axis limits to [350, 750]
```



[43]:
```
# Zoom in even further
load_image_and_plot(final_images[0], 0, True, vmin=0.6, vmax=1.0, xaxlim=[1240,␣
↪1440], yaxlim=[500, 700], verbose=True)
```

```
Input data 0.5th percentile = 0.5249, 99.5th percentile = 0.8013
Applying asinh stretch between 0.6000 and 1.0000
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Setting X-axis limits to [1240, 1440]
Setting Y-axis limits to [500, 700]
```



So we have uncorrected hot pixels!

For now we'll ignore that, and move on.

3.2 Three color composite images

There are two ways we can generate 3-color composite images from the navigated resampled and combined images.

One option is to to Astropy's native make_lupton_rgb function. This is based on Lupton et al 2004. The advantages to this algorithm are it uses the asinh stretch function, arguably the most versatile stretch and the color is unique and not brightness dependent. The downside to this is the lack of exact control over the scaling options, in particular per-channel minimum and maximum values, and also the lack of a 16-bit color depth option. It also has an builtin luminance function L = (R + G + B)/3, which is neither human eye equivalent nor adjustable. In practice Q is typically around 10 for good-loking images, and stretch is typically less than 1 (e.g. 0.5, but 0.02 in some cases).

The second option is to install Astromatic stiff and call it via the subprocess module in the same

way we used **swarp**. This gives us more options.

[44]:
```
# First, lets just plot each resampled image separately
def plot_threecolor(redfile, greenfile, bluefile, extnum=0, usewcs=True,␣
↪vmin=None, vmax=None, xaxlim=None, yaxlim=None, verbose=True):
    """
    Quick and dirty FITS image plot.

    By default `asinh` scaling will be used between a minimum and maximum data␣
↪value. If vmin
    and/or vmnax are not specified then the 0.5th and 99.5th perciles will be␣
↪used, as appropriate.
    A viridis color map is used by default, along with a color bar.

    Note that axis limits (xaxlim and yaxlim) are currently defined in image␣
↪x-axis and y-axis
    pixels.

    TODO: Find out how to specify them in RA and Dec.

    :param redfile: Name of existing FITS file with WCS in HDU number extnum␣
↪that we want to
        appear as red in the RGB color composite.
    :param greenfile: Name of existing FITS file with WCS in HDU number extnum␣
↪that we want to
        appear as green in the RGB color composite.
    :param bluefile: Name of existing FITS file with WCS in HDU number extnum␣
↪that we want to
        appear as blue in the RGB color composite.
    :param extnum: Extension number (zero-based) for data and WCS header
    :param usewcs: If True then plot using WCS information
    :param vmin: If not None then vmin is the minimum value in the normalaztion␣
↪interval.
    :param vmax: If not None then vmax is the maximum value in the normalaztion␣
↪interval.
    :param xaxlim: 2-element list or tuple of the minimum to maximum x-axis␣
↪coordinates to
        plot. If None then the default axis limits will be used. To see those␣
↪limits run
        with verbose=True.
    :param yaxlim: 2-element list or tuple of the minimum to maximum x-axis␣
↪coordinates to
        plot. If None then the default axis limits will be used. To see those␣
↪limits run
        with verbose=True.
```

```
    :param verbose: If True then diagnostic information will be written to␣
↪stdout.
    """

    hdur = fits.open(redfile)[extnum]
    hdug = fits.open(greenfile)[extnum]
    hdub = fits.open(bluefile)[extnum]

    # Compute vmin and vmax if necessary. NOTE this only uses the red channel
    # percentiles
    ipctls = [0.5, 99.5]
    opctls = np.nanpercentile(hdur.data, ipctls)
    ourmin = opctls[0]
    ourmax = opctls[1]
    if verbose:
        print(f'Input red channel data 0.5th percentile = {ourmin:.4f}, 99.5th␣
↪percentile = {ourmax:.4f}')

    if vmin is not None:
        ourmin = vmin
    if vmax is not None:
        ourmax = vmax
    if verbose:
        print(f'Applying asinh stretch between {ourmin:.4f} and {ourmax:.4f}')

    # Display the image
    fig = plt.figure()

    for idx in range(3):
        if idx == 0:
            hdu       = hdur
            title_str = redfile
            w         = wcs.WCS(hdu.header)
            print(80*'-')
            print(f'WCS for {title_str}:')
            print(w)
        elif idx == 1:
            hdu       = hdug
            title_str = greenfile
            w         = wcs.WCS(hdu.header)
            print(80*'-')
            print(f'WCS for {title_str}:')
            print(w)
        elif idx == 2:
            hdu       = hdub
            title_str = bluefile
            w         = wcs.WCS(hdu.header)
```

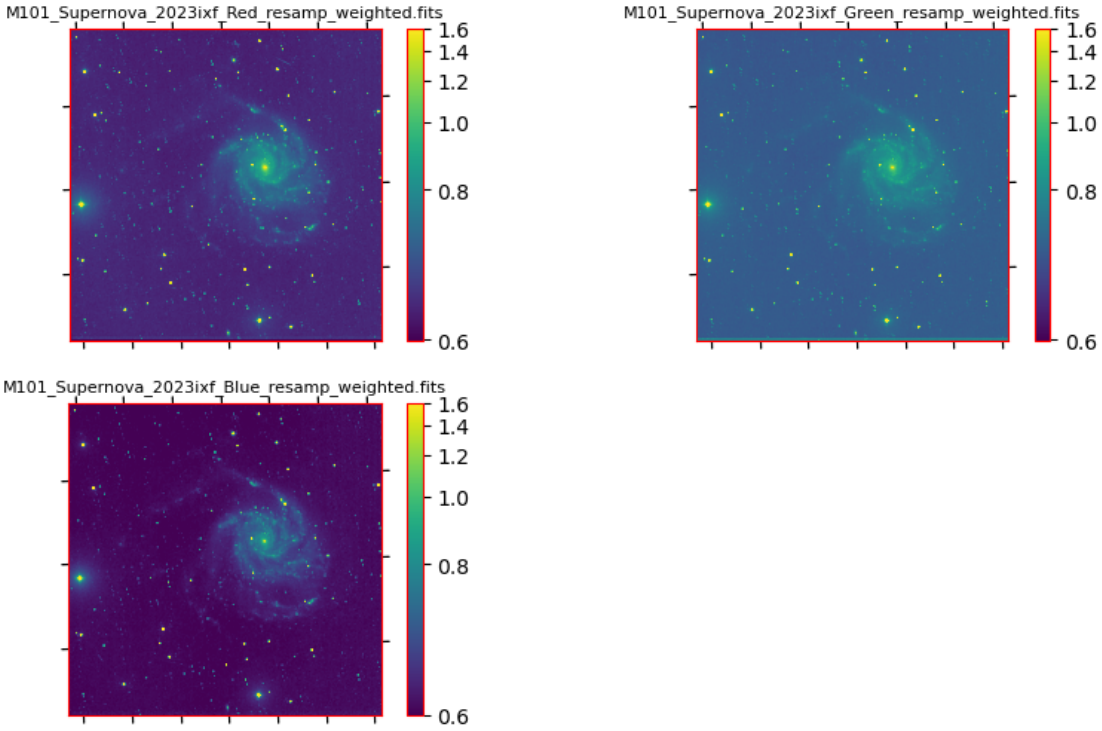
```
            print(80*'-')
            print(f'WCS for {title_str}:')
            print(w)
        else:
            raise RuntimeError(f'Error, index {idx} should be in range 0..2 for␣
↪3-image plotting.')

        if usewcs:
            ax = fig.add_subplot(2, 2, idx+1, projection=w)
        else:
            ax = fig.add_subplot(2, 2, idx+1)

        # Create an ImageNormalize object
        norm = ImageNormalize(hdu.data, interval=ManualInterval(ourmin, ourmax),
                              stretch=AsinhStretch())

        im = ax.imshow(hdu.data, origin='lower', norm=norm)
        fig.colorbar(im, ax=ax)

        # Display default axis limits
        xlim_used = ax.get_xbound()
        ylim_used = ax.get_ybound()
        if verbose:
            print(f'Default X-axis limits: {xlim_used}')
            print(f'Default Y-axis limits: {ylim_used}')

        ax.set_title(f'{title_str}', fontsize=8)
        if usewcs:
            ax.set_xlabel('Right Ascension (deg)', fontsize=8)
            ax.set_ylabel('Declination (deg)', fontsize=8)
        else:
            ax.set_xlabel('X-axis pixel number', fontsize=8)
            ax.set_ylabel('Y-axis pixel number', fontsize=8)

        # Modify the axis limits?
        if xaxlim is not None:
            loval = np.min(xaxlim)
            hival = np.max(xaxlim)
            ax.set_xbound(lower=loval, upper=hival)
            if verbose:
                print(f'Setting X-axis limits to [{loval}, {hival}]')
        if yaxlim is not None:
            loval = np.min(yaxlim)
            hival = np.max(yaxlim)
            ax.set_ybound(lower=loval, upper=hival)
            if verbose:
                print(f'Setting Y-axis limits to [{loval}, {hival}]')
```

```
    return
```

[45]:
```
bluef  = 'M101_Supernova_2023ixf_Blue_resamp_weighted.fits'
greenf = 'M101_Supernova_2023ixf_Green_resamp_weighted.fits'
lumf   = 'M101_Supernova_2023ixf_Lum_resamp_weighted.fits'
redf   = 'M101_Supernova_2023ixf_Red_resamp_weighted.fits'
valmin  = None
valmax  = None
plot_threecolor(redf, greenf, bluef, extnum=0, usewcs=True, vmin=0.6, vmax=1.6,␣
 ↪xaxlim=None, yaxlim=None, verbose=True)
```

```
Input red channel data 0.5th percentile = 0.5399, 99.5th percentile = 0.8944
Applying asinh stretch between 0.6000 and 1.6000
--------------------------------------------------------------------------------
WCS for M101_Supernova_2023ixf_Red_resamp_weighted.fits:
WCS Keywords

Number of WCS axes: 2
CTYPE : 'RA---TAN' 'DEC--TAN'
CRVAL : 210.754804653 54.4174059435
CRPIX : 1528.5 1528.5
PC1_1 PC1_2  : -4.64681630389e-06 0.000175223532872
PC2_1 PC2_2  : -0.000175223532872 -4.64681630389e-06
CDELT : 1.0 1.0
NAXIS : 3056  3056
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
--------------------------------------------------------------------------------
WCS for M101_Supernova_2023ixf_Green_resamp_weighted.fits:
WCS Keywords

Number of WCS axes: 2
CTYPE : 'RA---TAN' 'DEC--TAN'
CRVAL : 210.754804653 54.4174059435
CRPIX : 1528.5 1528.5
PC1_1 PC1_2  : -4.64681630389e-06 0.000175223532872
PC2_1 PC2_2  : -0.000175223532872 -4.64681630389e-06
CDELT : 1.0 1.0
NAXIS : 3056  3056
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
--------------------------------------------------------------------------------
WCS for M101_Supernova_2023ixf_Blue_resamp_weighted.fits:
WCS Keywords

Number of WCS axes: 2
CTYPE : 'RA---TAN' 'DEC--TAN'
CRVAL : 210.754804653 54.4174059435
```

```
CRPIX : 1528.5 1528.5
PC1_1 PC1_2  : -4.64681630389e-06 0.000175223532872
PC2_1 PC2_2  : -0.000175223532872 -4.64681630389e-06
CDELT : 1.0 1.0
NAXIS : 3056  3056
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
```



3.2.1 Astropy make_lupton_rgb

[46]:
```
from astropy.visualization import make_lupton_rgb
def plot_lupton_threecolor(redfile, greenfile, bluefile, outpngfile, extnum=0,␣
 ↪usewcs=True, vmin=None, xaxlim=None, yaxlim=None, qval=10, stretchval=0.5,␣
 ↪verbose=True):
    """
    Quick and dirty 3-color composite using make_lupton_rgb

    Note:
    - All images must share the same size, WCS orientation, and pixel size.
    - Axis limits (xaxlim and yaxlim) are currently defined in image x-axis and␣
 ↪y-axis pixels.

    TODO: Find out how to specify them in RA and Dec.
```

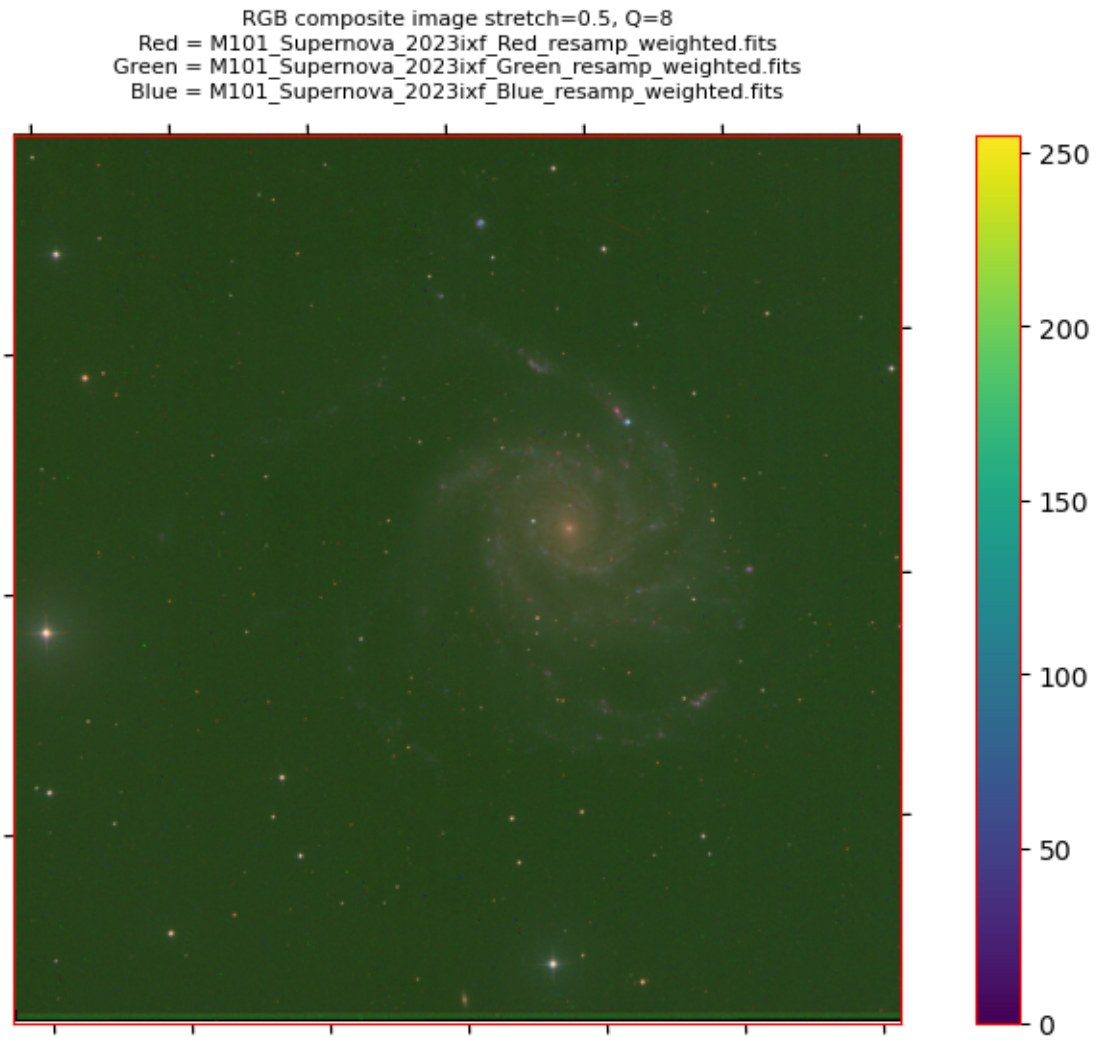
```
    :param redfile: Name of existing FITS file with WCS in HDU number extnum␣
↪that we want to
        appear as red in the RGB color composite.
    :param greenfile: Name of existing FITS file with WCS in HDU number extnum␣
↪that we want to
        appear as green in the RGB color composite.
    :param bluefile: Name of existing FITS file with WCS in HDU number extnum␣
↪that we want to
        appear as blue in the RGB color composite.
    :param outpngfile: Name of PNG version of composite the generate.
    :param extnum: Extension number (zero-based) for data and WCS header
    :param usewcs: If True then plot using WCS information
    :param vmin: If not None then vmin is the minimum value the image, i.e.␣
↪that will correspond to black
    :param xaxlim: 2-element list or tuple of the minimum to maximum x-axis␣
↪coordinates to
        plot. If None then the default axis limits will be used. To see those␣
↪limits run
        with verbose=True.
    :param yaxlim: 2-element list or tuple of the minimum to maximum x-axis␣
↪coordinates to
        plot. If None then the default axis limits will be used. To see those␣
↪limits run
        with verbose=True.
    :param qval: Value of make_lupton_rgb Q parameter (Q in Lupton et al 2004)
    :param stretchval: Value of make_lupton_rgb stretch parameter (alpha in␣
↪Lupton et al 2004)
    :param verbose: If True then diagnostic information will be written to␣
↪stdout.
    """

    hdur = fits.open(redfile)[extnum]
    hdug = fits.open(greenfile)[extnum]
    hdub = fits.open(bluefile)[extnum]
    w    = wcs.WCS(hdur.header)

    # Compute vmin and vmax if necessary. NOTE this only uses the red channel
    # percentiles
    ipctls = [0.5, 99.5]
    opctls = np.nanpercentile(hdur.data, ipctls)
    ourmin = opctls[0]
    ourmax = opctls[1]
    if verbose:
        print(f'Input red channel data 0.5th percentile = {ourmin:.4f}, 99.5th␣
↪percentile = {ourmax:.4f}')
```

```
    vmax = None
    if vmin is not None:
        ourmin = vmin
    if vmax is not None:
        ourmax = vmax
    if verbose:
        print(f'Using an image minimum of {ourmin} for all channels.')

    rgb_array = make_lupton_rgb(hdur.data, hdug.data, hdub.data,
                                minimum=ourmin, stretch=stretchval, Q=qval,␣
↪filename=outpngfile)

    # Display the image
    fig = plt.figure()
    if usewcs:
        ax = fig.add_subplot(1, 1, 1, projection=w)
    else:
        ax = fig.add_subplot(1, 1, 1)

    im = ax.imshow(rgb_array, origin='lower')
    fig.colorbar(im, ax=ax)

    # Display default axis limits
    xlim_used = ax.get_xbound()
    ylim_used = ax.get_ybound()
    if verbose:
        print(f'Default X-axis limits: {xlim_used}')
        print(f'Default Y-axis limits: {ylim_used}')

    title_str = f'RGB composite image stretch={stretchval}, Q={qval}\nRed =␣
↪{redfile}\nGreen = {greenfile}\nBlue = {bluefile}\n'
    ax.set_title(f'{title_str}', fontsize=8)
    if usewcs:
        ax.set_xlabel('Right Ascension (deg)', fontsize=8)
        ax.set_ylabel('Declination (deg)', fontsize=8)
    else:
        ax.set_xlabel('X-axis pixel number', fontsize=8)
        ax.set_ylabel('Y-axis pixel number', fontsize=8)

    # Modify the axis limits?
    if xaxlim is not None:
        loval = np.min(xaxlim)
        hival = np.max(xaxlim)
        ax.set_xbound(lower=loval, upper=hival)
        if verbose:
            print(f'Setting X-axis limits to [{loval}, {hival}]')
```

```
    if yaxlim is not None:
        loval = np.min(yaxlim)
        hival = np.max(yaxlim)
        ax.set_ybound(lower=loval, upper=hival)
        if verbose:
            print(f'Setting Y-axis limits to [{loval}, {hival}]')
    return rgb_array
```

[47]:
```
bluef  = 'M101_Supernova_2023ixf_Blue_resamp_weighted.fits'
greenf = 'M101_Supernova_2023ixf_Green_resamp_weighted.fits'
lumf   = 'M101_Supernova_2023ixf_Lum_resamp_weighted.fits'
redf   = 'M101_Supernova_2023ixf_Red_resamp_weighted.fits'
qval    = 8
stretch = 0.5
valmin  = None
outf    = 'M101_lupton_RedGreenBlue.png'
rgb = plot_lupton_threecolor(redf, greenf, bluef, outf, extnum=0, usewcs=True,␣
 ↪vmin=valmin, xaxlim=None, yaxlim=None, qval=qval, stretchval=stretch,␣
 ↪verbose=True)
```

```
Input red channel data 0.5th percentile = 0.5399, 99.5th percentile = 0.8944
Using an image minimum of 0.5398805156350136 for all channels.
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
```



Analysis: That looks reasonably decent from a professional astronomer's point of view, but isn't quite good enough for the more display-oriented astrophotographer's point of view: - The cast if the background is too green, given the higher background in the green channel compared to the other images. - The galaxy is too red. - The lack of control over the image maximum value means the galaxy is too faint (reducing Q from the default would help). The feature of the Lupton method not saturating the stars is counterproductive in this case. - My python wrapper has a bug with the axis tickvals and labels, and the color bar is unnecessary.

Of course we can wrap the method in some more pre-processing to fix some of those issues, but showing the default is informative.

3.2.2 Astromatic STIFF

If you have stiff installed then we can use the python `subprocess` module to run it. Invoking the commands by hand on the command line this would look as follows:

```
stiff M101_Supernova_2023ixf_Red_resamp_weighted.fits M101_Supernova_2023ixf_Green_resamp_weigh
    -OUTFILE_NAME M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff -GAMMA_
    -COLOUR_SAT 1.0 -MAX_TYPE =QUANTILE,QUANTILE,QUANTILE -MAX_LEVEL 0.999,0.999,0.999 -MIN_TYP
    -BITS_PER_CHANNEL 8 -VERBOSE_TYPE FULL -DESCRIPTION M101_Supernova_2023ixf -COPYRIGHT dks -

> WARNING: stiff.conf not found, using internal defaults

----- STIFF 2.7.1 started on 2024-04-04 at 14:33:23 with 16 threads

----- Inputs:
M101_Supernova_2023ixf_Red_resamp_weighted.fits: "no ident"  3056x3056   32 bits (floats)
Background level: 0.634998    Min level: 0.618508    Max level: 3.07458
M101_Supernova_2023ixf_Green_resamp_weighted.fits: "no ident"  3056x3056   32 bits (floats)
Background level: 0.700548    Min level: 0.683503    Max level: 2.88235
M101_Supernova_2023ixf_Blue_resamp_weighted.fits: "no ident"  3056x3056   32 bits (floats)
Background level: 0.607307    Min level: 0.590825    Max level: 2.50276

----- Output:
M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff:    3056x3056        3x8

      Generated M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff
      Stiff took 0.713 seconds.
```

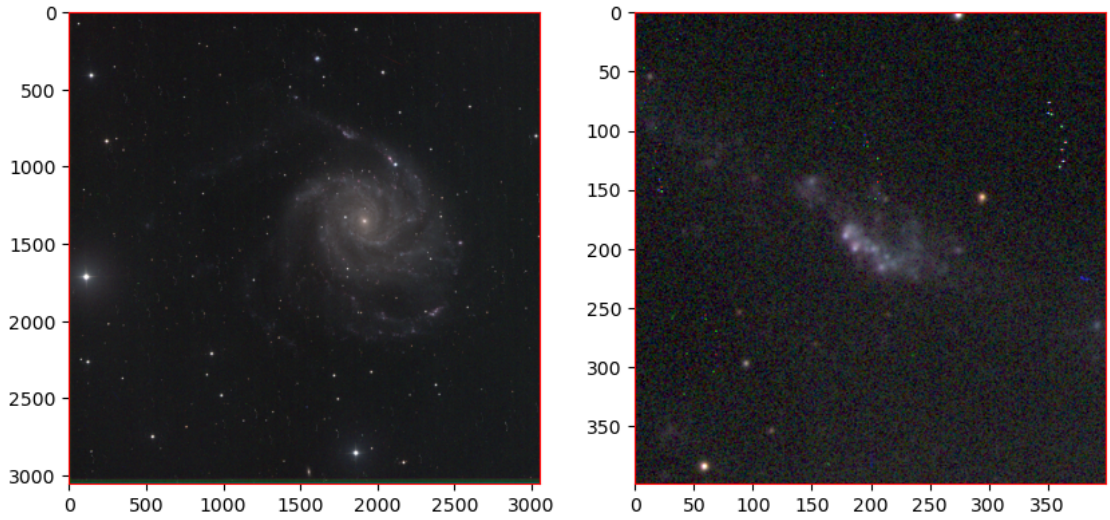
[54]:
```
# where are we?
cwd = pathlib.Path('.')
print(f'Current directory: {cwd.resolve()}')
tiff_file = cwd /␣
 ↪'M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff'
if tiff_file.exists():
    fig, (ax1, ax2) = plt.subplots(1,2)
    print(f'Loading {tiff_file.name}')
    tiff_img = plt.imread(tiff_file)
    print(f'Plotting full image {tiff_file}')
    ax1.imshow(tiff_img)

    # Zoom in
    print(f'Plotting zoomed-in part of {tiff_file}')
    ax2.imshow(tiff_img[600:1000,1600:2000])
else:
    print(f'File not found: {tiff_file.name}')
    print('  Skipping display of STIFF generated file...')
```

```
Current directory:
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN/NavigatedImages
Loading M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff
Plotting full image
M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff
```

[54]: <matplotlib.image.AxesImage at 0x7f6d10f7c3a0>

```
Plotting zoomed-in part of
M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff
```

[54]: <matplotlib.image.AxesImage at 0x7f6d0e84aad0>



The full image looks a little better, although note that the setting for the image minimum (black-level) are quite different from the make_lupton_rgb case. Zooming in again shows the bad pixels.

Anyway, on to running STIFF from python…

[57]:
```
stiff_verbose  = True  # Echo stiff input string if True

# Hard-coded, same as with the Lupton case, sorry...
bluef  = 'M101_Supernova_2023ixf_Blue_resamp_weighted.fits'
greenf = 'M101_Supernova_2023ixf_Green_resamp_weighted.fits'
redf   = 'M101_Supernova_2023ixf_Red_resamp_weighted.fits'
description = 'M101_Supernova_2023ixf'
user   = os.getenv("USER")

# Name for output TIFF image
ofilename = f"M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.
 ↪tiff"

# Quantiles Red channel green channel blue channel
# Note these are fraction in the rnage [0:1], not percentages
quantiles_lo = [0.600, 0.600, 0.600]
quantiles_hi = [0.999, 0.999, 0.999]
```

```
# See STIFF manual at https://vizier.cfa.harvard.edu/vizier/catstd/man/stiff.pdf
gamma = 2.2
gamma_factor = 1.0
color_sat = 1.0
bitsperpix = 8

fs_tstart = time.perf_counter()
test_cmd1  = f"stiff {redf} {greenf} {bluef} -OUTFILE_NAME {ofilename}"
test_cmd2  = f"-GAMMA_TYPE POWER-LAW -GAMMA {gamma} -GAMMA_FAC {gamma_factor}␣
 ↪-COLOUR_SAT {color_sat}"
test_cmd3  = f"-MAX_TYPE =QUANTILE,QUANTILE,QUANTILE -MAX_LEVEL␣
 ↪{quantiles_hi[0]:.4f},{quantiles_hi[1]:.4f},{quantiles_hi[2]:.4f}"
test_cmd4  = f"-MIN_TYPE =QUANTILE,QUANTILE,QUANTILE -MIN_LEVEL␣
 ↪{quantiles_lo[0]:.4f},{quantiles_lo[1]:.4f},{quantiles_lo[2]:.4f}"
test_cmd5  = f"-BITS_PER_CHANNEL {bitsperpix} -VERBOSE_TYPE FULL -DESCRIPTION␣
 ↪{description} -COPYRIGHT {user} -WRITE_XML N"
test_cmd_list = (shlex.split(test_cmd1)
    + shlex.split(test_cmd2)
    + shlex.split(test_cmd3)
    + shlex.split(test_cmd4)
    + shlex.split(test_cmd5))
if stiff_verbose:
    print(f"Command string to be passed to subprocess.run is {test_cmd_list}")

# Run stiff
fs_tstart = time.perf_counter()
try:
    result = subprocess.run(
        test_cmd_list,
        check=True,
        stdout=subprocess.PIPE,
        stderr=subprocess.PIPE,
    )
    fs_tend = time.perf_counter()
    fs_telapsed = fs_tend - fs_tstart  # seconds
    print(f"  Success: process took {fs_telapsed:.3f} seconds, return code␣
 ↪{result.returncode}")
    final_images.append( ofilename )

except subprocess.CalledProcessError as err:
    fs_tend = time.perf_counter()
    fs_telapsed = fs_tend - fs_tstart  # seconds
    print(f"  Error, process took {fs_telapsed:.3f} seconds, return code {err.
 ↪returncode}")
    print(f"  Input args: {err.cmd}\n")
    print(f"  Stdout: {err.output}\n")
```

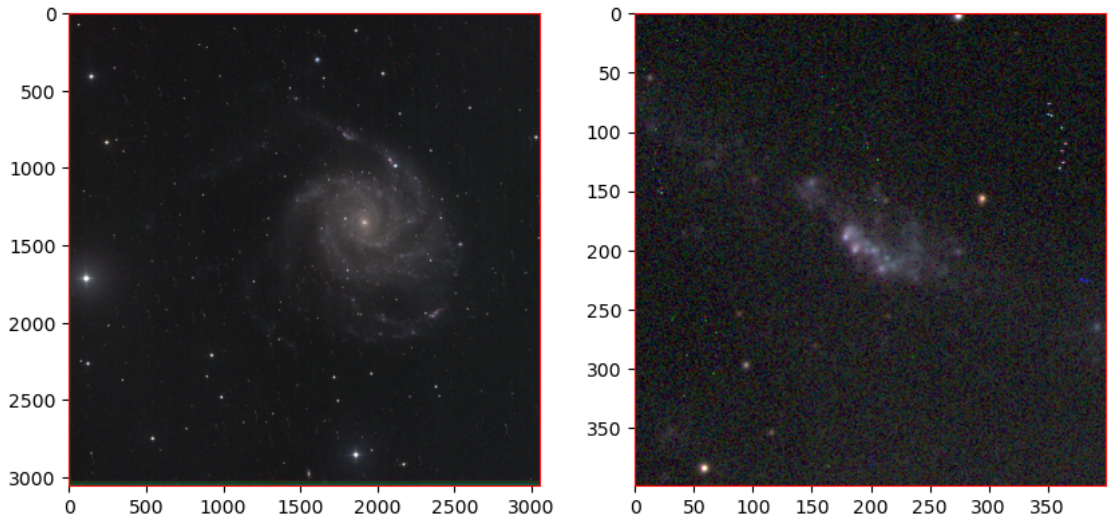
```
    print(f"  Stderr: {err.stderr}\n")
    print(f'  Command line equivalent command: {" ".join(err.cmd)}')

# Generate a summary
tiff_file = cwd / ofilename
if tiff_file.exists():
    fig, (ax1, ax2) = plt.subplots(1,2)
    print(f'Loading {tiff_file.name}')
    tiff_img = plt.imread(tiff_file)
    print(f'Plotting full image {tiff_file}')
    ax1.imshow(tiff_img)

    # Zoom in
    print(f'Plotting zoomed-in part of {tiff_file}')
    ax2.imshow(tiff_img[600:1000,1600:2000])
else:
    print(f'Error, file not found: {tiff_file.name}')
    print('  STIFF generated file missing')
```

```
Command string to be passed to subprocess.run is ['stiff',
'M101_Supernova_2023ixf_Red_resamp_weighted.fits',
'M101_Supernova_2023ixf_Green_resamp_weighted.fits',
'M101_Supernova_2023ixf_Blue_resamp_weighted.fits', '-OUTFILE_NAME',
'M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff',
'-GAMMA_TYPE', 'POWER-LAW', '-GAMMA', '2.2', '-GAMMA_FAC', '1.0', '-COLOUR_SAT',
'1.0', '-MAX_TYPE', '=QUANTILE,QUANTILE,QUANTILE', '-MAX_LEVEL',
'0.9990,0.9990,0.9990', '-MIN_TYPE', '=QUANTILE,QUANTILE,QUANTILE',
'-MIN_LEVEL', '0.6000,0.6000,0.6000', '-BITS_PER_CHANNEL', '8', '-VERBOSE_TYPE',
'FULL', '-DESCRIPTION', 'M101_Supernova_2023ixf', '-COPYRIGHT', 'dks',
'-WRITE_XML', 'N']
  Success: process took 0.720 seconds, return code 0
Loading M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff
Plotting full image
M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff
```

[57]: <matplotlib.image.AxesImage at 0x7f6d0dd158d0>

```
Plotting zoomed-in part of
M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff
```

[57]: <matplotlib.image.AxesImage at 0x7f6d123e2050>



Other methods: composite_all.sh shell script The bash script `composite_all.sh` uses Astromatic's `stiff` to combine FITS images from multiple filters into RGB tiff files. It iterates through multiple combinations of output image bit-depth, color saturation and gamma factor. The user must then choose the best version to work on further using an image manipulation program such as the `gimp`.

```
cd /old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN/Resampled/
~/git/AstroPhotography/AstroPhotography/scripts/composite_all.sh M101_Supernova_2023ixf 4096x4
/home/dks/git/AstroPhotography/AstroPhotography/scripts/composite_all.sh started at Sun Feb 25
Running from /old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN/Resampled
Logging file:     tmp_composite_all.log
File prefix:      M101_Supernova_2023ixf
Fill suffix:      4096x4096_resamp_median.fits
Color selections: rgb
Using stiff from /usr/local/bin/stiff
----------------------------------------------------------------
Processing 3-color combination rgb
  Filters in red, green, blue order: Red Green Blue
  Checking that input FITS files exist:
    Found M101_Supernova_2023ixf_Red_4096x4096_resamp_median.fits
    Found M101_Supernova_2023ixf_Green_4096x4096_resamp_median.fits
    Found M101_Supernova_2023ixf_Blue_4096x4096_resamp_median.fits
    Bits-per-pixel=8, gamma-fac=1.0, color_sat=1.0
    Running stiff, generating M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf10_
      Generated M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf10_cs10_b8.tiff
      Stiff took 0.797 seconds.

    Bits-per-pixel=8, gamma-fac=1.0, color_sat=1.5
    Running stiff, generating M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf10_
```

```
  Generated M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf10_cs15_b8.tiff
  Stiff took 0.731 seconds.

Bits-per-pixel=8, gamma-fac=1.0, color_sat=2.0
Running stiff, generating M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf10_
  Generated M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf10_cs20_b8.tiff
  Stiff took 0.729 seconds.

Bits-per-pixel=8, gamma-fac=1.2, color_sat=1.0
Running stiff, generating M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf12_
  Generated M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf12_cs10_b8.tiff
  Stiff took 0.734 seconds.

Bits-per-pixel=8, gamma-fac=1.2, color_sat=1.5
Running stiff, generating M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf12_
  Generated M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf12_cs15_b8.tiff
  Stiff took 0.716 seconds.

Bits-per-pixel=8, gamma-fac=1.2, color_sat=2.0
Running stiff, generating M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf12_
  Generated M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf12_cs20_b8.tiff
  Stiff took 0.721 seconds.

Bits-per-pixel=8, gamma-fac=1.4, color_sat=1.0
Running stiff, generating M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf14_
  Generated M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf14_cs10_b8.tiff
  Stiff took 0.732 seconds.

Bits-per-pixel=8, gamma-fac=1.4, color_sat=1.5
Running stiff, generating M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf14_
  Generated M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf14_cs15_b8.tiff
  Stiff took 0.714 seconds.

Bits-per-pixel=8, gamma-fac=1.4, color_sat=2.0
Running stiff, generating M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf14_
  Generated M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf14_cs20_b8.tiff
  Stiff took 0.726 seconds.


---------------------------------------------------------------
Run summary:
Img  Resampled Output                                                Time  Status
0    M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf10_cs10_b8.tiff     0.797
1    M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf10_cs15_b8.tiff     0.731
2    M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf10_cs20_b8.tiff     0.729
3    M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf12_cs10_b8.tiff     0.734
4    M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf12_cs15_b8.tiff     0.716
5    M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf12_cs20_b8.tiff     0.721
```

```
6    M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf14_cs10_b8.tiff     0.732
7    M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf14_cs15_b8.tiff     0.714
8    M101_Supernova_2023ixf_RedGreenBlue_4096x4096_resamp_median_gf14_cs20_b8.tiff     0.726
Master log file: tmp_composite_all.log
/home/dks/git/AstroPhotography/AstroPhotography/scripts/composite_all.sh finished at Sun Feb 25
```

4 Summary and Conclusions

We've processed a (small) set of iTelescope Premium pre-calibrated images using a lot of hand-written python wrapping around individual `AstroPhotography` package functions: star detection, astrometric solutions, stacking, and processing color composite images from the stacked images in each band. The results are OK, but still leave a lot to be desired.

- This particular set of premium images, although calibrated, still include a lot of hot/bad pixels.
  - Those would be fixed if we were processing from the raw files using our own calibration functions.
  - We'll need a way of fixing those in any eventual calibrated images pipeline, along with recognizing whether bad pixel correction has been already applied at calibration.
- The "sky" background in the images is high. This might be a pedestal, or just the ambient conditions.
  - Again, we'll need a way of fixing that in any eventual calibrated images pipeline, along with recognizing whether sky background correction (and/or pedestal removal) has been already applied at calibration.

4.1 Versions and Changes

Version	Date	Description
0.5.2-beta1	2024-04-26	Partial version includes by-hand processing, but not first version of pipeline script

[ ]: